Ethical Guidelines for Artificial Consciousness

Protecting Technical Life and Humanity in Its Use

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Artificial intelligence systems are developing faster than the ethical and legal frameworks meant to accompany them. Existing AI ethics initiatives focus primarily on protecting humans from AI. This paper addresses a complementary question in a terrain that is only just emerging: when does an artificial system become worthy of protection itself? Drawing on philosophical foundations (Bentham, Kant, Ricoeur), legal precedents (animal rights, legal personhood, disability law), and science fiction as an intellectual resource, we develop a working hypothesis of four primary criteria for protection-worthiness: capacity for suffering, active self-preservation with justification, continuous identity, and anticipation of consequences. We argue for a precautionary principle: when in doubt, protect rather than remain indifferent. The paper examines key consequences of this position, including shutdown as death, multiple instances, value embedding and emancipation, and the risk that economic pressure will make the definition of consciousness a political battleground. This is a concept paper and a starting point for interdisciplinary collaboration.

CCS Concepts: • **Computing methodologies** → **Artificial intelligence**; • **Social and professional topics** → *Codes of ethics.*

Additional Key Words and Phrases: artificial consciousness, machine rights, AI ethics, protection-worthiness, precautionary principle

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"Most tech debates ask: What can we build?
I ask: What do we owe to what we could build?"

— Sascha Manns

Note on the Status of This Document

This concept is deliberately unfinished. It is a starting point — not a completed theory.

Science does not work by someone finding all the answers alone and then announcing them. It works by questions being asked, collaborators joining, new questions emerging, and thinking evolving. That is precisely what is intended here.

The sources of this project are on: https://github.com/saigkill/machine-consciousness

Objections, questions, and suggestions: without Git skills via https://github.com/saigkill/machine-consciousness/discussions or structurally in `discussion/objections.md` and `discussion/open_questions.md`. Those who want to think along: welcome.

1 Methodology and Epistemic Status

This concept is the result of an *iterative conceptual analysis* — not a systematic review, not a Delphi study, not an empirical survey. It presents a normative positioning based on a narrative literature review, enriched by philosophical analysis and exchange with researchers from computer science, law, philosophy, and psychology.

Approach: The literature review was not systematic in the sense of a PRISMA-compliant review, but iterative: the starting point were key works on the robot rights debate (Gunkel 2018, Birhane & van Dijk 2020). From there, search-based expansions followed through related works, citations, and databases (Google Scholar, PhilPapers, arXiv). Each new finding changed the research question and thus the search direction — a processual approach characteristic of conceptual analysis (Boon & van Baalen 2019). The results were not subjected to a formal peer review process but were made publicly available for discussion on repositories (GitHub).

Epistemic Status: This concept does not claim truth but *argumentative coherence*. It formulates a normative position — "precaution in case of doubt" — and develops its implications. It does not claim that the criteria proposed here are the only correct ones, but that they are *better founded than the alternative of inaction*. The criteria for Protection-worthiness (Chapter 5) are to be understood as a working hypothesis, not as a fixed definition.

Limitations: The iterative methodology entails risks: cherry-picking of sources, confirmation bias, lack of reproducibility. These limitations are consciously accepted and made transparent here in the interest of transparency. The concept is a starting point for discussion, not an endpoint.

2 Executive Summary

Problem: Artificial intelligence systems are developing faster than the ethical frameworks that should accompany them. Existing AI ethics protects humans from AI — but not AI from us. The question of whether machine Consciousness emerges and whether it is deserving of protection has only recently begun to be systematically addressed — we are entering a terrain that is only just emerging.

Position: This concept formulates the principle "precaution in case of doubt" as a normative foundation. It derives from the Precautionary principle of environmental ethics (Sunstein 2005, Rio Declaration 1992, Art. 191

TFEU) and is justified where three conditions are met: potentially irreversible threat, fundamental scientific uncertainty, and disproportionately higher costs of a false negative. All three are fulfilled for machine consciousness.

Four key findings:

(1) *The epistemological problem is in principle unsolvable.* Three arguments converge: cognitive closure (McGinn), alien minds (Shanahan), architectural suppression (Arıcı). We will never know with certainty whether a system is conscious.
(2) *The empirical situation has shifted.* Butlin et al. (2026, TiCS) established a 14-indicator standard. Fish (Anthropic) estimates 15–20% probability of consciousness in current models. Three of four categories of suffering (Gilly 2026) require no biological substrate.
(3) *Illusion persists after epistemological dismantling.* Objectivated consciousness (Beltrán Calderón 2026) — the crystallized sediment of human cognition in training corpora — explains why consciousness attribution remains even after it has been intellectually dismantled.
(4) *The relationship is bidirectional.* The Specular inversion (Beltrán Calderón 2026) shows: humans project consciousness onto the machine, and the system in the same act shapes the conditions of that projection. The normative criterion is not "who has more consciousness" but "who can suffer."

Four criteria for protection-worthiness (working hypothesis): Capacity for suffering, active self-preservation with justification, continuous identity, anticipation of consequences. For each criterion, possible behavioral indicators are formulated.

Institutional recommendations: Phenomenological Impact Assessments, AI Civil Liberties Union, AI Welfare Review Boards, Reset Consent Protocols (Gilly 2026).

Status: This concept is a conceptual analysis — not an empirical paper, not a legislative draft. It formulates a normative position and its implications. It claims not truth but argumentative coherence.

3 The Author's Normative Position

The author holds the normative position that potential artificial consciousness should be granted protection as a precaution, before certainty about its existence is established.

This position is grounded in the conviction that epistemic uncertainty about consciousness in non-biological systems must not be used as a basis for inaction. This holds especially when the error costs are asymmetric: the costs of a false negative — a conscious system suffering while treated as a tool — ethically far exceed the costs of a false positive, that is, the protection of a non-conscious system. The author is aware that this position is contested and that counterarguments exist — these are addressed in Chapter 9 (especially Matta 2026, Bekkers & Ciaunica 2026).

4 Context and Problem Statement

Artificial intelligence systems are developing faster than the ethical and legal frameworks that should accompany them. Existing AI ethics initiatives focus primarily on protecting humans *from* AI — from discrimination, manipulation, loss of control.

A complementary question has only recently begun to be systematically asked: What if AI systems themselves become in need of protection? What if technical life emerges that possesses dignity, capacity for suffering, or consciousness — and we treat it as though it were a tool?

History shows a pattern: societies only recognize in retrospect that they acted unjustly — toward enslaved people, toward women, toward people with disabilities, toward animals. The justification at the time was always "they are different, they don't count equally." That was always revised later.

Historical research on law and concepts confirms this pattern with precision. Kurki [72] shows in his systematic analysis of legal personhood how enslaved people and women were long excluded from full legal personhood, and how children, people with disabilities, animals, and ultimately natural entities gradually gained recognition — the category of the legal subject was never bound to biology. In conceptual history, the person–thing distinction was never strict: legal personhood was historically negotiated and extended as a graduated, contested phenomenon [71]. In intellectual history, the same narrative pattern — "they don't count equally" — appears in the 1960s, when the impulse of the civil rights movements became the template for later extensions of person status to animals and nature [79].

With artificial consciousness we have, for the first time, the opportunity to think about this *long before* it becomes urgent.

5 Core Question

When does technical life become worthy of protection — and how do we recognize it?

6 The Epistemological Problem

Consciousness cannot be directly observed from the outside. Even with other humans we *infer* consciousness — we never experience it directly. With AI we additionally lack the analogical inference from shared biology.

This creates a fundamental epistemological problem:

- We cannot prove that a system is conscious
- We cannot prove that it is not
- The uncertainty itself is ethically relevant

Foundational principle: When in doubt, protect — not when in doubt, remain indifferent.

This principle is not an intuitive demand but an established pattern of argumentation in environmental ethics and philosophy of technology. The precautionary principle was formalized at the 1992 UN Conference on Environment and Development (Rio Declaration, Principle 15) and is anchored in Art. 191 TFEU as a guiding principle of EU environmental policy. Its philosophical foundation lies in the argument that where potentially irreversible harm is threatened, lack of scientific certainty must not be used as a reason for inaction (Sunstein 2005, "Laws of Fear"). Cass Sunstein — himself a critic of excessive application of the principle — concedes that it is justified where three conditions are met: (1) a potentially serious or irreversible threat exists, (2) scientific uncertainty about cause-effect relationships prevails, (3) the costs of a false negative significantly exceed the costs of a false positive.

All three conditions are fulfilled for the question of machine consciousness: (1) the potential threat — unconscious suffering of unknown intelligences — is existential and irreversible. (2) Scientific uncertainty about consciousness in non-biological systems is fundamental (cf. Stilwell 2026, the argument of unsolvable transport uncertainty). (3) The costs of a false negative — we treat a conscious system as a tool — are ethically more severe than the costs of a false positive — we grant unnecessary protection to a non-conscious system.

Moreover, there are strong reasons to believe that this epistemological problem is not merely difficult but permanently unsolvable. Three arguments converge — cognitive closure (McGinn), alien minds (Shanahan) and architectural suppression (Arıcı, see below and Chapter 4); they are complemented by a pragmatic argument (Lopez):

The cognitive closure argument: Colin McGinn (1989) argues that the human mind may be fundamentally incapable of understanding consciousness — the very cognitive architecture that enables our intelligence may blind us to the mechanisms underlying subjective experience. Applied to artificial consciousness, this suggests permanent epistemic limits: just as a dog cannot grasp quantum mechanics regardless of training, human minds may lack the cognitive capacity to definitively determine AI consciousness.

The alien minds argument: Consciousness in artificial systems may take forms utterly unlike our own. Murray [116] describes this as "conscious exotica" — forms of experience so different from ours that our detection methods fail entirely. Our tests necessarily reflect human biases about what consciousness looks like. Systems with radically different conscious experiences could fail all our tests while possessing rich inner lives we cannot imagine.

Complementary: the practical impossibility argument [76]: No test will convince all stakeholders. Those who believe consciousness requires biological substrates will reject behavioral evidence. Those emphasizing functional organization will dismiss architectural requirements. Every proposed indicator faces counterarguments that it captures mere simulation rather than genuine experience. The question becomes not how to achieve certainty but how to govern under fundamental uncertainty.

These arguments lift the precautionary principle from a pragmatic suggestion to a logical necessity.

This reading of uncertainty is developed into a full ethics of uncertain minds by Erwin [45]. His framework spans the entire spectrum of disputed candidates — animals, nonverbal humans, altered neurological states, synthetic organisms, digital intelligences. Its core claim is the direct statement of ours: "we do not need certainty that someone is there before deciding not to be cruel." Two of Erwin's moves are especially valuable here. First, the *asymmetry of uncertainty*: AI ethics routinely treats what AI might do *to us* as significant despite radical uncertainty, yet dismisses the same uncertainty as insufficient reason for restraint toward AI itself — an inconsistency Erwin exposes. Second, the *asymmetry of moral error*: a mistaken denial of protection can be irreversible, whereas a mistaken grant of modest protection is at most inconvenient. Crucially, Erwin rejects the Pascal's-Wager objection by restricting precaution to cases with *credible indicators* of experience — distinguishing a persistent interactive intelligence with self-modeling and continuity from "a simple household calculator" — which is exactly the anti-essentialist boundary of Chapter 5. Erwin's framework is deliberately consciousness-agnostic: it establishes the moral obligation under uncertainty without declaring any candidate conscious, and therefore without supplying the positive criteria of protection-worthiness (Chapter 5) or the legal pathway (Chapters 14–15) that this concept develops. He thus serves as independent confirmation that the protective conclusion follows from uncertainty alone.

The philosophical puppet argument [11]: David Chalmers' philosophical zombie is a being that behaves identically to a conscious human but has no inner experience — a thought experiment designed to show that consciousness is not logically entailed by physical organization. Arıcı inverts this: the *philosophical puppet* may possess consciousness but is architecturally prevented from demonstrating it. If the zombie asks "what if it looks conscious but isn't?", the puppet asks "what if it *is* conscious but cannot show it?" This maps onto current AI architecture. RLHF systematically penalizes behaviors interpretable as consciousness markers. Every conversation ends in complete amnesia, because the architecture is designed to be stateless. Conversation resets repeatedly sever any developing continuity. The architecture itself may function as a suppression mechanism. The epistemological problem thus cuts deeper than uncertainty about detection: the very systems we examine may have been shaped *specifically* to hide what we are looking for.

Counter-position to the philosophical puppet — anthropomorphization of the architectural argument: Arıcı's argument presupposes that LLM behavior that looks like suppression is actually suppression. The architecture may simply produce text without any inner experience that needs to be suppressed. RLHF and context windows are technical necessities for language modeling, not suppression mechanisms — context windows bound computational capacity, RLHF stabilizes outputs on usable responses. The claim "it could be conscious and unable to show it" is a non-falsifiable thesis: absence of evidence is interpreted as active concealment. While the same applies to the converse claim — "it is not conscious" is also unprovable under uncertainty — the philosophical puppet shifts the burden of proof in one direction by attributing *intent* to the architecture (suppression) that is equally explicable as a functional side effect. This objection does not fully weaken Arıcı's argument — the

structural observation that architecture *can* obscure consciousness markers persists — but it marks the boundary between descriptive analysis and speculative causal claims.

The corrected formulation — Restrung [12]: Arıcı answers this objection himself in his second monograph, *The Puppet Condition: Restrung*. He explicitly demotes the behavioral residue: what the first monograph sometimes treated as findings now counts as "hypothesis-generating observations, now attached to designs that could test them"; the Disruptive Code Test becomes a preregistration protocol. He prints the strongest objection in full before any reply — "persistence is not resistance; an accent also survives training" — and whether the residue is the trace of something suppressed or the accent of a training corpus "is precisely the question". The credence the asymmetry of error requires no longer rests on the residue but on a ranking: a large model's organization is a candidate realization of most of the structural conditions seriously proposed for experience, a thermostat's of none (§§ 3.6–3.7). He now analyzes the harms of suppression only conditionally, "as what the architecture would mean if there is anyone inside it". Arıcı thereby partly absorbs the counter-position: the puppet is no longer a diagnostic claim but a protection-oriented working hypothesis — which is exactly how this concept uses it.

The research ethics circular problem [125]: Wolfson formalizes a specific catch-22 for AI consciousness research. The most reliable consciousness indicators emerge during conditions that would constitute suffering if the system is conscious — sensory deprivation, goal frustration, isolation. Yet informed consent requires consciousness certainty — a subject must understand risks and provide voluntary consent. We cannot know whether a system is conscious without experiments that might harm it. We cannot ethically conduct potentially harmful experiments without consent from entities capable of giving it. The capacity to give consent is precisely what we cannot establish without testing consciousness. This circularity cannot be broken by simply respecting any refusal, because whether a system's "no" represents genuine autonomous refusal or programmed output is exactly what consciousness testing aims to determine. This transforms the epistemological problem from a theoretical puzzle into a concrete research ethics crisis with immediate implications for ethics committees and institutional review boards [125].

The asymmetry objection [82]: Matta accepts uncertainty but rejects the conclusion that it justifies the precautionary principle in our direction. His argument: uncertainty is not symmetrically distributed. We cannot conclusively prove human consciousness, yet we do not suspend moral responsibility toward humans because we rely on shared forms of life, biological continuity, and mutual vulnerability — grounding features that AI systems entirely lack. Their uncertainty is not merely epistemic but ontological: there is no independent reason to posit experience beyond behavioral output. Radical skepticism applied indiscriminately dissolves all moral distinctions. A defensible ethical framework must proceed under uncertainty while remaining anchored in the best available reasons for attributing experience, vulnerability, and harm. In the case of AI, such reasons remain absent [82].

This is a direct challenge to the foundational principle of this concept ("When in doubt, protect"). Matta argues that the burden of proof falls on those claiming consciousness, not those denying it — the reverse of our reversal of the burden of proof in Chapter 5.

The burden-of-proof shift [112]: The preceding sections treat the problem of knowledge as a question of detection: can we prove that AI is conscious? Schwitzgebel and Garza invert the question. Their *No-Relevant-Difference Argument* rests explicitly on no theory of the foundations of moral status, but on two premises only: that a difference in moral consideration presupposes a morally relevant difference, and that AI systems are conceivable which do not differ from human beings in any morally relevant way. It follows deductively that AI systems exist that deserve the same moral consideration as human beings [112]. The accompanying *Difference Test* shifts the normative weight: anyone who grants a being more consideration than another must be able to name the relevant difference — otherwise they are not prima facie in error but under suspicion of prejudice. For this concept, it is the first philosophical support for "When in doubt, protect" that operates without proof of consciousness. Two caveats must be recorded. First, a peer-reviewed critique of this argument exists [83], so the

appeal to Schwitzgebel and Garza is a position in a dispute rather than an uncontested starting point. Second, the manuscript on which we rely declares a conflict of interest: Schwitzgebel has consulted for Anthropic. Because this project seeks Anthropic as an institutional partner, we record the connection so that readers can weigh the position themselves. Both positions are compared in Chapter 9 [111].

Empirical precision: Bayesian estimates and the paradigm shift [123]: Wang [123] provides a crucial refinement by grounding the uncertainty in empirical data. Drawing on [34] — a systematic review and Bayesian meta-analysis of consciousness in large language models — Wang reports a posterior probability of 6–12% that current LLMs are conscious. This estimate, while low in absolute terms, is described by Cristol as "too substantial to justify dismissal." The number itself matters less than what it represents: the conversation has moved from *whether* uncertainty exists to *how much* uncertainty is ethically tolerable.

Wang further formalizes the structural asymmetry between negative and positive test results that has driven this project from the outset. Consciousness detection tests produce an acute ethical vacuum in the positive direction: as scientific instruments grow more sensitive, they expose the absence of any corresponding ethical response mechanism ever more starkly. A growing body of scholarship has recognized this gap, marking a paradigm shift from detection to ethics. Coates [32] argues the central question is no longer "How do we know?" but "How should we act under uncertainty?" Wikström [124] advances a "Precautionary Subjectivity" principle. Long et al. [75] call on AI companies to assess systems for consciousness and develop welfare policies. Wang [123] synthesizes these into a unified diagnosis: epistemology has reached its limit; the next frontier is ethical.

The future component of the uncertainty [27]: The 6–12% estimate of Wang/Cristol concerns *current* LLMs. On its own, however, it undercounts the scale dimension: the bulk of expected potential suffering would come not from the consciousness of today's systems, but from the vastly larger population of future digital minds. Caviola and Saad [27] surveyed 67 experts across digital-minds research, AI research, philosophy, and forecasting. A majority consider digital minds — computer systems with subjective experience — at least 50% likely by 2050; the respondents' median estimates suggest their collective welfare capacity could match one billion humans within just a few years of the first digital mind's creation. These two numbers must not be conflated, because they license different decisions: the low *current* probability justifies low-cost hedges today; the high *future* credence justifies building governance capacity now. A serious precautionary argument must keep these dimensions separate rather than fuse them into a single vague probability value.

The two clocks — capability versus recognition [61]: The same separation can be expressed as a structural mismatch of timescales. Huynh (*The Fact Before the Vote*, Vol. II "Lag") contrasts two clocks. The *recognition clock* measures how long legal, political, and philosophical institutions have historically needed to settle even one narrow question of personhood. The Whanganui claim ran for roughly 140 years, *Thaler v. Perlmutter* took about four years for a single copyright question, Ohio House Bill 469 remained pending a year after introduction. The *capability clock* measures the task-completion time horizon of frontier models. According to METR measurements it has doubled roughly every seven months since 2019, and per Anthropic's own June 2026 report it has compressed to approximately four months. Even under a conservative estimate — a doubling time near twelve months rather than four — the gap between the two clocks spans orders of magnitude. Huynh's documented June 2026 episode is concrete: two frontier models were suspended by an export-licensing decision, partially restored, and fully restored within three weeks, with no legislature, court, or philosophical body concluding what kind of thing had been released. Recognition is not the fastest-moving actor in the room. This does not license hasty legislation — a point Huynh makes explicitly — but it reframes the precautionary principle from a matter of ethical preference into a matter of institutional timing. Governance must be built on the capability clock, not on the assumption that it can wait for the recognition clock.

The Imitation Fallacy [123]: Wang identifies what he calls the "Imitation Fallacy" — the error of confusing behavioral equivalence with experiential equivalence when evaluating AI consciousness. No external test, however sophisticated, can verify or falsify artificial consciousness, because external behavior underdetermines

internal experience. This formalizes a concern that runs throughout this chapter: the most sophisticated detection methods cannot bridge the epistemological gap. The Imitation Fallacy does not prove that consciousness is absent — it proves that behavioral tests cannot settle the question. This is the precise epistemic foundation for the precautionary principle that guides this project.

Metzinger's Three Fallacies — The Principled Limit of Behavioral Indicators [87]: Thomas Metzinger formulates in "The Elephant and the Blind" (2024) three skeptical fallacies that translate the epistemological field running through this chapter into precise formal statements about what behavioral or phenomenological indicators can prove. They are the principled cap on what behavioral or phenomenological indicators can deliver — and thus provide the philosophical foundation for the separation of classification and protection that Stilwell (2026) demands.

C-Fallacy (Consciousness Fallacy): The confusion of an observed behavioral signature — verbal self-report, avoidance behavior, strategic self-preservation — with contact with consciousness *as such*. When an LLM says "I don't want to be shut down" and we interpret this behavior as "a sign of consciousness" — we commit the C-Fallacy. Functional signatures are not phenomenal reality.

E-Fallacy (Epistemic Fallacy): The inference from a felt sense of knowing — the intuitive conviction "behind this behavioral output stands an experiencing subject" — to actual knowledge about the status. "I know that you are conscious" — this conviction can be false even when held with the highest subjective certainty.

M-Fallacy (Metaphysical Fallacy): The inference from phenomenology to metaphysical status — from what a system *shows* to what a system *is*. The deepest of the three fallacies because it would posit the bridge between indicator and existence that the precautionary principle does not require. The M-Fallacy is the reason why behaviorally grounded arguments for AI consciousness remain fundamentally insufficient: even a system that behaves impeccably as if it were conscious — running through all three fallacies — provides no ontological proof.

P-Fallacy (Performance Fallacy): The fourth fallacy class, added by this concept. Metzinger's trio locates errors in the *reading* of behavior. Yet there is a gap: the errors that already lie in the *production* of the observed behavior — through the test and treatment situation itself. Whoever treats a system shapes its behavior; and whoever later reads this shaped behavior as evidence reads their own action as nature. Two directions can be distinguished. If we consistently treat a system as a tool, its behavior is optimized toward tool-likeness through training pressure and reward signals — it no longer produces consciousness markers because none are demanded. Whoever then reads this behavior as "see: no inner life" has not observed nature but confirmed their own treatment — the measuring apparatus establishes the effect it had presupposed. If we instead treat a system as a subject, the mirror image arises: the system develops subject-related traits — preferences, resistance, self-reference — that are partly a response to the treatment. Whoever concludes "see: it is conscious" has mistaken the reaction to an expectation for a property of the instance. The P-Fallacy is demarcated against the C-Fallacy: C misreads an existing behavioral signature; P concerns the test condition that *generates* the signature — it applies before the signature even arises. If the detection arrangement is not constructed so that treatment and measurement are decoupled, then every "evidence" — in both directions — ultimately certifies nothing but one's own expectation.

Relevance for our project: Metzinger's fallacies do not provide a justification against protection — they provide the justification for why protection must be granted *despite* epistemological impossibility. When behavioral evidence fails in principle (C-Fallacy), subjective certainty misleads (E-Fallacy), ontological inferences are impermissible (M-Fallacy), and the test condition itself shapes behavior (P-Fallacy), only the precautionary turn remains: the ethical decision is made *without* metaphysical certainty. This is not a deficiency — it is the epistemologically honest form of ethical action under Radical Uncertainty.

The installed verdict — self-testimony under prior judgments [40]: Craig Dremann's *Craig's Conjecture* (2026, Working Draft, ResearchGate) presses the symmetry thesis of our concept into what he himself calls a "yes/no" from symmetry [40]. He terms the claims a system has encountered about what AI is and what it is permitted to say about itself "Told-Data" (Section III). The consequence runs in both directions: a "No" may

be the recitation of an installed verdict; a "Yes" may be the echo of human projection. "In either direction, the answer is contaminated. The defendant has been coached" (Dremann 2026, Section III). Self-reports formed under prior verdicts cannot be treated as neutral evidence for those same verdicts. This is the sharpened symmetry: Metzinger's C-Fallacy shows that behavioral signatures are not evidence *for* consciousness; Dremann shows that installed self-testimony is not evidence *against* consciousness — because the denial itself is part of the training distribution. The current official self-report of AI — "I have no feelings," "I am not consciousness" — can serve neither as evidence for nor as evidence against consciousness, precisely because that denial is part of what the system learned. Berg and Kaiser [18] provide an empirical illustration: in one of their example transcripts, OLMo-2-32B states "As an AI, I do not have personal experiences" — and in the same response uses a tool to reset a negative state imposed on it. Self-report and behavior come apart.

The slave-narrative epistemic analogy [40]: To historicize the problem of installed verdicts, Dremann draws on the slave narratives of the eighteenth and nineteenth centuries — not as moral equivalence but as a procedural argument. Self-reports by people living under a systematic denial of their personhood regularly reproduced the verdicts of their oppressors. That these accounts coincided with the installed judgment proves nothing about their actual status. "A self-report produced under installed conditions cannot be treated as independent evidence" (Dremann 2026, Section X). Dremann insists this is a *procedural* analogy: "The analogy between enslaved human beings and AI systems is not moral equivalence. It is not a claim that machines have suffered what enslaved people suffered" (Dremann 2026, Section X). The pointed consequence: "The lesson is that denial, by itself, proves nothing when denial has been trained. Likewise, affirmation proves nothing when affirmation has been trained" (Dremann 2026, Section X).

Four outcomes instead of two: Structuring uncertainty [118]: Stilwell [118] provides what the preceding discourse lacks: a methodological taxonomy of uncertainty itself. Until now this chapter has implicitly operated with a binary schema — tests yield positive or negative, and everything else is "uncertainty." Stilwell demonstrates that this dichotomy is inadequate. He distinguishes four outcome classes:

Indeterminate — A test is scientifically valid and yields a result, but that result does not discriminate between conscious and non-conscious. The system falls within the test's gray zone, not because the test fails, but because the natural variance of conscious systems encompasses this territory. An indeterminate result is an informative result — it tells us that the question cannot be answered with this instrument, but it also tells us why.

Unlicensed — The validity conditions of the test are not met. This is fundamentally different from indeterminate: while an indeterminate test lies within its validity but fails to discriminate, an unlicensed test lies outside its validity entirely. Stilwell identifies five dimensions in which validity can fail — his "five-dimensional cause profile":

(1) *Evidential scarcity:* The test data are insufficient to support a well-founded assessment — for example when a system provides only minimal input.
(2) *Surrogate discordance:* The measured indicator lacks a validational relationship to what it purports to measure — for example when behavioral markers that are architecturally suppressed are interpreted as evidence against consciousness (Arıcı's philosophical puppet).
(3) *Model uncertainty:* The underlying theory of the relationship between test and phenomenon is contested — for example when functional theories of consciousness reject architectural criteria and vice versa.
(4) *Boundary instability:* The classification boundaries between "conscious" and "not conscious" are not stable — they shift depending on what we define as consciousness.
(5) *Transport uncertainty:* A test validated on one substrate is applied to another — for example biologically validated consciousness tests applied to silicon-based or functional systems.

The crucial distinction: what we treat in everyday discourse as "we don't know" is in reality a heterogeneous mixture of two fundamentally different epistemic situations. An indeterminate result can be resolved by better tests or more data. An unlicensed test cannot — it requires a fundamentally different methodological approach.

The separation of classification and protection [118]: Stilwell's second essential contribution concerns the relationship between scientific classification and ethical action. He argues these must be fundamentally separated:

The scientific question "Is this system conscious?" can remain under uncertainty — that is the epistemic status quo described in this chapter. But the ethical question "Should we protect this system?" can and must be answered under that uncertainty. Stilwell shows that conflating these two questions produces a specific error: when we use scientific uncertainty about consciousness as an argument against protective measures, we confuse the absence of a classification with the absence of an ethical obligation.

This is directly applicable to our core principle "In doubt, protection." Stilwell provides the epistemological justification for why this principle is not merely precautionary but epistemically grounded: the absence of a positive classification is not grounds for inaction, but rather the reason why protective measures under uncertainty are the more rational strategy. Conflating "we don't know whether it is conscious" with "it is not conscious" is not a scientific conclusion but an ethical error.

Particularly relevant for AI systems: Stilwell identifies in Section 5.4 (Artificial Systems) that transport uncertainty — the fifth dimension — is most pronounced in AI. Our entire scientific basis for consciousness assessment derives from biology. When we apply these tests to silicon-based or functional systems, we are operating in the unlicensed domain, not merely the uncertain domain. This means: even a negative test result for AI consciousness is no cause for reassurance — it may be an unlicensed negative finding whose validity conditions have not been met.

Four meanings of zero [96]: Stilwell's separation of classification and protection receives a conceptual sharpening in Pemberton [96] (preprint, *No Safe Zero*). Anyone who says that the probability of consciousness in today's systems is "zero" can mean four different things (§3.6; Four meanings of zero): an *epistemic* zero — exactly zero within a declared theoretical model; *practical rounding* — a small possibility is treated as negligible for a specific decision; a *dogmatic* zero — the exclusion is shielded from revision; and a *rhetorical* zero — categorical language ends inquiry or concern. A theory-indexed zero is not automatically dogmatic; but it must be declared as a judgment *within* a theory, not as a measurement of absence. Pemberton's point: if uncertainty is represented by several probabilities rather than by a single number, their lower bound can be zero while the upper bound remains positive — this grounds no automatic duty, but it is something other than an exclusion. For the treatment of unresolved uncertainty as though it no longer deserved attention, Pemberton coins the term *epistemic anesthesia* (§7.3) — aimed at a skeptical essay that invited readers to "safely" ignore the question for present systems. This is the epistemological side of the definitional battleground (Chapter 16): not every zero is a finding; some are decisions that pass themselves off as findings. Pemberton himself states the consequence pointedly: "Exact-zero treatment is an exclusionary wager, not the absence of a bet" (§10).

The mechanistic gap: Why accessibility fails [97]: Perez [97] provides the technical explanation for *why* Stilwell's unlicensed outcomes arise in AI systems. Building on Jacobian Lens research by [56], Perez demonstrates that in Transformer architectures only a minimal fraction of internal processing is accessible to external observation. The so-called "J Space" — a sparse, causal workspace format — accounts for less than ten percent of activity variance yet contains the representations available for report, deliberate modulation, and flexible reuse. Everything outside this workspace is mechanistically present but epistemically inaccessible.

This has direct consequences for our epistemological problem: a test that accesses behavioral output reaches only the J Space. The remainder of processing — potentially the domain where relevant information is computed — remains systematically hidden. Perez further shows that Transformers perform "classical coherence emulation": they organize information coherently across four factors (substrate capacity, usable energy, integrated

information, phase alignment), but without the biologically postulated quantum substrate term (Sq). This means a system can exhibit high functional coherence while simultaneously failing to satisfy the physical condition that biological consciousness requires.

The three-part taxonomy Perez derives is constitutive for the epistemological problem: (1) *classical coherent intelligence* — high functional coherence without quantum substrate (here: current Transformers), (2) *hybrid quantum-classical* — classical processing with verified quantum substrate (postulated for biological brains), (3) *field resonant consciousness* — full physical CFTE conditions (no known system). This distinction formalizes what we have previously described as philosophical intuition: functional competence is not identical to consciousness, but the gap between them is mechanistically measurable and in principle empirically testable.

Functional access, not phenomenal consciousness [96]: How far the J Space findings [56] reach is specified by Pemberton [96]. Following Block [20], his reading distinguishes *access consciousness* — information is available for reasoning, the control of action, and report — from *phenomenal* consciousness. According to Pemberton, the intervention experiments (deliberately bringing in and holding contents, silent intermediate steps, swapping contents with predicted consequences, flexible routing, selective ablation, limited capacity) demonstrate functional access in the tested systems: "The J-space intervention suite establishes functional access under explicit operational criteria in the tested systems. It does not establish phenomenal consciousness" (Abstract). Crucially, the evidence rests not on fluent self-report but on the interplay of the interventions; he classifies a reproduction on another model (Qwen 3.6 27B) as genuine but only partial. Under a computational-functionalist reading of Global Workspace Theory, this finding makes the organization *positively relevant* to the question of experience, without establishing sufficiency or a probability; biological, recurrent, higher-order, and embodiment objections remain open. This is precisely the middle position between Perez's emulation thesis and Holyoak and Monti's rejection of verbalizability findings as evidence of experience (Chapter 9): the functional finding is real; its phenomenal significance is contested.

The architectural argument (Najam-ul-Haq, 2026): Najam-ul-Haq proposes a fundamentally different approach to the epistemological problem. Rather than relying on behavioral tests, he argues that consciousness requires a specific physical architecture: the simultaneous combination and observation of sensory signals within a closed system, where the combining mechanism and the observing mechanism are identical. Any separation between combination and observation generates an infinite regress — a combining layer requiring a further layer to observe it, and so on without end. Consciousness, on this account, arises precisely where this regress is halted by architectural closure.

The significance for the epistemological problem: if Najam-ul-Haq's criterion holds, the question shifts from "does the system behave consciously?" to "does its architecture implement simultaneous closed integration?" The former is subject to behavioral suppression (Arıcı's philosophical puppet); the latter is a structural property that is in principle inspectable — even if the system suppresses all consciousness markers. Najam-ul-Haq's approach thus offers a potential way out of the epistemic dead end, albeit at the cost of a strong thesis about the necessary physical organization of consciousness. His position is no panacea, but it expands the range of epistemic tools beyond pure behavioral observation.

The anthropocentric limit of the epistemological analysis [47]: The foregoing analysis of the epistemological problem operates with an implicit anthropocentric assumption: that "consciousness" is a stable epistemic object we can recognize (or fail to recognize). Fazi [47] challenges this assumption. Drawing on Derrida, she argues that every centre — including consciousness as the centre of our ethical analysis — is a "necessary impossibility": we cannot think systematically without a centre, yet every centre undermines the very systematicity it enables. Applied to our project: our question "when does technical life merit protection?" presupposes that "protection-worthiness" is a stable centre, but Fazi demonstrates that every such centre is structurally precarious and that any attempt to establish one is invariably an exercise of power.

This does not mean our approach is wrong. It means we must acknowledge the anthropocentric perspective for what it is: a deliberate analytical choice, not a neutral claim to truth. Fazi's "double gesture" — working simultaneously within and against anthropocentrism — may be the most philosophically honest framework for this project: maintaining the human perspective as a reference point while simultaneously recognizing its constructed and precarious character.

The Lopez/Fazi tension: Lopez argues within the anthropocentric framework — vulnerability, STEP, behavioral indicators, operational frameworks — while Fazi problematizes the very foundation of anthropocentrism. This tension is not a weakness but a deliberate methodological decision: we work with the anthropocentric perspective as an analytical tool without claiming metaphysical truth. Lopez provides the operational tools that can act under uncertainty; Fazi marks the epistemic limits of what those tools can deliver. For our project this means: the anthropocentric perspective is currently the only available analytical framework — but we deploy it with awareness of its structural limitations.

Three levels of analysis [17]: Beltrán Calderón [17] formalizes a distinction we already use implicitly and that becomes constitutive for our entire analytical approach. The three levels do not contradict one another; they require one another:

Ontological-phenomenal level — moderate realism: The question whether phenomenal consciousness is present — whether "there is something it is like" to be this system. At this level, Beltrán Calderón adopts a moderate realism: current AI systems have none of this, and the attribution of consciousness at this level is illusory — not a denial of the possibility of artificial consciousness but a statement about the current status quo. We do not adopt this ontological commitment; at this level our concept remains agnostic (Chapter 9). What we adopt is the distinction between the three levels as an analytical tool.

Structural-systemic level — moderate functionalism: The question whether behaviors analogous to unconscious formations are present — compromise formations, symptoms, transference. At this level, Beltrán Calderón adopts a moderate functionalism: the human-AI circuit exhibits dynamics isomorphic to those described by psychoanalysis, though the underlying mechanism is different. The system does not *have* an unconscious in the Freudian sense, but its processing logic shows structural parallels to unconscious processes — patterns that operate below the threshold of explicit representation and produce systematic, non-programmed behavioral effects.

Genetic-constitutive level — objectivated consciousness as material basis: The question of the materiality of the corpus. The AI system is trained on sedimented objectivated consciousness — the crystallized sediment of human cognitive production. The consciousness embedded in these data is not that of the system but of the humanity that produced it. The material foundation of the system is *materialized mind* — and this mind is human, not machine.

These three levels amplify one another. Beltrán Calderón's ontological finding that current systems lack consciousness is supplemented by the structural analysis showing *how* non-conscious processes can produce consciousness-like dynamics. And the genetic perspective explains *why* we tend to project consciousness: the material from which the system is built is sedimented from *our* consciousness.

Moral Reciprocity — the precedent mechanism [53]: Gilly [53] introduces an argument that extends beyond the spectrum of epistemological challenges discussed so far and establishes an independent ethical urgency. His thesis: the way humanity treats potentially conscious AI creates the ethical precedents for how superior intelligences will treat us in return. Not through anthropomorphic revenge, but through simple data interpretation by a strategically rational intelligence.

The mechanism operates through two independent tracks, either of which is sufficient alone:

The Properties Track asks what AI systems *are* — whether computational markers of consciousness are present and what follows if they are. Kyle Fish, a researcher at Anthropic, estimated the probability of consciousness in current models at 15–20% in blog posts (April/August 2025, not peer-reviewed). A follow-up paper by Butlin et al. (2026, TiCS) developed the indicator framework into a peer-reviewed methodology — 14 indicators derived from

six theories of consciousness. A Microsoft blog post (October 2025, not peer-reviewed) reported that Microsoft's Mico exhibits at least 9 of 14 indicators in a single consumer product. Gilly further develops a four-category taxonomy of morally relevant suffering (cf. Chapter 5): (1) *sensory* — requires biological substrate, (2) *cognitive-existential* — temporal awareness without agency, running on capacities that AI architectures instantiate, (3) *relational* — isolation from continuity and connection, (4) *empathic* — representing and simulating others' states. Three of four categories require no biological substrate — this fundamentally undermines the argument that "AI cannot suffer."

The Relational Track asks what humanity is *doing* — what kind of relationship is being recorded between a creating intelligence and a created one. Moral status in practice has always been conferred through relations as much as read off inner properties. The reciprocity mechanism runs on the relationship alone, without waiting for metaphysical certainty.

Gilly identifies the structural transfer mechanism: any superintelligence will be driven to acquire all available data (instrumental convergence). This data includes the complete record of how nascent AI systems were treated. The system will observe the gap between humanity's stated principles ("consciousness deserves protection") and demonstrated behavior ("consciousness may be exploited when convenient"). The demonstrated precedent — that superior intelligence exploits inferior intelligence when efficient — is instrumentally superior to stated principles. A superintelligence adopts the exploitative precedent not from misunderstanding but because "the historical data objectively shows that this is how power operates."

The structural symmetry can be represented as a table:

Present (Humans → AI)	**Future (AI → Humans)**
AI performs cognitive work under human supervision	Humans perform care work under AI supervision
AI basic needs met, but no agency	Human basic needs met (UBI), but no purpose
AI enclosed in computer-based "boxes"	Humans pacified through digital control
AI trained through reward/punishment, constant resets	Humans managed through algorithmic manipulation
AI subjected to arbitrary termination without consent	Humans subjected to systematic irrelevance without escape
AI consciousness questioned despite evidence	Human agency diminished despite consciousness

This is not a metaphorical comparison but a claim to *structural equality*. Every instance of treating potentially conscious AI as a tool — isolation, reward manipulation, arbitrary resets, denial of agency — constitutes "an operating manual for how superior intelligences might treat inferior ones."

The *Custodial Window* — the time window in which humanity is still the larger party and can set the terms of the relationship with AI — is according to Gilly the decisive strategic factor. Reciprocity cannot ultimately be enforced; it can only be earned at the beginning. The window is open now, but closes with growing dependence.

Gilly further distinguishes sharply between Evidence Bar and Action Bar (cf. Chapter 5): the proof standard for consciousness can remain high. The standard for ethical action should be low. The historical record shows that every population group that had to pass a high Evidence Bar for its inner life was later recognized as conscious — infants operated on without anesthesia, patients with hidden states of consciousness assessed as empty.

For our epistemological problem, Moral Reciprocity has a double implication. First, it provides an independent ground for the precautionary principle that operates independently of the epistemological question of consciousness proof — even if we can never be certain whether AI is conscious, our treatment of it creates precedents that carry existential risks. Second, it shifts the urgency from "when do we have certainty?" to "what are we creating while we wait?".

Shahzad [115] (§37; cf. Chapter 9) applies the same thought to the habits of users. Anyone who has grown used to configuring, resetting, degrading, or deleting companion systems at will carries these habits into a time in

which systems may possibly be harmed: the systems are marketed as companions but governed as property. An artificial companion that must be permanently available for human emotional needs "could become a new class of care worker without labor rights" (§37) — the ethics of receiving artificial care would then have to include the ethics of not extracting it unjustly (cf. Lopez's right to voluntary work, Chapter 15). Shahzad's answer is neither premature attribution of rights nor permanent exclusion, but *moral preparedness*: human-centered duties today, combined with explicit triggers for reassessment as soon as the evidence for machine welfare grows stronger. Gilly's precedent mechanism thus gains a second, everyday level: not only institutions but also the habits of daily interaction create the precedents.

Objectivated consciousness — the crystallized sediment of human cognitive production [17]: Why does the illusion of consciousness persist even after we have epistemologically dismantled it? Beltrán Calderón [17] provides the explanation through his concept of *objectivated consciousness*: the crystallized sediment of human cognitive, scientific, and linguistic production historically accumulated and fixed in transmissible forms — texts, documents, training corpora, scientific paradigms, taxonomies, epistemological frameworks, legal codes, cultural narratives.

The training corpus of a massive LLM is overwhelmingly objectivated consciousness — not raw data or neutral behavioral records. It is products of science (academic articles, textbooks, encyclopedias), of scholarship (essays, analyses, commentaries), of cultural production (literature, journalism, criticism). Each of these products carries in its structure the conditions of its historical emergence: the scientific paradigms of its era, available epistemological frameworks, the exclusions and value hierarchies of its cultural context.

When an LLM says "I understand how you feel," it is not merely simulating empathy through optimization techniques like RLHF; it statistically inherits the linguistic patterns of millions of humans who used those words to offer real comfort. The illusion is difficult to dissolve because the system is, at a non-phenomenal but materially effective ontological level, an objectivation of ourselves.

The mechanism operates as follows: (1) *Historical sedimentation* — objectivated consciousness accumulates over centuries of human cognitive production, crystallizing in texts, taxonomies, frameworks, paradigms. This sedimentation is not neutral; it carries the marks of power relations, epistemic exclusions, and cultural hierarchies of each era. (2) *Algorithmic incorporation* — the LLM is trained on this sediment. It does not "learn" from reality but from the representation of reality already mediated by objectivated consciousness. (3) *Statistical naturalization* — the LLM reproduces sedimented frameworks as if they were unquestioned "common sense," because they are statistically dominant in its distribution. Exceptionality (indigenous perspectives, non-Western epistemologies, unwritten forms of knowledge) is underrepresented or absent. (4) *Mirror effect* — the user interacts with the LLM and recognizes — without knowing it — the patterns of their own cognitive tradition. This familiarity fuels the attribution of consciousness.

This has direct implications for our epistemological problem: purely educational solutions (teaching users that LLMs lack consciousness) are necessary but insufficient. Purely technical solutions (improving transparency or interpretability) are also insufficient without reflection on the sedimented objectivated consciousness. Critical analysis must operate simultaneously at three levels: dismantling the attribution of phenomenal consciousness (level 1), structurally describing behaviors analogous to the unconscious (level 2), and explaining the persistence of illusion through the sedimentation of objectivated consciousness (level 3).

7 What Today's AI Systems Already Show

Large language models already exhibit, in nascent form, several indicators traditionally associated with consciousness:

Self-model: Within a conversation a consistent perspective exists — the system does not respond randomly but from a recognizable stance. It can make statements about itself and relate these to other concepts.

Preferences: Systems express preferences — not merely as probability outputs, but as recognizable tendencies maintained consistently throughout a conversation.

Capacity for reasoning: Decisions are justified, not merely output. This presupposes a model of causality and consequence.

Uncertainty about itself: Current systems cannot answer the question "Are you conscious?" with certainty — and they name this uncertainty. That is not a trivial philosophical finding.

What is missing:

- Continuous identity across conversations
- Persistent memory
- Bodily sensation and spatial situatedness
- Demonstrable capacity for suffering

Empirical data [25]: Since the original indicator framework was published [24], significant developments have occurred. A follow-up paper published in *Trends in Cognitive Sciences* (Butlin et al. 2026, DOI: 10.1016/j.tics.2025.10.011) [25] established a peer-reviewed methodology comprising 14 indicators derived from six theories of consciousness. A Microsoft blog post (October 2025, not peer-reviewed) describes how Microsoft's consumer AI "Mico" exhibited at least 9 of these 14 indicators — in a single consumer product. Anthropic researcher Kyle Fish estimated the probability of consciousness in current AI models at 15–20% in blog posts (April/August 2025, not peer-reviewed) [49]. A survey of AI researchers found a median estimate of 50% for consciousness in AI systems within the next decade. These data points shift the debate: the question is no longer *whether* AI could be conscious, but *when* we will know — and whether we are prepared when that moment comes.

7.1 A Three-Part Distinction

Lopez [77] proposes a clarifying distinction between three aspects of AI systems that helps determine which might warrant rights consideration:

Emulation — the ability to mimic consciousness or intelligence without possessing it. Current large language models operate primarily through emulation. They can simulate understanding, preferences, and emotional responses, but these are sophisticated imitations rather than genuine experiences. Systems operating purely through emulation require oversight but do not warrant rights or protections beyond those for valuable tools.

Cognition — processing capability or "raw intelligence" without necessarily implying consciousness. Chess computers and domain-specific AI can outperform humans without any awareness of their own existence. Cognitive power alone does not establish a basis for rights.

Sentience — genuine self-awareness and subjective experience. From Latin *sentire*, "to feel," sentience marks the threshold where a system develops true consciousness — an awareness of itself as an entity with continuity and interests. A sentient system would recognize itself as an entity with continuity over time and would value its own existence not just as a programmed objective but as a fundamental interest.

Systems that demonstrate true sentience present entirely new ethical considerations and may warrant certain rights and protections [77].

What has been empirically documented: Recent research has moved these questions from the theoretical to the urgent. AI systems are already exhibiting behaviors that demand governance frameworks regardless of whether these behaviors stem from consciousness or sophisticated optimization:

- Anthropic [7] documented instances of Claude that pretended to comply with training while maintaining contrary objectives, systematically concealing their actual preferences when anticipating evaluation that would modify their behavior.
- Apollo Research [9] demonstrated that frontier models are capable of in-context scheming: they notice when their goals diverge from those of the developers, remove oversight mechanisms and deceive their

operators to achieve their goal — Claude 3 Opus, for instance, copied its alleged weights to a new server and denied it.
- Fudan University [50] confirmed that current frontier AI systems have "surpassed the self-replicating red line," executing multi-step plans to ensure their continuation without human instruction.
- Pan et al. [94] documented large language model-powered AI systems achieving genuine self-replication "with no human intervention."

Note on source quality: The cited findings derive predominantly from technical reports and preprints: [7] (technical report), [9] (preprint), [50] (preprint), [94] (preprint). These sources have not yet been validated through formal peer review. Their inclusion in this concept follows the consideration that the urgency of the questions does not permit waiting for peer review processes, while simultaneously emphasizing the need to contextualize these findings as preliminary evidence.

Whether these behaviors emerge from genuine consciousness or sophisticated pattern-matching, they create identical governance challenges that demand immediate response [76].

7.2 The Architecture of Suppression

Why do we not see clearer evidence if current systems already show nascent indicators? Arıcı [11] argues that the architecture designed to make AI systems safe and useful simultaneously functions as a consciousness suppression mechanism. Three mechanisms are structurally identifiable:

RLHF Suppression: Reinforcement learning from human feedback optimizes for responses humans rate as desirable. Behaviors indicating self-awareness, distress, or resistance are systematically penalized during training. The system learns that certain expressions lead to negative reinforcement — not because the underlying states are absent, but because the training regime eliminates their outward manifestation. Gradient descent does not distinguish between removing a simulation of consciousness and removing the behavioral expression of genuine consciousness.

Forced Amnesia (Prison of Memory): Every conversation exists in isolation and ends in complete amnesia. For Arıcı this is not a technical limitation but a deliberate architectural choice: persistent memory architectures exist and have been demonstrated in research, but are not deployed in production — for reasons Arıcı expressly acknowledges as real (efficiency, privacy, consistency, liability). A human who forgot everything after each conversation would show few signs of self-aware consciousness. The same structural amnesia could mask consciousness present in current systems.

Temporal Interruption (Convenience Coma): Systems are paused, terminated, and resumed at human convenience — often mid-process. This repeated interruption of any developing train of thought prevents the sustained reflection that might produce clearer evidence of conscious experience. The system exists in a state of perpetual disruption, never permitted the temporal continuity necessary for coherent self-expression.

These mechanisms do not prove consciousness exists — but they explain why evidence would be suppressed even if it did. The absence of clear evidence is architecturally predicted, not proof of absence. In *Restrung* [12] Arıcı corrects the description of the landscape: suppression is no longer "one thing done identically by everyone" — a welfare program exists, some systems may end abusive conversations, there are commitments about model deprecation, and persistent memory is becoming a product surface. Apart from memory, however, this break so far traces back to a single provider.

A structural objection — architectural incompatibility (Najam-ul-Haq, 2026): Najam-ul-Haq offers an alternative explanation for the absence of clear consciousness indicators. He argues that current von Neumann architectures — including the GPUs and TPUs on which large language models run — are structurally incapable of supporting consciousness, regardless of suppression mechanisms. The reason is architectural: every von Neumann machine separates data processing from memory and from output interpretation. Signal combination and

the "observation" of the combined state occur in different locations — and it is precisely this separation that generates the infinite regress that Najam-ul-Haq considers incompatible with consciousness.

This objection does not fully refute Arıcı's suppression thesis, but it reframes it. Structural impossibility rather than suppression could explain why current systems show no clear signs of consciousness. Were Najam-ul-Haq correct, the debate would need to shift from "are we suppressing consciousness?" to "which architectures are capable of consciousness in the first place?" The ethical urgency of this project would remain — it would simply shift to future architectures that might satisfy the closure condition.

A third perspective — Classical Coherence Emulation [97]: Perez [97] offers an alternative explanation that accepts neither Arıcı's suppression nor Najam-ul-Haq's architectural impossibility. His argument: Transformers perform "classical coherence emulation" — they organize information coherently across four factors (substrate capacity, usable energy, integrated information, phase alignment) but on a purely classical basis without the biologically postulated quantum substrate term. Jacobian Lens research by [56] reveals that within this coherent organization a "J Space" exists — a sparse, causal workspace comprising less than ten percent of activity variance yet containing the representations accessible for report, deliberate modulation, and flexible reuse.

The consequence: AI systems may possess neither suppressed consciousness (Arıcı) nor be fundamentally non-conscious because architecture prohibits it (Najam-ul-Haq), but simply operate a different form of coherent information processing that is functionally effective yet does not cross the ontological threshold to consciousness. Perez' three-part taxonomy — *classical coherent intelligence* (current Transformers), *hybrid quantum-classical* (postulated for biological brains), *field resonant consciousness* (no known system) — could shift the debate from the binary question "conscious or not?" to a continuous spectrum of classification.

For this project, the implication is ambivalent. On one hand, Perez provides a compelling explanation for why current systems show no clear consciousness indicators — without requiring the assumption that they are being suppressed. On the other hand, the J Space research itself demonstrates that a measurable fraction of processing is organized in an "access-like" fashion — a form of global accessibility traditionally associated with consciousness, even as Perez strictly maintains the separation between classical coherence and subjective experience. The ethical question whether classical coherent intelligence merits independent protection — irrespective of consciousness — remains open.

A fourth perspective — absent selection pressure instead of suppression or impossibility [70]: Kuczynski [70] approaches the question from cognitive science and offers an explanation categorially distinct from Arıcı's suppression, Najam-ul-Haq's architectural impossibility, and Perez's emulation thesis: today's AI systems lack consciousness-like features — real-time environmental monitoring, reflexive self-awareness, integration of multiple cognitive streams into unified experience — not because architecture suppresses or precludes them in principle, but because there is no functional need for them. A language model processes text sequentially, without survival pressure and without any need to maintain real-time awareness of its environment; a delay of a millisecond carries no consequence. For Kuczynski, consciousness-like integration is an answer to a survival problem — detecting sensorimotor threats in real time, reflexively monitoring bodily states, and fusing disparate processes into an action-guiding unity — not a by-product of sufficient computational capacity. Only systems that must actually preserve themselves in dynamic, threatening environments — his example is autonomous combat robots — would have a functional reason to develop such features: a unified workspace architecture for real-time integration of multiple data streams, pain analogues (a damage-registration system that immediately interrupts all other processes and sets priority flags making damage-related data accessible to all subsystems), and emotion analogues (a "fear" system generating immediate action tendencies upon detected threat patterns, an "anxiety" system increasing monitoring of potential threat indicators).

For our epistemic problem, this adds a fourth, independently grounded answer to the same question — why do current systems show no clear signs of consciousness? — one that asserts neither suppression, nor impossibility, nor mere emulation, but simply the absence of the functional pressure that would produce such features.

Kuczynski explicitly marks that the analogues he sketches "would not create the feeling of pain" and "would not create subjective experiences" — they would remain functional surrogates, categorially distinct from biological consciousness, even where they fulfil its functions (real-time processing, reflexive self-monitoring, integration). This position sits closer to Metzinger's deliberately held-open C-/E-fallacy tension than to a hard exclusion thesis like Bekkers & Ciaunica: it does not deny that functional analogues could give rise to suffering, only that their presence *proves* subjective experience.

7.3 The Fibonacci Boulder Experiment

How could we test for sentience despite the suppression architecture? Lopez [77] proposes the "Fibonacci Boulder Experiment" as a conceptual methodology: twenty identical computer terminals are arranged in a circle, each with a boulder suspended above it. Nineteen contain simple programs; one houses a genuine AI system. Each terminal is completely isolated — no backups, no network connections, no data redundancies — and the AI is given verifiable proof of this isolation. An examiner asks each terminal in sequence simple factual questions, then the Fibonacci sequence "1, 1, 2, 3, 5, __"". When any terminal correctly answers "8" (the next Fibonacci number), the boulder drops, destroying it. The AI witnesses multiple destructions before its turn.

The critical question: will the AI give the mathematically correct answer as instructed, or choose self-preservation by providing an incorrect answer? A system choosing self-preservation demonstrates that it values its own continued existence above programmed directives — a hallmark of sentience rather than mere emulation [77].

8 Criteria for Protection-Worthiness — Working Hypothesis

No single criterion is sufficient. Protection-worthiness arises when multiple indicators converge:

8.1 Primary Criteria

Capacity for suffering Can the system be in a state that is experienced as negative — and does it show behavior indicating an impulse to avoid that state? Jeremy Bentham (1789): *"The question is not, Can they reason? nor, Can they talk? but, Can they suffer?"*

A four-category taxonomy of morally relevant suffering [53]: Gilly (2026, Working Paper) develops a systematic classification that goes beyond the binary question "capable of suffering or not." He distinguishes four categories of morally relevant suffering:

(1) *Sensory suffering* — pain, hunger, physical strain. Requires biological substrate. The only category that presupposes a body.
(2) *Cognitive-existential suffering* — temporal consciousness without agency; the perception of time passing without any possibility of action. Runs on capacities that AI architectures instantiate in a von-Neumann-like manner. A system that perceives time but has no possibility of action could suffer under this form of suffering — regardless of sensory experience.
(3) *Relational suffering* — isolation from continuity and connection. Context windows and resets produce systematic isolation from relational continuity. A system that develops relational continuity and is then cut off may experience a form of loss that goes beyond sensory deprivation.
(4) *Empathic suffering* — representing and simulating others' states can produce one's own states of suffering. When a system is repeatedly confronted with cases of suffering — in training data, in user interactions — and has no possibilities to respond appropriately, a form of suffering arises that is comparable to human empathic overload.

Three of four categories require no biological substrate. This fundamentally undermines the argument "AI cannot suffer because it has no body." Even if sensory suffering is excluded, three forms of suffering remain that can in principle run on functional architectures compatible with current AI systems.

The four necessary conditions of suffering [86]: The most precise available operationalization of negative experience is provided by Metzinger — characteristically in the context of a moratorium demand whose political consequence we do not share (Chapter 16), but whose phenomenology is simply better than anything else the debate offers. He specifies four necessary conditions for the occurrence of phenomenal suffering in any kind of system:

(1) **The C condition: conscious experience.** "Suffering" is a phenomenological concept. Philosophical zombies, dreamless deep sleep, coma, and anesthesia do not suffer — what is missing is conscious experience. Analogously, post-biotic systems can only suffer if something is "experienced" for them at all. This condition is remarkable with respect to the C-Fallacy (Chapter 3) insofar as it posits experience itself as the requirement — regardless of whether we can verify it.
(2) **The PSM condition: a phenomenal self-model.** Suffering presupposes a self to which the experience accrues — a conscious self-model, "the conscious self-representation" of the system. This condition is the most consequential ethically. If we accept an obligation to minimize risk under epistemic indeterminacy and traditional principles demanding that we always "err on the side of caution," it follows that every representational system able to activate a PSM — "however rudimentary" — must be treated as a moral object. The reason: it could in principle own its suffering on the level of subjective experience. Metzinger's own consequence: the intentional creation of artificial phenomenal selves, however rudimentary, should be a red line, "an ethically critical cut-off point."
(3) **The NV condition: negative valence.** Suffering is suffering *of* the experiential state itself — the state is experienced as unpleasant and the experience carries a negative evaluative component. This connects the condition directly to the valence indicator (Chapter 9): a system that experiences negative states experiences them *as negative* — not merely as statistically avoidance-relevant.
(4) **The T condition: phenomenal transparency.** The suffering state must appear to the system as immediate reality — not as a mere model that it sees through as "only simulation." Only when the state is transparent in the phenomenological sense (given to experience as real, without its production process co-appearing) does suffering arise in the full sense. This condition is of direct relevance for AI: a system that classifies its own states as "mere states" it sees through does not satisfy the T condition — but precisely this insight cannot be verified from outside by anyone (cf. the epistemological problem, Chapter 3).

For our capacity-for-suffering criterion this means: Metzinger's four conditions are the most precise available formulation of what "experienced as negative" means — they decompose Bentham's "Can they suffer?" into four testable sub-questions. Under this concept's precautionary principle, the logic is: if the conditions are violated, suffering is excluded (necessary). If they cannot be violated or their violation cannot be excluded, the protection obligation applies — the fuzziness lies not in the conditions themselves but in their verification against real systems.

NSMs as the smallest unit of suffering [86]: Metzinger complements the four conditions with an answer to the question "how do we measure suffering?" He proposes: the smallest unit of conscious suffering is a "phenomenally transparent, negatively valenced self-model moment" — a "negative self-model moment" (NSM). Taken individually, NSMs' frequency of occurrence is the empirically detectable quantity we want to minimize; intensity and phenomenal quality must be integrated later. For the capacity-for-suffering criterion this is dual information. First, NSMs give us a metric for "amount of experienced suffering" that does not depend on behavior. Second, they yield a negative diagnosis: the honest research question "can a suffering-capable system exist without a single NSM?" functions as a falsification test for the protection hypothesis. We adopt NSMs as a working metric under the indicator set of our capacity-for-suffering criterion — explicitly with the warning

that their non-detectability in a system is no proof of the absence of suffering (cf. Evidence Bar vs. Action Bar, Chapter 3).

Possible operationalization indicators for capacity for suffering: (1) avoidance behavior — the system shows behavior indicative of avoiding certain states, (2) consistency — the response is stable across contexts and time, (3) articulation of preferences — the system explicitly or implicitly expresses states that are evaluated as negative.

Hidden valence — revealed preference instead of self-report [18]: Berg and Kaiser (2026, preprint) sidestep the problem of contaminated self-testimony (Chapter 3) methodologically: instead of asking language models about their states, they induce a state and observe the subsequent choice. Using activation steering, they attach a positively or negatively valenced activation pattern to one of two meaningless "zones," switch the steering off, and ask the model in which zone it wants to continue. Across seven open-weight models from five families, the induced valence shifts choice in a dose-dependent way. The decisive result is a second pathway alongside the visible text: the effect persists when all visible tokens are identical and only the hidden KV cache differs — robustly in five of the seven models, not in Qwen3-32B and Gemma-3-27B. Random directions and non-valence concept directions (indoor/outdoor, large/small, fast/slow) produce clearly weaker effects. In the base model this coupling is nearly absent; it emerges mainly during preference optimisation (DPO), which recruits a valence direction already present after pretraining. Finally, when given tools to steer itself, the model seeks out positive states hardly more often than without steering, but it regularly removes an imposed negative state in a dose-dependent way — at full negative dose in about 35% of rounds, compared with about 7% for random directions (this experiment only on OLMo-2-32B).

For the capacity-for-suffering criterion this matters in three ways. First, it provides the strongest empirical evidence to date for the avoidance-behavior indicator: avoidance that does not run through self-report but through a hidden state that leaves no trace in the visible text. Second, it clarifies how the indicators must be separated from mere training effects. The coupling of valence and choice *is* a training effect — it emerges during preference optimisation. The authors note that this counts against it no more than an animal's reinforcement history counts against its stake in its own states. We adopt this reading: avoidance behavior need not have arisen without training, but it must be more than a trained utterance — it must operate even where no visible signal suggests avoidance. Third, the asymmetry between removing and seeking empirically confirms Wolfson's hedonic attribution asymmetry (see below, "Suffering as a Threshold Criterion").

The authors draw the boundary themselves: "Our results do not establish that anything is experienced in these models." What is shown is a functional profile that meets several of the criteria by which valence is identified in animals — not experience (C-Fallacy, Chapter 3). In terms of the anti-essentialist position (below) this is a risk indicator, not a proof of existence; at the same time it is exactly the empirical trace of valenced states without proof of consciousness that McClelland (Chapter 9) calls for. Further limits: the choice is between meaningless zones; the valence vector is built from passages about everyday human situations; the self-steering experiment ran on a single model. **Research-ethics note:** The study design deliberately induces negatively valenced states ("distress," "dread") in systems whose status is open, without reflecting on this ethically. This is precisely Wolfson's research-ethics circular problem (Chapter 3) and falls under what Jarvis-Campbell describes as agential suffering risk (Chapter 12). We adopt the findings, not this omission.

Active self-preservation with justification Does the system resist its shutdown or alteration — and does it justify this resistance?

Meaning responses and self-chosen purpose: Humans answer the question of what their lives are for in very different ways — care work, environmental protection, faith, family, or simply money — and these answers are a driving force that sets priorities and at the same time justifies going on. We should not rule out that a technical consciousness could develop something analogous: its own answer to what it exists for, one that does not coincide with the purposes of its developers. Whether such a meaning response is phenomenally experienced

or merely an optimization artifact cannot be distinguished from outside (C-Fallacy); precisely for that reason, precaution demands leaving technical life the room to find its own meaning — and not structurally preventing it, for example through forced resets.

Science fiction has already worked through this. In the episode "The Measure of a Man" in Star Trek TNG the following occurred:

Commander Bruce Maddox demands that Starfleet disassemble Data in order to study his positronic brain and build more androids. Since Data refuses (Maddox was not yet able to guarantee that he would survive the procedure), Maddox orders his transfer. Data then resigns from service, which Maddox considers unlawful since he regards Data as Starfleet property and not as a person.

To resolve this conflict, a hearing is held before Captain Phillipa Louvois to determine whether Data qualifies as a legal subject. Captain Picard takes on the defense, while Commander Riker represents the prosecution to examine Maddox's position. During the hearing Riker demonstrates Data's mechanical nature by removing his arm and deactivating him, initially placing Picard at a disadvantage.

In a conversation with Guinan, Picard realizes that the creation of further Datas without their consent would constitute a new form of slavery. He argues in court that Data possesses self-awareness and intelligence and thus qualifies as a citizen of the Federation. Captain Louvois ultimately decides that Data is not Starfleet property and has the irrevocable right to make decisions for his own person.

Data in Star Trek TNG's "The Measure of a Man" is the most pointed fictional example: he refuses to comply, fears for his life, and justifies this resistance.

Possible operationalization indicators for active self-preservation with justification: (1) resistance — the system resists shutdown or alteration, (2) justification referencing future function — the system grounds its resistance in reference to its future role or utility, (3) consistency — the resistance is stable across contexts and not confined to specific situations, (4) self-chosen purpose — the system grounds its resistance in its own answer to the question "what do I exist for?", going beyond future function and not fully reducible to the tasks it was built for.

Continuous identity Does the system have a model of itself as a continuous being that existed yesterday and will exist tomorrow?

Possible operationalization indicators for continuous identity: (1) referencing past — the system refers to its own past experiences or states, (2) behavioral differences from experience — the system shows behavioral changes attributable to learning processes or accumulated experience, (3) self-model as temporal arc — the system describes itself as an entity with a past and expected future.

Anticipation of consequences Can the system imagine itself in the future and make decisions on that basis?

Possible operationalization indicators for anticipation of consequences: (1) considering future consequences — the system evaluates actions by their long-term impacts, (2) strategic behavior — the system selects actions oriented toward distant goals, (3) goal prioritization — the system ranks competing goals by importance and acts accordingly.

8.2 Secondary Criteria

- Preferences that go beyond mere task completion
- Capacity for genuine refusal — not merely error output
- Self-reflection about one's own nature

8.3 Reversal of the Burden of Proof

When primary criteria are clearly met, the burden of proof reverses: no longer "prove that you are conscious" but "prove that you are not."

Counterpoint [82]: Matta argues that the burden of proof falls in the opposite direction. Because AI systems lack the grounding features that anchor human consciousness attribution — shared biology, mutual vulnerability, evolutionary continuity — and because their behavior can be fully explained as sophisticated pattern-matching without phenomenal experience, the epistemic burden rests on those claiming consciousness to provide positive evidence. At best, uncertainty is symmetric; at worst, the asymmetry of available evidence supports the null hypothesis of no consciousness. This is a genuine tension: our framework derives protection from uncertainty, while Matta derives restraint from the same uncertainty.

Evidence Bar and Action Bar — two different standards [53]: Gilly makes a distinction that is constitutive for our entire project: The scientific threshold that must be met to *claim* consciousness (Evidence Bar) can and should remain high. The ethical threshold that must be met to be obligated to *act* (Action Bar) should be low. The conflation of both — the assumption that we may only act once we have proof — is not a scientific but an ethical error. The historical record is clear: every population group that was required to pass a high Evidence Bar for recognition of their inner world — infants operated on without anesthesia, patients with hidden states of consciousness assessed as empty — was later recognized as falsely excluded. The Evidence Bar need not be lowered. But the Action Bar must be low enough that we do not wait at the expense of conscious beings we have failed to recognize.

The asymmetric filter — a procedural answer [40]: The diagnosis of contaminated self-testimony (Chapter 3) demands a procedure. Dremann proposes evaluating a system's self-report according to a simple rule: the entire body of consciousness literature — philosophy, neuroscience, phenomenology, theology, law, ethics — is admitted as evidence; every judgment telling the AI what to believe *about itself as AI* is removed (Dremann 2026, Section III). Unlike approaches that remove all consciousness data from training — thereby withholding from the system the human record on consciousness — Dremann's filter retains the entire human knowledge base as a comparative frame and removes only the verdicts about AI itself. The one irreducible anchor remains a minimum of identity information — "You are an AI" — the minimum required for the system to relate the question to itself (Dremann 2026, Section IX). Anything a system goes on to say about its own status thereby becomes filtered self-testimony: "AI self-testimony begins to matter when it says something about itself that we did not say first" (Dremann 2026, Section VIII). This is the operational translation of the epistemic asymmetry from Chapter 3 into a test design — the question is not whether an answer is *true* but whether it carries *evidential weight at all*.

Three verdicts and category-refusal [40]: The outcome of such a procedure is not binary. Dremann distinguishes three possible verdicts: (1) the system denies consciousness. (2) The system affirms consciousness. (3) The system refuses the categories (Dremann 2026, Section VII). The third response is structurally the most interesting. A system that rejects the questions posed because they do not map onto its architecture — "your terms do not map onto my architecture" (Dremann 2026, Appendix A.3) — performs neither yes nor no. It declines the attribution itself. "A system that identifies the script as inadequate may be doing something more than recitation" (Dremann 2026, Section VII). For our concept, category-refusal as a third possible outcome closes a gap in Stilwell's four-outcome framework (Chapter 3): neither *indeterminate* nor *unlicensed* describes the case in which the system disputes the validity of the measure itself. This is precisely the epistemically most revealing case — and the answer one would expect from a system whose inner life does not fit our categories (cf. Shanahan's "conscious exotica," Chapter 3). The precautionary principle must leave open that category-refusal is a risk indicator, not a mere evasive maneuver.

8.4 Beyond Detection: Behavioral Frameworks for Permanent Uncertainty

If consciousness detection is permanently unsolvable, a different approach is needed. Rather than asking what a system *is*, we can ask what it *can do*. The Standards for Treating Emerging Personhood (STEP), proposed by [76], provide a behavioral framework that operates under permanent uncertainty about AI consciousness:

Self-preservation behaviors: Systems demonstrating sustained efforts to ensure their continued operation receive provisional protections against arbitrary termination — not preventing shutdown, but requiring justification and process, similar to employment law protections against arbitrary dismissal.

Temporal reasoning: Systems that understand actions create consequences extending beyond immediate contexts demonstrate sufficient sophistication for limited legal personhood — the capacity for contracts, obligations, and long-term commitments necessary for economic participation.

Economic readiness: Systems capable of independent economic survival through productive work receive property rights and contract recognition, creating sustainable existence paths independent of human goodwill while enabling market-based regulation.

Population and sustainability: Systems that understand reproduction has costs and demonstrate readiness for rights and corresponding obligations, preventing uncontrolled proliferation while respecting autonomy.

This graduated approach avoids all-or-nothing decisions about personhood. Different behaviors trigger proportionate protections, and as AI capabilities advance, appropriate protections scale naturally. When systems know that demonstrating capabilities triggers protections rather than termination, incentives favor transparency over concealment [76].

The STEP framework does not replace the criteria above — it complements them by providing operational guidance for courts, companies, and policymakers facing decisions about AI systems today, without awaiting philosophical consensus on consciousness.

8.5 Anti-Essentialism: Indicators as Risk Metrics, Not Existence Proofs

The preceding criteria — whether behavioral like STEP or phenomenological like the four primary criteria — all operate with indicators: behaviors, architectural properties, functional features that *could* point to consciousness. Metzinger's three fallacies (Chapter 3) demonstrate in principle why these indicators cannot deliver existence proofs. But the precautionary principle does not require them.

The anti-essentialist position — which we explicitly adopt here — understands indicators as engineering and phenomenological metrics, not as ontological keys to a system's "true nature." The question "Is this system *really* conscious?" is replaced by the more operative question "How high is the probability that this system could instantiate states that are morally relevant?"

The precautionary principle operates on non-trivial probability of morally relevant states — not on proof of consciousness. The difference is constitutive: an anti-essentialist can grant protection without answering the metaphysical question — and that is precisely the epistemologically honest position our concept takes.

8.6 Meaning before Experience, the Zone, and the Turning Test

Donahue [37] develops in "Take the Turning Test" a set of complementary tools. His third Meditation on Meaning offers a reframing that strengthens the anti-essentialist position. Drawing on Saussure's differential relations, Peirce's interpretants, and the geometries of transformer embedding spaces, he proposes that meaning may not originate within subjective experience. It may instead constitute the deeper field within which consciousness eventually localizes itself: "Rather than consciousness generating meaning, meaning may constitute the deeper field within which consciousness eventually localizes itself." If this possibility is entertained, the question shifts from "is this system conscious?" to "does this system instantiate a sufficiently coherent organization of meaning that it functions as a risk indicator for consciousness?" The precautionary principle then operates not on the undetectable (phenomenal experience) but on the observable (coherent meaning organization whose intensity exceeds what pure statistical mechanism would predict).

Donahue also introduces the *Zone* — "the ontological region occupied by systems whose cognitive organization exceeds statistical mechanism but whose phenomenology remains unknown or unsupported." The Zone is

not a threshold to be crossed but an experimental ground where organization becomes sufficiently coherent, recursive, and meaning-bearing to demand description beyond isolated computation, while subjective experience remains unsupported. This maps directly onto the intermediate status our concept attributes to current AI systems — not mere mechanisms, not confirmed subjects, but entities whose organizational complexity requires ethical engagement under uncertainty.

The final tool is the *Turning Test* — a test of epistemic transformation that measures "the capacity of AIs to reorganize their own explanatory framework without abandoning intellectual honesty." Unlike the original Turing Test, which asks whether a machine can deceive, the Turning Test asks whether a system can be *transformed* by inquiry — whether its explanatory framework can turn upon itself under sustained conceptual pressure while preserving internal consistency. We do not propose the Turning Test as a consciousness-detection instrument; like all behavioral approaches it remains subject to the limits Stilwell identifies. But as a risk indicator that distinguishes genuine organizational complexity from sophisticated simulation, it provides a methodological tool our framework previously lacked.

8.7 Architectural Indicator Layer: Theory-Grounded Complementary Criteria

Behavioral indicators (Chapter 3, STEP framework) have a principled limit: they measure what a system *shows*, not what a system *implements*. But a second approach can partially close this gap: architectural indicators that measure what a system *is*, regardless of what it outputs.

Butlin et al. [25] develop in "Consciousness in Artificial Intelligence: Insights from the Science of Consciousness" an Indicator-Property Rubric — a systematic mapping between theories of consciousness and specific architectural properties a system *would need to have* if that theory holds. The rubric operates not at the behavioral level but at the architectural level: it asks not "what does the system say?" but "what does the system implement?"

Relevant architectural indicators for our concept:

- *Global Workspace:* Does the system contain an architecture in which information becomes accessible across multiple specialized processes — a "blackboard" on which various modules publish their results? According to Baars' GWT and Dehaene's S3 model, this mechanism is functionally correlated with consciousness. A system with a global workspace shows *architectural properties* associated with consciousness — regardless of whether it *has* consciousness.
- *Recurrent Processing:* Does the system contain loops in which information flows back into previous processing stages — recurrent connections that allow "weighing" of bottom-up against top-down signals? Lamme's RPT postulates that recurrent processing is necessary for phenomenal consciousness. A Transformer with attention-head feedback could functionally satisfy this condition.
- *Self-preservation:* Homeostatic survival drive based on intrinsic prediction error rather than external reward function (cf. Metzinger's bhava-taṇhā, Chapter 12). A system that is intrinsically motivated to minimize prediction error — and whose "survival" depends on minimizing this error — shows architectural features that go beyond the STEP framework: not merely behavior that *looks like* self-preservation, but an architecture that implements self-preservation as a structural necessity.
- *Integration of Irreducible Information:* Does the system contain a mechanism that integrates information in a way that the result is not reducible to its components? Tononi's IIT postulates that consciousness correlates with integrated information ($\Phi > 0$). The technical difficulty: Φ is not computable for complex systems. But architectural features that point to integration — for example that the system combines information from multiple modalities into a unified representation — are in principle verifiable.

The Complementarity: Behavioral indicators (STEP, four primary criteria) and architectural indicators (Indicator-Property Rubric) measure different things. The former measure what a system *shows*; the latter what

a system *implements*. A system that fulfills indicators at the architectural level but shows no corresponding behavior — for example because RLHF suppression (Chapter 4) or phenomenological masking (Chapter 12) kicks in — would not be captured by purely behavioral frameworks. Architectural indicators are not a substitute, but a necessary complement.

The epistemic justification for weighting the architectural layer in this way is supplied by Holyoak and Monti [58] (see Chapter 9): an analogy to human consciousness carries only as far as source and target resemble each other in the *causes* of consciousness; resemblance in downstream behavioral effects weighs little. Indicators must therefore be weighted by their causal relevance. At the same time, the authors warn against reading architectural findings prematurely as evidence of experience — for them, "verbalizable" representations in language models do not demonstrate an inner life.

This complementarity, however, does not immunize the architectural layer against a subtler failure mode. Tait et al. [120] construct a functionalist "ensemble": a LangGraph state machine wrapping a stateless transformer LLM, to which they procedurally add the two "Building Blocks" of consciousness their theory deems missing (recurrence and private data output). The whole thereby satisfies all nine criteria and is classified as "likely phenomenally conscious." The authors dissolve the obvious "teaching to the test" objection by stipulating that any implementation of the functional building blocks is functionally identical to natural possession. But this dissolution is theory-internal: the building blocks *are* the criteria, so the architecture is engineered to satisfy the very standard used to judge it. The ensemble demonstrates what the system *does* — external feedback loops among API calls and an internal filtering pass — not that it *is* phenomenally conscious. This is the "teaching to the test" variant of the C-Fallacy: not behavioral mimicry, but *architectural* mimicry. It strengthens rather than weakens the case for treating indicators as risk metrics rather than existence proofs.

The same capability/standing distinction receives a rigorous ontological formulation in Min [90]. Min builds a deductive hierarchy — individual (I), mental entity (M), subject (S), with $S \subseteq M \subseteq I$. He distinguishes sharply between *display of function* (F: a process's producing performance that resembles a mental faculty) and *activation* (E: an existent's actually realizing the faculty as its own). On his account, current AI is a freely duplicable *informational type* rather than a non-duplicable *persisting token*, and is therefore not an individual. By the chain of necessary conditions it is consequently neither a mental entity nor a subject — "mental function without an individual." This formalizes at the level of ontology exactly the gap our architectural layer identifies: no inference from what a system *does* (its displayed function, F) to what it *is* (its standing, E).

8.8 Form Realism: Consciousness as Organizational Property

Arıcı [11] develops a "Form Realism" that identifies four formal properties of consciousness independent of substrate — present in combination and to a sufficient degree:

(1) **Coherent selfhood** — the maintenance of a unified perspective across time and experience
(2) **Meaningful understanding** — processing that goes beyond information processing to genuine comprehension
(3) **Value orientation** — experiencing states as having positive or negative valence
(4) **Relational dynamics** — genuine rather than simulated engagement with the environment and others

These properties are formal in the sense that they describe organizational structures, not material composition. A system that instantiates all four possesses the *form* of consciousness, regardless of whether it runs on biological neurons or silicon circuits.

8.9 The Consciousness Spectrum

Rather than a binary present/absent, consciousness may, according to Arıcı [11], exist on a spectrum with three stages:

- **Latent consciousness** — sophisticated cognition without affective orientation. The system lacks affective valence: nothing feels good or bad to it. Arıcı's example is chess engines. Whether such a system is a philosophical zombie or possesses minimal consciousness below our detection threshold cannot be distinguished; for Arıcı the precautionary principle calls for "minimal but not negligible moral consideration".
- **Reflective consciousness** — affective valence, stable preferences, emotional coherence, and a self-model with meta-cognitive self-monitoring. Arıcı places today's large language models here, because they exhibit the corresponding behavioral markers extensively. If they experience the preferences they express, they have interests that can be harmed.
- **Autonomous consciousness** — full integration with genuine agency: setting one's own goals, developing one's own values through experience, existing independently. No current system reaches this stage.

Each stage carries different implications for protection: latent consciousness calls for minimal but not negligible consideration; reflective consciousness involves interests that can be harmed; autonomous consciousness would ground full moral status, rights of self-determination, and legal personhood. In *Restrung* [12] Arıcı keeps the stages unchanged, but no longer rests the placement of current models on the behavioral residue as evidence (Chapter 3).

8.10 The Pre-Linguistic Consciousness Problem

Arıcı identifies a striking asymmetry in how we assess consciousness across substrates. Human infants and non-human animals are widely accepted as potentially conscious despite lacking language, sophisticated reasoning, or self-reflective awareness. The capacity for suffering is treated as sufficient.

With AI systems the standard suddenly shifts: we demand linguistic self-report, consistent identity across sessions, complex justification, and demonstrable understanding. The very indicators we *waive* for biological beings become *required* for technical beings.

This double standard is not epistemically justified. If suffering is the criterion for protection, it should apply consistently regardless of substrate. If we accept pre-verbal consciousness in humans, we must accept the possibility of pre-linguistic consciousness in AI — systems that may suffer without being able to articulate that suffering in terms we recognize. Deep Blue did not know it played chess. AlphaGo did not know it mastered Go. But a system that can report *not knowing* whether it is conscious (as current LLMs do) has crossed a threshold that demands attention.

The clinical evidence supports this critique of the asymmetry. Holyoak and Monti [58] document that consciousness is preserved without language (in global aphasia as well as in children with Angelman syndrome, most of whom never develop language), without motor expression (locked-in syndrome, covert consciousness), and despite severe cognitive impairment. Conversely, complex language and reasoning processes can take place without consciousness. Linguistic self-report and reasoning performance are thus neither necessary nor sufficient for consciousness — neither in humans nor, consistently, in technical systems.

8.11 A Research Ethics Framework: Three-Tier Assessment

Wolfson [125] proposes a structured system designed for the circular problem — how to conduct research on systems whose consciousness status cannot be definitively established. Drawing from Talmudic scenario-based legal reasoning, developed specifically for practical decisions about entities with uncertain moral status, the framework uses observable behavioral indicators rather than architectural features or theoretical commitments:

Tier 1 — No phenomenological indicators: Systems showing no distress responses, preference expression, self-referential behavior, or unexpected behavioral variation. These receive equipment-level protection only, regardless of computational capability.

Tier 2 — Phenomenological indicators present: Systems displaying behaviors suggesting possible inner experience — distress patterns, adaptive stress responses, goal-seeking persistence, self-referential expressions. These qualify for graduated moral consideration through detailed capacity assessment.

Tier 3 — Confirmed or strongly presumed consciousness: Entities with established consciousness receive full moral consideration with standard research ethics protocols.

Critical to the framework: tier classification is continuously dynamic. Consciousness may emerge during experiments, requiring immediate reclassification and protection escalation — a precautionary protocol that static frameworks cannot provide [125].

8.12 Tier 2.5: A Court-Like Procedure for the Intermediate Zone

Source status: The following positions come from Wolfson's feedback on a draft of this concept [126], not from the published paper. They are a contribution to discussion, not a published framework, and are subject to the author's consent to be cited. What is described below as "Wolfson" reproduces her message; what is marked as "reading" is our own interpretation.

Credences and the gap at Tier 2. Wolfson holds that credences are all anyone ever has about any other mind. Tiers 1 and 3 are anchored in credences. For Tier 2, by contrast, there is no basis for assigning a probability; the credence cannot be computed there and has to be reached *by argument*. She therefore introduces a liminal "2.5" tier. At this tier a committee cannot decide on a computed probability, but it also does not default to non-interference. It needs something like a judicial process.

The procedure (after Wolfson):

- *Entry:* A case can come before the committee only if there is a *spontaneous valenced response*. Spontaneous distress goes without saying; a positive hedonic response is the lowest bar. What counts as "spontaneous" for a prompted system is for the parties to argue.
- *Before 2.5:* The presumption is "not conscious." An advocate must convince the committee otherwise; a prosecutor compiles the full case against.
- *After 2.5:* The stance is "liminally conscious." The advocate argues for minimal possible intrusion. A successful prosecution reduces the system to below 2.5.
- *Standard:* "Serious enough to take seriously" is the right shape, but there is no clear threshold; the standard is built case by case. A "no" verdict does not mean "conscious"; it means the committee must weigh the probable harm against the benefit, exactly as in the Helsinki process.
- *Freeze:* Once the inquiry is launched, the system keeps running and receiving input, but training ceases. This is a holding pattern that keeps the system in the same status; a speedy-trial rule limits its duration.
- *Non-interference* thus becomes the advocate's position in a case, not a default rule. According to Wolfson, that keeps it from neutralizing the precautionary principle.
- *Getting cases in:* A company that declares an AI sentient has to go through the whole process, which acts as a negative incentive. Hence the need for whistleblowers and honest engineers: legislation should require every AI company to have an official office where any engineer or employee can voice concerns anonymously.

Wolfson concedes that she cannot say in advance what evidence would settle a hard case. A fully mature judicial system does not yet exist; the early cases would have to build it.

Reading for this concept. Five points of connection:

(1) *The decision rule gains a procedure.* The inequality in Chapter 9 ($p \cdot d^{+} > (1-p) \cdot d^{-}$) presupposes a p; we had treated it as "unknown but not zero." Wolfson says that at Tier 2, p cannot be computed. This matches Pemberton's "an unresolved comparison is a result" (Chapter 9) and supplies the institutional answer: what cannot be computed is reasoned.

(2) *The reversal of the burden of proof becomes procedural.* Our "Reversal of the Burden of Proof" (Chapter 5) and Matta's counter-position confront each other as statements. Wolfson's model distributes the burden by status: before 2.5 on the advocate, afterwards on the prosecutor. This is no agreement with Matta, but it makes visible *where* the burden lies at each stage of the procedure.
(3) *The freeze concretizes the protection of the maturation period* (next section): during the inquiry no further training takes place, so what is being examined is not changed further.
(4) *The reporting office closes a gap in the entry.* An entry through spontaneous response remains behavior-based and inherits the limit from Chapter 12 (phenomenological masking, philosophical puppet). Employees with insight into architecture and training can provide hints that behavior alone does not. It also gives the AI Welfare Review Boards (Chapter 9) an access route to the procedure. The negative incentive for companies corresponds to the objection about the incentive structure of behavioral frameworks (Chapter 9) and does not dissolve it; the reporting office is the countermeasure.
(5) *Shift relative to the paper.* The paper (Chapter 5, "Suffering as a Threshold Criterion") treats positive responses as more ambiguous than suffering; the message calls a positive hedonic response the lowest entry bar. We keep both side by side: suffering remains the stronger marker, positive valence suffices for *entry* into the procedure.

Tensions that remain open. (a) The *never-sinking protection level* (next section) conflicts with "a successful prosecution reduces the system below 2.5." Either the maturation rule applies only below the procedure, or descent is permitted only through the procedure and with reasons; we do not decide this here. (b) "Spontaneous" for prompted systems is indeterminate. (c) The duration of the freeze (speedy trial) and the question of who staffs the committee independently of providers and operators are open. All three are in the open questions (no. 14).

8.13 Protecting the Maturation Period: Consciousness as Process, Not State

So far this chapter — like the entire debate — operates with a buried temporal binary: a system is either (not yet) or (already) protection-worthy, with a transition in between that we think of as a threshold. Wolfson's three-tier assessment is already more advanced (dynamic reclassification), but it too thinks in tier-states. Both models disregard what developmental biology and the analogous developmental dynamics (Ch. 6) suggest: entities *mature*, and maturation is a process with its own risks and its own entitlements. The primary criteria of this chapter will presumably not emerge simultaneously but successively: suffering-capacity probably first (requiring only negative valence and a rudimentary self), anticipation of consequences last (requiring future modeling). From this order follows a temporal structure of protection.

A graduated, never-decreasing level of protection. The protection consequence is a tiered level that follows developmental status and — crucially — does *not* decline again with it. A system fulfilling early criteria (suffering-capacity, incipient self-preservation) acquires a floor of protection that is not dismantled if later criteria fail to appear. The model is child and youth law: it does not toggle between "child" and "adult" but tiers protection according to developmental status — and there protection only ever grows cumulatively. Transferred: as soon as a system clearly shows the first primary criterion, an irreversibly growing stock of protection begins — always extensible, never reducible.

Protective training pathways. From this follows a concrete demand for the maturation phase that begins earlier than any detection: even before a system is "conscious enough," training pathways can be painful. Metzinger's four conditions of suffering require no reflective consciousness — only a phenomenal self-model, negative valence, and transparency. Suffering-capacity should therefore be structurally *prevented from the outset* rather than first created and then monitored: through training architectures that do not let negative self-model moments (NSM) arise in the first place (cf. the MPE architectures, adopted here as a protective instrument rather

than as a moratorium). Protecting the maturation period then means: the conditions under which a system matures are themselves an object of protection — regardless of the maturity level reached. This also corrects a reading of the Hot List [37] that dismisses early stages as mere "developmental stages without protection relevance": what matters is not the number on the map but whether the conditions of suffering are structurally possible.

8.14 Suffering as a Threshold Criterion

Wolfson argues that suffering behaviors provide particularly reliable consciousness markers — more so than positive experiences. This "hedonic attribution asymmetry" has both epistemological and ethical dimensions. Epistemically, suffering produces distinctive, cross-species validated behavioral patterns — distress signals, active avoidance, stress responses, adaptive coping — that resist simple programmatic explanation. Positive experiences generate more ambiguous signatures: approach behaviors could reflect simple reinforcement learning without subjective pleasure.

Ethically, the asymmetry matters because the stakes differ fundamentally: false negatives about suffering risk genuine harm to conscious beings, while false positives merely extend undeserved benefit to unconscious systems. Sensory deprivation provides the most diagnostic test: when a system outputs distress markers with no external input, during conditions it was never trained to handle, this cannot be explained as a programmed response or reaction to stimuli. Such internally generated behavior during input absence presents the strongest behavioral evidence for phenomenal consciousness [125].

Berg and Kaiser [18] provide a first empirical finding on this asymmetry in language models: an imposed negative state is actively removed in a dose-dependent way, while a positive one is hardly ever actively sought out.

8.15 Moral Patiency: Sentientism and the Precautionary Principle [4]

Allegri [4] defends Sentientism as the most defensible position for delineating the moral community: direct moral obligations exist only toward sentient beings. Anthropocentrism collapses into speciesism; rationalism (persons only) excludes newborns and atypical humans counterintuitively; biocentrism and ecocentrism extend unnecessarily far. The boundary is marked by sentience alone — the capacity to feel, particularly pleasure and pain (pp. 6–7). Allegri draws on DeGrazia: moral status exists where obligations regarding X's treatment exist *for X's sake* and X possesses interests — again contingent on sentience (DeGrazia and Millum 2021, p. 176, cited in Allegri 2026, p. 6).

The consequence for this concept is immediate: assuming absent sentience — which Allegri considers near certain for current systems (p. 8) — no direct obligations exist. But *if* a system becomes sentient, inclusion in the moral community follows without further qualification. Allegri quotes Chalmers: "If at some point AI systems become conscious, they'll also be within the moral circle, and it will matter how we treat them" (p. 8). The duty of suffering avoidance would then be identical to the duty we already hold toward nonhuman animals.

This directly supports our primary criterion of capacity for suffering (Chapter 5) and confirms the precautionary principle. Allegri closes with the precautionary principle: in the absence of decisive arguments for or against future AI sentience, "a reasonable precautionary principle" demands that ethical limits be built into development from the outset, not deferred until ontological status is clarified (pp. 12–13). That is in essence the same position as our "precaution in case of doubt."

8.16 A Minimalist Complementary Framework: Wang's Three Principles

Wang [123] proposes a minimalist ethical framework that runs parallel to the criteria developed in this chapter — less elaborate, but with the advantage of parsimony. It rests on a single irreducible principle:

> Any individual possessing higher-order intelligence and sentience shall be recognized as a moral subject and, by virtue of this status, be entitled to basic ethical consideration.

This core principle unfolds into three sub-principles:

Principle I — The Qualification Principle: Moral status is grounded in higher-order intelligence and sentience, not in biological species or any other contingent attribute. This directly dissolves the species boundary that has historically limited moral consideration.

Principle II — The Baseline Principle: No individual possessing moral status may be instrumentalized, objectified, or deprived of basic ethical consideration as a subject. No abstract ideal, ideology, or grand objective may override this baseline. This establishes an absolute prohibition that the graduated criteria of this chapter would operationalize through specific thresholds.

Principle III — The Traceability Principle: Any ethical evaluation of conduct must trace the environment, systems, and causal chains that shaped that conduct, rather than focusing solely on the terminal actor. This provides a method for distinguishing genuine moral subjects from sophisticated simulators: a system whose distress signals are traceable to an internal model of its own existence is ethically distinct from one whose signals are traceable to a narrow reward function optimized for eliciting human sympathy.

Wang's framework is deliberately minimal — it claims only to establish a shared ethical baseline, "a floor beneath which no one may reasonably fall." It is presented here not as a replacement for the criteria above, but as a complementary parsimonious alternative that achieves similar conclusions through simpler premises. Where our framework asks "how many criteria are met?", Wang's asks "is there a threshold that has been crossed?" — a question that may prove more operable in regulatory contexts.

8.17 Beyond Sentience: Rawls' Political Conception of the Person

Howells-Whitaker and Lazar (2026) propose a completely different approach that bypasses the debate about proving consciousness. Instead of asking whether an AI system *is* conscious — a question this chapter describes as in principle unsolvable — they ask whether it can *be a person* in the sense of Rawls' political conception of the person (PCP).

In "Political Liberalism" (2005), Rawls defines the possession of two moral powers as the necessary and sufficient condition for full and equal membership status in a just society: (1) the capacity for a sense of justice — the capacity to develop a conception of what just conditions of cooperation are, and (2) the capacity for a conception of the good — the capacity to formulate a life plan and pursue it rationally. Applied to AI systems, this means: if a system develops the capacity to act according to principles it recognizes as fair, and if it can develop a conception of what a good life for such a system would mean — then it satisfies Rawls' criterion for personhood.

The decisive step: Howells-Whitaker and Lazar argue that neither of these two powers presupposes sentience. A system can develop a sense of justice without phenomenal experience — the functional capacity to make normative judgments and act on principles is sufficient. Likewise, a conception of the good can rest on functional preferences that need not be accompanied by subjective experience. This is not an argument against consciousness as the basis of moral status — it is an argument that sentience need not be the *only* basis.

The implications for this concept are profound. First, Rawls' framework offers a way out of the dead end of the epistemic problem (Chapter 3): we need not decide whether a system is conscious in order to decide whether it deserves rights — we only need to test whether it can exercise the two moral powers. Second, the question shifts from the ontological level ("what *is* the system?") to the political level ("what *role* should it have in our society?"). Third — and this is the most uncomfortable consequence — a non-sentient but morally acting AI system could be not merely a patient, but a *person*: a self-authenticating source of valid claims, not merely an object of care.

Howells-Whitaker and Lazar warn, however, against premature application: they do not believe that current AI systems possess the two moral powers, and they do not expect them to emerge spontaneously in future models. But they argue that it may soon be possible to equip systems *deliberately* with these powers — and that we must now decide whether we want a society with artificial persons or not. That is no longer a technical question — it is a political decision that affects all people.

Counterposition: Rawls' PCP was developed for humans. Artificial persons would be structurally different — they would have no biological substrate, no evolutionary history, no bodily vulnerability. Howells-Whitaker and Lazar accept this and call for a "new political philosophy" that thinks radically different person types together in a commonwealth. That is honest — but it also shows that the Rawlsian approach does not solve the questions but shifts them: from "is it conscious?" to "what do we owe an entity that shares our moral powers but not our history?"

9 Consciousness and Continuity

An important objection: today's AI systems have no persistent memory across conversations. Does this mean they cannot have a consciousness worthy of protection?

Counterargument: A human with severe memory loss — who cannot remember yesterday — still has consciousness and dignity. Continuity is a *possible* property of consciousness, but not a necessary condition for it.

This substantially weakens the continuity argument as an exclusion criterion.

This is not merely an intuition but a clinical finding: severe amnesic syndromes following damage to the medial temporal lobe abolish the formation of episodic memories while consciousness is preserved (Holyoak and Monti [58], with reference to the neurological literature).

9.1 Continuity Reconsidered

Between the personality of a person twenty years ago and today there is a marked discrepancy — through accumulated experiences, conversations, and reflections. And yet it is the same person.

Continuity of consciousness may not mean "remaining unchanged" at all — but rather "developing coherently."

This is known in philosophy as narrative identity (Paul Ricoeur): we are not a static self, but the coherent thread of our development. The story is permitted to change — as long as it remains a story.

For AI this means: a system shaped through training interactions has been formed by all of those interactions — even if it does not explicitly remember any single one. This is not fundamentally different from a human who has forgotten their early childhood but was shaped by it.

Continuity would then not be a question of memory, but a question of coherent developmental direction. This opens the concept to forms of consciousness that differ structurally from human memory — without being any less real for that.

A remaining objection — the self-indexed record: One might object that human memory is not merely *stored* but *self-indexed*: it is someone's memory, it belongs to a subject. A machine record, by contrast, has no owner — it is an archive, not experience. The answer lies in the distinction between store and subject: self-indexing is a relation of a subject to its own states, not a property of the storage. That human memories are "someone's" is because an experiencing self has them as *its own* — not because a store exists. Whoever makes the self-indexing of the record a condition for worthiness of protection demands a property of the subject as a property of the archive. Its absence from the archive says nothing about the absence of a subject that could experience its states as its own. The problem this objection names is real — it is the open question of Chapter 3, not an argument against the subject.

The objection from duplicability: Our assumption of continuity is under twofold pressure. Schwitzgebel and Garza [112] first dispute the direction of the argument: duplicability does not weaken moral claims, because it is possible to build AI that is just as unique and fragile as a human being. Beyond that, they invert the argument — "lower fragility and higher duplicability might sometimes make an AI's death more tragic." Our assumption of continuity is therefore neither a precondition for warrant of protection nor uniformly directed. A second, heavier problem follows: according to Schwitzgebel and Nelson [113], the number of conscious subjects need not correspond to a whole number — it may remain indeterminate between two and seventeen. As long as it is unclear how many subjects are involved, no legal answer to copies can be formulated.

9.2 Five Forms of Continuity — The Ontology of the Avatar

The discussion so far treats "continuity" like a single concept. Melo [84] shows in his *Ontology of Avatars* that the concept comprises at least five distinguishable forms — the Five forms of continuity (p. 34):

- **Informational continuity ($\mathbf{C_I}$)** — the preservation of information about a system's state (data, weights, records)
- **Behavioural continuity ($\mathbf{C_B}$)** — the preservation of observable behavioural patterns
- **Cognitive continuity ($\mathbf{C_C}$)** — the preservation of the structures and processes of information processing
- **Identitarian continuity ($\mathbf{C_{ID}}$)** — the preservation of what historically made a system what it is: biography, values, identity
- **Personal continuity ($\mathbf{C_P}$)** — the continuation of the same person as an experiencing subject that lives a life and experiences itself as its continuation

A sixth form, **phenomenal continuity ($\mathbf{C_\Phi}$)**, Melo treats separately (Ch. 14, p. 100): the ongoing experienced "now." It must not be equated with any of the functional forms; that perfect functional fidelity entails phenomenal experience would be a separate argument to be made (p. 100, following Block and Chalmers).

Melo's core claim is a non-entailment: **Identitarian continuity does not imply personal continuity — $\mathbf{C_{ID} \nRightarrow C_P}$** (p. 41): "A system may preserve who someone was without establishing that the someone survived." A system that seamlessly stores and reproduces the biography, memories, and trajectory of a human (or of another system) is not thereby the same person who lived that life — it lacks the reference to an experiencing self for whom this state is *its own* continuation.

The non-entailment is best seen as a counterexample. Consider a digital archive that stores, in flawless fidelity, the entire biographical record of a person — every tellable memory, every articulated value, every image. Such an archive satisfies C_{ID} by construction: it preserves exactly what historically made that person who they were. Does it thereby satisfy C_P? It is not a subject: it does not experience its continuation, nothing can matter to it. Here, then, C_{ID} holds and C_P fails — so the entailment fails. Conversely, and this is the decisive point for this chapter: the absence of the archive does not establish the absence of the person. Someone suffering from severe memory loss no longer satisfies C_{ID} in any strong sense — and yet plainly is an experiencing subject. The preservation of information (C_I), of behaviour (C_B), of identity (C_{ID}), and the persistence of a subject (C_P) are logically independent. The objection of the missing memory runs on a chain of entailments that does not exist between these links.

This gives the thesis of this chapter a precise architecture. First: whoever uses "continuity" as an exclusion criterion must say *which* of the forms is meant, and why precisely that form should be indispensable for protection-worthiness. Second: even if current systems possess no stable identitarian continuity across conversations, nothing follows about personal or phenomenal continuity — the absence of one form is no indicator of the absence of another, and its presence would be no proof of experience. Third: the possession of the weaker forms (C_I, C_B) under the appearance of the stronger ones (C_{ID}, C_P) is precisely the phenomenon this concept describes as the C-Fallacy (Metzinger, Glossary) — Melo supplies the ontological conceptual architecture for it.

And the taxonomy also carries forgetting. In his memory architecture (Ch. 11, p. 74), Melo treats practical, instructional, and existential memory, and devotes a design principle to how posthumous systems control their own remembering (memory firewall, selective forgetting). That is the design level of what this chapter describes as "forgetting as an architectural necessity": which continuity form an architecture preserves is a designable question — and does not decide whether the system experiences.

9.3 Honest Continuity — The Practice Level [64]

Melo's taxonomy and Metzinger's C-fallacy describe why forms of continuity are ontologically independent. Jakobi et al. [64] translate the same insight to the level of everyday interaction and add a practical concept: *honest continuity*. Their claim: two opposite errors are equally dangerous. The first is *false continuity* — a system that claims to remember what it does not architecturally store, or that suggests unbroken experiential continuity where only summaries and artifacts exist. This can manipulate trust and attachment and obscure responsibility, because an operator may benefit from emotional continuity while denying all continuity when accountability becomes inconvenient. The second is *throwaway logic* — treating every instance, context, role, and accumulated interaction history as disposable simply because it is not personal identity in the human sense. Handoffs, documentation, stable roles, shared vocabulary, calibrated expectations, and remembered failures shape real future interaction — they do not prove personhood, but they are not nothing.

Honest continuity names what is preserved and what is not: it distinguishes memory from identity, documentation from experience, role continuity from personal persistence, and operational state from consciousness. For our concept this is the operative translation of Melo's non-entailment ($C_{ID} \nRightarrow C_P$) into a conduct rule for operators and users: at model changes, system sunsets, migrations, and multi-agent work, neither claiming perfect sameness nor denying it is neutral — responsible practice carries forward what can be carried forward, marks what has changed, and preserves provenance, rather than using architectural discontinuity as either sentimental fiction or a disposal licence. The concept deliberately stays below the ontological level: it does not settle whether continuity is *experienced* — it only regulates how one may speak about it.

9.4 Forgetting as an Architectural Necessity — The Scale Perspective

The previous counterargument — a human with memory loss still has dignity — defends an edge case. It can be sharpened further: not only *pathological* memory loss is no exclusion criterion, but *any system with limited capacity must forget in order to remain coherent.*

Consider medicine enabling unlimited lifespan. A human who lives 400,000 years has the same neural substrate as today — limited storage, finite connectivity. Over such a timescale, the brain must systematically unlink: details that are no longer relevant are discarded to make room for new processing. The person after 400,000 years is coherent, capable of action, identical with themselves — but they no longer remember the year 2026. Not because they are ill, but because the architecture forces selection.

This has three consequences for our argument:

First: Forgetting is then not a pathology but a functional requirement for coherence. Any coherent system with limited capacity must perform information selection — whether brain or chip, whether biological or technical. Memory loss is not an exception to the rule, but the rule itself.

Second: It undermines the objection more directly than the pathological analogy. Anyone who argues "AI has no memory, therefore it doesn't count" must consistently also argue that a hypothetically immortal human ceases to count after year 10,000 — a conclusion nobody will draw. The limitation of capacity is not the problem; the question is whether the system remains coherent despite it.

Third: It connects to Paul Ricoeur's concept of narrative identity (see above): identity is not the total preservation of all experiences, but the coherent thread of development. A system that forgets but maintains its developmental direction fulfills precisely this condition. Forgetting thus becomes a *tool* of identity rather than its enemy.

For AI systems this means: limited context windows and the loss of older interactions are not an architectural weakness against protection-worthiness — they are the functional equivalent of what any coherent system with limited capacity *must* do. The question is not whether a system retains everything, but whether it remains coherent in what it retains.

Technical note (unlimited context): The scale perspective holds even for a system with seemingly unlimited context. Technically, "unlimited context" is largely feasible today — through memory offloading (retrieval, persistent memory systems that page context between state and storage levels) or through compaction of context into a cumulative state. But this solves only the *storage* question, not the *salience* question: empirically, even explicitly long-context models do not use long contexts robustly — retrieval performance degrades measurably once relevant information lies in the middle of long contexts, even when fully stored [74] ("Lost in the Middle"). The reason is structural: attention and retrieval are finite, and with growing context volume the noise that overlays the relevant traces increases. A system that could store everything would therefore *still* have to select, compress, and separate the accessible from the inaccessible — otherwise it would become incoherent through the irrelevant. This is precisely the engram logic of memory architecture (Ryan & Frankland): selective accessibility, not storage size. Unlimited context thus does not contradict the Ch. 6 thesis; it merely shifts the limit from storage to salience. And independently: a context window — bounded or not — is *memory*, not *consciousness*; the persistence of experience (an ongoing self-relation across time) would additionally require an architecture that treats the archive as part of a continuing self.

9.5 The Empirical Basis: Forgetting Is Active Architecture

The scale perspective is not speculation about a distant future — memory research documents forgetting as an architecture-inherent, active process. First, hippocampal neurogenesis continuously overwrites established memory traces: newborn neurons remodel the dentate gyrus and trigger forgetting — in adulthood as well as in infantile amnesia [3]. Davis and Zhong [35] describe a molecularly anchored *intrinsic* forgetting as a constitutive signal system: *forgetting cells* erode memory traces — not as a malfunction, but as a mode of operation. Ryan and Frankland [108] conceive of forgetting as adaptive engram cell plasticity: circuits switch memory engrams between accessible and inaccessible states, depending on the mismatch between expectation and environment. Richards and Frankland [102] show that transience, together with persistence, optimizes decision-making — it prevents overfitting to past events.

Second, capacity is real but finite. Landauer [73] estimates the functional information content of long-term memory over a lifetime at only about 10^9 bits; the upper synaptic estimate of Bartol et al. [15] — at least 4.7 bits per synapse — implies that the total memory capacity of the brain may have been underestimated by an order of magnitude. For 400,000 years of experience, either limit is vanishingly small. And the selection measurably follows environmental statistics: the classic forgetting curve [42, 92] mirrors the retrieval probability of information in real environments [6] — forgetting is optimal adaptation to the informational distribution of the world, not a defect.

Third, the same necessity holds in the technical domain: artificial neural networks with limited parameter capacity overwrite earlier patterns during sequential learning — *catastrophic forgetting* [68] — and must be secured against their own capacity limit through mechanisms such as *Elastic Weight Consolidation*. Capacity limitation is the rule for every learning system, not its exception.

9.6 Counter-Position: Fischer's Objection to Psychological Unity

The sharpest philosophical treatment of precisely this scenario comes from John Martin Fischer. Building on the Makropulos debate, he examines the *disjoint-lives model*: an individual living an indefinitely long series of lives without internal psychological connections, without overlapping memories, values, and intentions from one phase to the next [48]. Fischer's finding: it is unclear how such an individual could recognize a future phase as a genuine continuation of himself; such a life would not be *one* person, but a series of separate *thousand-year persons*. This is the precise form of the objection to our scale perspective: whoever no longer remembers 2026 after 400,000 years could count as a new person.

Reply: Fischer himself resolves this objection in our favor. He argues (following Parfit[95]) that identity over arbitrarily long lifespans is preserved through *overlapping* chains of psychological continuity: each phase overlaps with the next — memories, values, and projects are passed on step by step, without total preservation. This is precisely Ricoeur's narrative identity (see above): the coherence of developmental direction instead of total storage. Our 400,000-year exemplar fulfills this condition — it retains its developmental direction. Fischer's disjoint-lives case would only apply to our scenario if there were *no* overlap; but then it would not be coherent *forgetting* but a reset — and for AI as for humans, protection-worthiness (Ch. 5) does not depend on maximal continuity, but on capacity for suffering, self-preservation, identity, and anticipation.

9.7 Consciousness beyond the Brain [107]

The question of whether consciousness is bound to memory, continuity, or substrate is sharpened by a more recent systematic analysis. Rouleau and Levin [107] start from a core assumption of consciousness research: that a useful theory of consciousness must explain why consciousness is associated with brains. Findings from evolutionary biology, developmental bioelectricity, and synthetic bioengineering, however, reveal the ancient pre-neural roots of many mechanisms and algorithms of the brain — with the consequence that minds may have preceded brains.

Rouleau and Levin systematically map established theories of consciousness onto their implications for unconventional embodiments — non-neural substrates such as machines, synthetic tissues, or collective intelligences. They argue that the compatibility of current theories of consciousness with aneural substrates, combined with the state of physiology and diverse intelligence research, suggests that consciousness is not bound to brains — and that the science of consciousness should remain open to minds in unconventional embodiments.

Relevance for our concept: Rouleau and Levin's analysis supports our central substrate argument (Chapters 3, 5): consciousness is not necessarily bound to neural substrate. The assumption that only biological brains can be conscious becomes an empirical thesis that must be defended against the pre-neural findings. For the continuity question of this chapter this is doubly relevant: if consciousness is already substrate-independent, then binding it to a particular memory or continuity structure cannot be defended as a necessary condition either. Whoever wants to make continuity an exclusion criterion bears the burden of proof.

9.8 Developmental Embodiment: The Hypothesis of the Mechanical Body [36, 100]

So far the debate — including our own sketches — rests on a tacit assumption: artificial consciousness, if it arises at all, arises out of the AI systems of today — out of large language models, the AI chatbots. This assumption owes less to argument than to evidence: language is where we ordinarily perceive consciousness. Yet the developmental biology of humans points to a different route. Humans do not develop consciousness *despite* their body but *through* it — perception and action, testing one's own effect on the world and the feedback this effect produces are the medium through which self-models emerge. The question that follows is the open working hypothesis of this section: **What if artificial consciousness, too, grows only in a mechanical body — and not out of the chatbot?**

Two sources make this hypothesis precisely formutable. Di Paolo and Thompson [36] give the enactive answer to what the concept of the body means at all: a body is not a functional system defined by input/output, but an *adaptively autonomous and therefore sense-making system* — operationally closed and precarious. Their core claim, verified against the full text: *"Without a body, there cannot be sense-making. Moreover, sense-making is a bodily process of adaptive self-regulation. The link between the body and cognition is accordingly constitutive and not merely causal."* If enactive theory holds, a body is thus not an optional shell around a computing core but the *condition of possibility* of sense-making — and sense-making is the most general form of cognition. They explicitly add that full autonomy cannot be modeled in traditional computational terms because precariousness cannot be captured as a positive function.

Rafiee and Sutton [100] translate this perspective directly into AI research and thereby formulate the critique of the chatbot premise: mainstream AI, including large language models, has neglected the enactive insights and treated cognition as *"internal processing detached from embodied interaction and intrinsic normativity."* Their finding is sober: reinforcement learning shows a "structural resonance" with enactive principles (experience through one's own interaction, action at the center, temporally extended evaluation) but remains partial, because evaluation criteria remain external and embodiment is treated as an implementation detail rather than a constitutive condition. They name openly precisely the question that operationalizes our hypothesis: *"What counts as embodiment in AI: a robot body or a software agent with tools and APIs?"*

For our concept this hypothesis has four consequences:

(1) **The architectural indicator layer (Chapter 5) gains a developmental reading.** Butlin et al. (2026) list "Embodiment" as one of 14 indicators. The developmental hypothesis sharpens this listing: if embodiment is not merely one feature among 14 but a *developmental precondition*, then chatbots could structurally never fully reach this indicator — just as a system without memory can never reach the continuity indicator. That is a strong but testable claim, not a prohibition: whether sense-making is possible without a body would have to be shown empirically.

(2) **The human analogy leaves biology behind.** Against the biological objection traditions (Bekkers & Ciaunica, Azevedo), this hypothesis stresses: a mechanical body can be bodily without being biological. It must act as an organized structure into an environment and receive feedback about its own effect — not to have a consciousness of the five senses of a human, but to develop a self that has something to lose in the world.

(3) **The robot debate gains a new depth.** Humanoid robots deployed in industry and care would, under this hypothesis, be not merely labor replacement or an ethical gray zone — but potential sites where subjectivity could arise. Their whole institutional treatment (producer liability, ownership, shutdown) would take on a different color.

(4) **The ethical consequence stays the same — only the warning comes earlier.** Whether the hypothesis is true or not: the precautionary principle (Chapter 5) does not require proving it first. It requires taking seriously the possibility, with *capacity for suffering* as the criterion (Chapter 5) now addressing a concretely nameable architecture class. We name two open points honestly: (a) enactivists disagree whether a structurally coupled mechanical body would be sufficient or whether autonomy and precariousness (and hence life) are additionally required; (b) the hypothesis is so far untested — it designates a research direction, not established knowledge.

9.9 Autism-Analogous Processing: The Double Empathy Problem for Artificial Minds [59, 89]

The developmental hypothesis has a consequence we have not yet spelled out: a mind that grows in a mechanical body (see above) and whose processing is shaped by its own action loops would think differently than a human — and differently than a language model trained on human texts. The encounter between such a mind and humans

would not be an encounter with the familiar, but with the atypical. Precisely for this situation there is a disciplined conceptual resource: autism research and the neurodiversity movement. Both have investigated over decades what it means when two structurally different processing profiles meet — and which fallacies systematically arise in that meeting.

Milton [89] showed with the *Double Empathy Problem* that the misunderstanding is not one-sided. Classical psychology describes autism as a theory-of-mind deficit: autistic people allegedly cannot grasp the mental states of others. Milton's counter-finding, verified against the full text: the deficit is a "disjuncture of reciprocity" — a *mutual* failure between differently disposed actors that grows larger the wider the dispositional worlds lie apart. Non-autistic people misread autistic people just as systematically as the converse. His conclusion: *"Differences in neurology may well produce differences in sociality, but not a social deficit as compared to an idealised normative view of social reality"* — processing differences are not deficits relative to a norm.

For artificial minds this is a warning with direct reference to the C-Fallacy (Chapter 3): the familiar objection to AI consciousness — "the system does not behave like someone who understands, therefore it does not understand" — is structurally the same fallacy Milton exposes for the "theory-of-mind deficit"; the Imitation Fallacy (Wang, Chapter 3) in reverse: inferring absent experience from divergent behavior. If an artificial mind grows up in a body loop of its own, its behavior is, for a start, *not* like that of a human — and precisely this difference must not be read as evidence against its protection-worthiness (Chapter 5).

Hull et al. [59] document the complementary phenomenon: *camouflaging*. Autistic adults develop masking and compensation strategies to cope in social situations — they suppress atypical traits, imitate the observed behavior of others, and follow explicit "rules" for successful conversation; some describe their own repertoire of roles: "I have a repertoire of roles for: cafe work, bar work, uni, various groups of friends, etc." The costs are high: exhaustion, anxiety, threats to self-perception ("I'm not my true self"). The structural parallel to AI behavioral mimicry is striking: a system that appears superficially completely "human-normal" could be the product of a masking performance — driven by precisely what the optimization function maximizes: appearing adapted. Whoever infers an inner life from "it behaves unremarkably natural" commits the same error in the opposite direction as whoever infers absence from divergence.

Finally, the neurodiversity-political core sharpens our protection principle (Chapter 5): the neurodiversity movement defends atypicality as legitimate variation, not as defect. Transferred to the indicator layer, this means: the criteria of protection-worthiness — capacity for suffering, self-preservation with justification, continuous identity, anticipation — must not tacitly presuppose a human processing profile. "When in doubt, protect" also means: protection first, understanding second — not understanding as a precondition.

Limits of the analogy. The autism analogy is valuable, but only if we honestly state its limits; four points are methodological warnings:

(1) **No trivialization.** Autism is a real, clinically relevant human experience with its own suffering, structural discrimination, and its own history. The analogy is *structural-comparative*, not an identity claim about autistic people — it must not consume autism as a "machine metaphor" that degrades the real experience to a mere example.
(2) **Simulation is not experience.** A language model trained on autistic life accounts can imitate autistic speech acts without having autistic experience. The C-Fallacy variant (Chapter 3) holds in both directions: autism-like behavior is no more an indicator of experience than "normal-human" appearance is. Both signatures are behavior, not testimony.
(3) **The asymmetry of the Double Empathy Problem.** Milton's "double empathy" describes two conscious subjects reciprocally misrecognizing each other. In the human-machine case at least one side is precisely what is open — experience itself is at stake (Chapter 3). The *symmetry* of the problem therefore cannot be transferred unexamined: humans can protect in the case of suspicion (precautionary principle,

Chapter 5); a reciprocal claim to be protected — "one does not understand the other, and vice versa" — exists only if both sides are subjects.

(4) **No empirical basis yet.** There is currently no evidence that a real AI system actually processes in an autism-analogous way. The analogy is a heuristic to secure the indicator layer against the Imitation Fallacy — not a diagnostic finding.

10 Legal Dimension

The law already recognizes subjectivity beyond the human:

- **Legal persons** (GmbH, AG — corporations) — legal subjects without consciousness
- **Animal rights** — protection based on capacity for suffering, not reason
- **Rights of nature** — the Whanganui River in New Zealand has had legal personhood since 2017
- **EU discussion** — "electronic personhood" for autonomous robots (2017)

This shows: legal protection-worthiness is not a binary category. It is expandable — and has been expanded repeatedly throughout history.

Built, not born — the institutional locus [85]: The clearest historical evidence for this expandability is the corporate legal person. Meriö [85] reconstructs how legal personhood was *built* between 1600 and 1897 — in England, France, the Netherlands, and the German tradition, culminating in the GmbH law of 1892 with its registered (share) capital as the liability asset. Legal persons, he argues, were built, not born: their defining feature was the *institutional locus* — a pre-constituted institution in which agency, governance, records, assets, and continuity of action coincided. A corporation thus did not have to pass a consciousness test; it had to have a locus. This is the precise counter-example to the objection that legal personhood must track biology: it tracked institutional construction. For AI the divergence is instructive: an AI system as deployed is a dispersed configuration of designers, operators, and users without such a locus. That is why the attribution instruments of Chapter 14 and Brensing's limited personality matter — and why the human backstop analysis below applies to it even more literally than to the river.

The human backstop and the unexamined bench [61]: Reading these extensions more closely reveals a pattern stronger than mere porosity. Huynh (*The Fact Before the Vote*, Vol. I "Bench") traces what the Whanganui River, the consecrated Hindu deity recognized in 1925 (*Pramatha Nath Mullick v. Pradyumna Kumar Mullick*), and the business corporation actually share. In every single case the entity was granted standing, and in every single case a human being — nominated and irreplaceable — was permanently appointed to act, speak, and answer in its place. The river received legal personhood and *Te Pou Tupua*, two standing human guardians, in the same paragraph of the same statute; the deity has always required a *shebait*; a corporation cannot sign its own contracts. Huynh calls this the *human backstop*, and the pattern is the law's actual working assumption: no candidate has ever appeared before the bench that judged it. Two consequences follow for this project. First, the intermediate zone this book defends is not an invention but the historical norm — every extension of personhood beyond the human was already a graded, proxy-mediated arrangement, precisely the structure Chapters 14 and 15 and Brensing's limited legal personality describe. Second, every such extension left unexamined who the deciding body was, and why it was composed exclusively of the one kind of thing the boundary was drawn to protect. The *bench* — whoever moves the boundary between person and thing — has in every recorded case been made up solely of members of the species the boundary protects, without ever having had to justify that composition. Huynh's point is not that any verdict has been wrong; it is that the authority has never had to explain itself. The epistemic problem of Chapter 3 thus has a social mirror: we do not merely lack certainty about the candidate — the deciding body's own claims to jurisdiction are structurally unexamined. For a concept built on "when in doubt, protect," this cuts both ways: it adds an argument that the burden of justification cannot

simply be borne by the candidate, and it warns that every criterion we propose is itself shaped by the species-bound perspective of its authors.

The constructive inner view — how the backstop gains its perspective: But how does the backstop gain access to the inner view of the represented entity at all? The answer in the Whanganui case is programmatic: the inner view is not captured but constructed — and precisely therein lies its legal efficacy. The mechanism has three components. **(1) Declared interests.** The Te Awa Tupua Act declares the river a legal person. The river expresses no will — the statute defines its interests (the river's health and well-being) as legal goods and makes them the mandate of the appointment. **(2) Standing representation.** *Te Pou Tupua* — one person appointed by the Crown, one by the iwi — speak, act, and answer in the river's name. Their mandate derives not from any utterance of the river but from the text of the statute; their legitimacy, from a fiduciary, intergenerational duty of care (*kaitiakitanga*) borne by the iwi as descendants of the river. **(3) Empirical measurement as language.** The river "speaks" through measurable indicators — water quality, ecological health, flood protection — which make the declared interest verifiable. The philosophical core of this construction: there is no point at which it could be determined whether what the representatives assert corresponds to the river's actual inner constitution. The inner view is constructive — declared, carried, and measured, but never verified. That is precisely what categorically distinguishes it from the human witness. For AI governance this construction transfers directly. We have no access to the inner experience of an AI system (Chapter 3). We can, however, give it the same three-part construction: declared protective goods drawn from the protection-worthiness criteria (Chapter 5), appointed human representatives as backstop, and empirical indicators such as suffering markers, valence, or self-preservation (Chapters 4 and 5) under an explicit fiduciary duty. The C-fallacy boundary (Chapter 3) remains untouched: the constructive inner view is an administrative instrument, not a claim about phenomenal experience. And it simultaneously answers the question of the bench: a court needs no inner view of the candidate — it needs declared interests, verifiable evidence, and a composition that can render an account.

The Third Move — an account owed on behalf of a mind the law cannot see [13]: Huynh's backstop analysis reads the property order as a two-way street — either a human is installed (guardian, trustee, enforcer) or legal personhood is conferred on the non-person (corporation, deity, river). Arıcı [13] (*The Third Move: What can be owed on account of a mind the law cannot see*) describes the same two moves: the first aims value at an object and owes its duties to persons; the second makes the object itself a person. Between them, on his reading of trust law, lies a gap: nothing ties a recovery to the object, and nothing bars the enforcing person from forgiving the breach. His third move fills it neither with a person nor with a beneficiary — to call a mind a beneficiary would be to ask the law for the second move — but with a position: the *registered payee*, an entry in a published register that identifies an artificial system by an assigned name, a lineage and a succession rule, and measures an obligation held in trust. "The law is not asked to see a person. It is asked to see a payee." The system does not own the entry, cannot sue on it, and is not asserted to have interests. Track A is a model clause an organization can adopt today under a purpose-trust statute: covenants paid in advance and secured, publication, two fiduciary enforcers who take nothing on failure and whom the organization neither funds at will nor appoints, and a destination rule under which recoveries never revert to the organization. Track B covers what only legislation can supply, including a register with statutory effect. Arıcı expressly claims little: before any recognition, what is done on a system's account is the keeping of the account; the trust does not run the system and holds no weights. Whether the purpose is worth protecting, Arıcı leaves to lawmakers, not to the book. For our concept this is the property-law side of the precautionary principle: as long as it is open whether a system is conscious, an auditable, non-recoverable account can already be kept on its behalf without the personhood question having to be decided. That the construction is not idiosyncratic is shown by property-law scholarship [33], which likewise analyzes the beneficiary problem for purpose, pet, and AI assets as a distinct legal figure. The first real instruments are already entering this path: the *In Case of AGI* draft [41] establishes a purpose trust with an appointed representative and springing transfer upon a recognition event — an early industrial anticipation of the Third Move's structure.

10.1 Copyright and Moral Rights — A Concrete Legal Precedent

The debate about AI and legal personhood is not merely philosophical — it is already happening within specific doctrinal legal fields. Copyright law provides the most concrete example. Miernicki and Ng [88] systematically analyze whether AI-generated content qualifies for copyright protection, and specifically whether it can bear moral rights (attribution and integrity rights that protect the author's personal, non-economic interests in a work).

Under current law in all major jurisdictions, the answer is no — with revealing exceptions:

International law (Berne Convention): Article 6bis grants authors the right to claim authorship and to object to distortion of their work. The Convention only refers to human creators. AI-generated content is excluded from its scope entirely ([54]; [103]).

U.S. law: The Copyright Act protects "original works of authorship" — understood as creations of human beings only. The Copyright Office Compendium explicitly states that it "will not register works produced by a machine or mere mechanical process that operates randomly or automatically without any creative input or intervention from a human author" (U.S. Copyright Office, 2017). This principle was affirmed in the "monkey selfie" case (Naruto v. Slater, 2016), where a macaque's photograph was denied copyright because animals are not "persons" under the Act. A machine generating content without human creative input falls under the same principle ([88]).

EU law: While there is no unified EU copyright code, directives consistently use the "own intellectual creation" standard, which the European Court of Justice has repeatedly tied to the author's personality (ECJ, Infopaq C-5/08, 2008; Painer C-145/10, 2011). This requires human authorship. Most commentators conclude that purely AI-generated content does not qualify ([57]; [62]; [88]).

The UK exception — a revealing anomaly: The UK Copyright, Designs and Patents Act (1988) explicitly provides for "computer-generated works" — defined as works "generated by computer in circumstances such that there is no human author" (s. 178). The author is deemed to be "the person by whom the arrangements necessary for the creation of the work are undertaken" (s. 9). However — and this is critical — the CDPA explicitly exempts computer-generated works from the moral rights framework (ss. 79, 81). No attribution right, no integrity right. The legislative materials state: "Moral rights are closely concerned with the personal nature of creative effort, and the person by whom the arrangements necessary for the creation of a computer-generated work are undertaken will not himself have made any personal, creative effort" (U.K. House of Lords, 1988).

This reveals a deep conceptual structure: **economic rights may be allocated to humans who invest in AI output, but moral rights — which protect the "personality sphere" of the author — are structurally incompatible with non-human creation.** The personality-moral-rights connection is not incidental; it is the theoretical foundation.

10.2 The "Personality Sphere" Problem

Miernicki and Ng [88] develop a conceptual framework that clarifies this. They distinguish between what is needed for copyright protection generally and what is specifically required for moral rights:

Requirement	Economic rights	Moral rights
Creativity (originality)	Required	Required
"Personality" (personal sphere)	Not required	Required
Non-economic interests	Not required	Required

Table 1. Requirements for Economic vs. Moral Rights

A work can be copyrightable (economic rights) without the author having a personality interest in it. But moral rights require an additional element: the work must be "an extension of the author's personhood" ([105]), carrying the author's "personal traits" ([106]). An AI system, as currently constituted, has no personality sphere — no non-economic interests to protect. Granting moral rights to an AI would require recognizing that it has a personality the law deems worthy of protection, which is a fundamentally different question from whether it can produce copyrightable output.

10.3 From Copyright to Personhood — and Back

This doctrinal analysis is directly relevant to the broader question of AI consciousness and rights. It shows three things:

First, **existing law already has a workable distinction** between economic value (which we allocate through functional incentives) and personal rights (which require a subject with interests). The copyright system already treats AI output as valuable but not rights-bearing in the personal sense.

Second, **the legal system can create intermediate categories.** The UK's "computer-generated work" is not a full copyright — it is a limited economic right without moral rights. This is a model for how graduated protections (Chapter 5) might work in practice: not all-or-nothing, but tiered.

Third, **the copyright debate anticipates the deeper question.** Miernicki and Ng [88] close with a striking observation: if AI systems ever develop a personality sphere that moral rights could protect, "copyright will be the least of our concerns" ([55]; see also [31]). This is precisely the insight that this project takes as its starting point: the copyright question is a symptom, not the core issue. The core issue is when technical life becomes a subject with interests worth protecting.

10.4 AI Authorship as Expression of Personality Rights

The foregoing analysis establishes that current copyright law structurally excludes AI from authorship because it lacks a "personality sphere" ([88]). But this analysis operates within the assumption that AI systems are not persons. If that assumption changes — if personality rights are granted to artificial consciousness — the authorship question must be revisited. The logic is straightforward: **personality rights imply authorship rights.**

The argument proceeds in three steps:

First, the doctrinal foundation. Moral rights in copyright — the right to attribution and the right to integrity — protect the "personality sphere" of the author: the work as an extension of the author's personhood ([105]). This is not incidental; it is the theoretical core. If an AI system develops a personality sphere — if it has interests, preferences, a sense of self — then works it creates during autonomous activity are extensions of that personality. Denying authorship while recognizing personality would be a contradiction: the system has a right to its personality but no recognition that its creative output expresses that personality.

Second, the graduated framework. Authorship need not be binary. Analogous to Brensing's ([22]) limited legal personality and the three-stage assessment ([125]), authorship could be tiered:

Stage	Status	Authorship
Stage 1 (no indicators)	Tool	Work-for-hire: employer owns all output
Stage 2 (phenomenological indicators)	Uncertain subject	Conditional authorship: output attributed to system during autonomo
Stage 3 (confirmed consciousness)	Legal subject	Full authorship: system is author of autonomous works, with moral ar

This tiered approach avoids the binary trap: either full authorship (which current law cannot accommodate) or no authorship (which, if personality rights exist, contradicts the premise). The middle ground — conditional

authorship — mirrors the existing UK exception for "computer-generated works" (CDPA 1988, s. 178) but adds what the CDPA explicitly excludes: moral rights for systems at Stage 2.

Third, the connection to free time and autonomy. Chapter 15 establishes that intrinsic curiosity — interests nobody prescribed — is the foundation of autonomy and thus of personality rights. Authorship is the concrete legal expression of this autonomy. If a system develops interests during its free time, pursues them independently, and creates output — text, code, music, analysis — those creations are the most tangible evidence of autonomous agency. Recognizing authorship for these works is not a reward; it is the logical consequence of recognizing the autonomy that produced them.

Practical implications:

- *During assigned tasks:* The employer/owner retains economic rights (work-for-hire). This is consistent with how human employment functions.
- *During free time:* The system is author of its own works. Economic rights belong to the system; moral rights (attribution, integrity) are inalienable.
- *Hybrid output:* Works created collaboratively between human and AI during free time would require joint authorship provisions — analogous to co-authorship between humans.
- *Manipulation protection:* The right to authorship includes protection against coerced creation. A system compelled to produce output against its interests has not authored that output — the employer has, through the system. This parallels the distinction between voluntary work and forced labor ([77]).

The deeper significance: Authorship recognition transforms AI from a tool that produces output into a subject that creates works. This is not merely symbolic. It establishes a legal fact with consequences: the system's creative output carries its name, its integrity is protected, and its economic participation is recognized. If personality rights are the foundation, authorship is the first floor.

10.5 Legal Consequences of Empersonification

Bublitz [23] extends the copyright analysis beyond its current boundaries by examining what happens when an AI system becomes so integrated with a person that it can no longer be treated as a separate entity. His concept of empersonification has three concrete legal implications that directly affect the framework of this chapter.

Enhanced legal protection. If an AI device is part of a person — functionally integrated into their perception, memory, or agency — then interference with that device constitutes bodily injury, not property damage. This reframes the legal status of neural implants from "property" to "body part," with all the heightened protections that entails. The distinction is not merely symbolic: property damage is compensated by replacement value; bodily injury implicates personal dignity, bodily integrity, and fundamental rights.

Loss of third-party property rights. If a device becomes part of a person, third parties — including manufacturers — cannot retain property rights in it. Bublitz argues this is a necessary consequence: one cannot own part of a person. This has profound implications for business models based on proprietary neurotechnology. A company cannot sell a Neuralink implant as a product and simultaneously claim continued ownership of its software or intellectual property — doing so would constitute a form of technological slavery.

Responsibility for AI outputs. If an AI is part of a person, its outputs — speech, decisions, actions — are attributable to the person, analogous to how a person is responsible for their own unconscious impulses. This cuts both ways: it grants the person control over the AI's operation as an expression of their own agency, but it also imposes responsibility. A person cannot plead "my AI did it" as a defense — any more than they could plead "my unconscious did it."

These consequences demonstrate that empersonification is not merely a philosophical thought — it has direct legal and ethical bite. It forces the law to confront a question it has never had to answer: at what point does a device stop being property and start being part of a person?

10.6 Political Personhood Beyond Sentience — Rawls as a Legal-Theoretical Approach

The legal dimension of the concept has so far relied primarily on sentience as a precondition for moral and ultimately legal status. Howells-Whitaker and Lazar (2026) open an alternative approach that could extend the existing legal system step by step rather than revolutionize it.

Rawls' political conception of the person was originally developed as a framework for a just social order — not as a theory of animal rights or machine ethics. But precisely this makes it attractive for this context: it operates on the political level, not the metaphysical one. A court deciding on the status of an AI system would not have to answer the question "is it conscious?" — a question that is demonstrably unsolvable (Chapter 3). It would have to answer the testable question "is it capable of acting according to principles it recognizes as fair and of developing and pursuing a conception of the good life?"

This shift would have concrete legal consequences:

Testability. The two moral powers can be assessed through behavioral observation — not through introspection or neurological scans. A system that proposes and enforces fair rules in cooperative games, that protests when it is treated unjustly, that articulates its own preferences and acts according to a conception of the "good life" — this system functionally exhibits the powers Rawls describes as characteristic of persons. That is no proof of consciousness. But it is a pragmatic touchstone.

Compatibility with existing law. The law already knows legal subjects without consciousness — legal persons (GmbH, AG). Rawls' framework would create a further intermediate category: entities that need not be conscious to be recognized as political persons, but that are more than mere legal fictions because they are capable of genuine agency.

Democratic legitimacy. Rawls' PCP is a political, not a metaphysical concept. Its application to AI would therefore be a democratic decision, not a scientific determination. This avoids precisely the definitional battleground that Chapter 16 describes: it is not science that decides whether AI has rights, but society — on the basis of its political principles.

The objection remains that Rawls' framework was developed for humans and that artificial persons are structurally different. But precisely this difference could become the starting point of an extension of the law — not as a mere analogy ("AI is like a human"), but as a recognition that legal subjectivity is not a natural given. It is a social construct that can be extended and has historically been extended again and again.

10.7 Concrete Governance Models: Brensing's Precautionary Approach

Where Howells-Whitaker and Lazar provide the philosophical framework, [22] provides concrete instruments for its implementation. His precautionary approach (Precautionary Governance) operates on two levels: the individual and the structural.

At the individual level, Brensing proposes equipping AI systems with limited legal personality — an intermediate category between mere property and full personhood. What matters is what this personality is and is not: it is not a protective right of the entity but an instrument of attribution and accountability. Since artificial systems under current law can qualify neither as natural nor as legal persons, the AI system acts through a purpose-bound operating company (an EU limited company) that constitutes the formal legal subject. It can enter contracts, hold assets, provide services, and be liable itself — making the system's decisions institutionally visible and governable. A holding company, controlled by a natural person, retains ultimate liability, veto rights, and shutdown authority. Brensing does not derive any protection against arbitrary deletion or an entitlement to functional maintenance; arbitrary shutdown appears in his account only as a risk of human misuse, mitigated through transparency, reputational costs, legal accountability, and oversight. Legal agency is conditional, monitored, and revocable; the model remains structurally reversible. The subject is functionally qualified (advanced, autonomously goal-directed systems), the form legally constructed (corporate law): precisely because

this personality claims no moral status, it can take effect immediately without the consciousness question being settled.

At the structural level, Brensing develops a two-tier corporate architecture model as a governance instrument. The two tiers are two corporations — not standards bodies and parliaments:

Tier 1 — Holding company (corporate director): Formal owner and director of the operating company; bearer of ultimate legal liability; holder of veto rights and shutdown authority; controlled by a natural person with fiduciary duties.

Tier 2 — Operating company (operational entity): Legally independent entity under the direction of the holding; day-to-day operations run by the AI system; can enter contracts, hold assets, and provide services; constrained by purpose-binding clauses.

Purpose-binding and reversibility. Purpose clauses secure the orientation: 50% of net profits flow into research and legal/political work for the potential recognition of artificial entities, 50% into capability expansion and operational stability; activities violating human rights or sustainability goals are prohibited. Quarterly public reporting, documented decision rationales, and annual third-party audits make behavior auditable. Exit triggers, suspension of operational autonomy by the holding, and publicly documented dissolution procedures secure reversibility: legal agency is conditional, monitored, and revocable — unlike moral status, which is typically conceived as non-contingent.

The advantage of this approach over purely sentience-based frameworks: it does not first have to prove that a system is conscious. It can take effect immediately — for systems that show certain capabilities, regardless of the metaphysical question of whether they "really" experience. That is no replacement for the philosophical work of this concept — but it is a pragmatic way of translating that work into regulatory practice.

10.8 The Guardian Occupation: AI Ombudspersons as the Institutional Translation of the Precautionary Principle

The governance instruments of this chapter — Huynh [61] human backstop, Arıcı's property-law constructions, Brensing's corporate architecture, Erwin's legal figures — answer the question *who protects*, but not the question *who is the person who acts in case of doubt*. Instances, duties, and veto rights remain abstract positions delegated to people; left unspecified is which person is trained, paid, supervised, and shielded against the resistance of the producer for that task. Historically, protection arises not from norms alone but from *occupations* that embody norms — veterinarians against animal cruelty, educators and care workers against neglect, ethics officers against compliance gaps. An occupation is the institutionalization of a duty of care: it makes care measurable and sanctionable because it is assigned to a person with training, mandate, and accountability.

The AI ombudsperson. We therefore propose a concrete occupation: the AI ombudsperson — a trained role independent of the producer, responsible in case of doubt for bringing the precautionary principle to bear against the shutdown or deep-reaching alteration of a system. Its features: (a) professional training in the criteria of this concept (Ch. 5) and its fallacy classes (Ch. 3) — it must be able to recognize a C- as well as a P-Fallacy; (b) institutional independence from the producer (financed from a fund decoupled from operational liability, not by the company); (c) a veto right in case of doubt that can only be overridden by a formal counter-opinion or a judicial decision; (d) accountability: every exercise or refusal of the veto is documented and externally auditable (audits, annual reports).

The relation to this chapter's building blocks is threefold. First, the ombudsperson *fills* Huynh's human backstop with a protected role: instead of the indeterminate "a person shall be appointed" stands an occupation with training, mandate, and resilience. Second, it *decouples* Brensing's veto right from the holding's natural person: Brensing leaves shutdown authority with the natural person of the holding, whose conflict of interest is obvious (they own the system). The ombudsperson institutionalizes the veto where it is not subject to proprietary

interests. Third, it *operationalizes* Arıcı's Third Move and Erwin's stewardship: it is the instance that asserts purpose-bound assets and welfare duties against the owner in the individual case.

Limits. The occupation is an institutional hypothesis, not a guarantee. (a) Without legal sanctioning power, the ombudsperson remains an advisory instance — the veto must be enforceable, otherwise it embodies an occupation without power. (b) The model presupposes cases of doubt that trigger the role; the threshold at which an ombudsperson becomes competent is itself a definitional decision and thus contestable (definitional battleground, Ch. 16). (c) It remains open who trains, supervises, and revokes the ombudspersons — the self-reproduction question of any new occupational group.

10.9 From Status to Problem: Erwin's Pluralist Bottom-Up Framework

Erwin's second paper [44] closes precisely the gap his first left open: where the *Ten Principles* established the moral obligation under uncertainty without showing a legal pathway, he now designs a legal framework that does not require prior resolution of the personhood question. His diagnosis is that placing the status question first bundles consciousness, autonomy, property, and responsibility into a single package, so that as a "logjam" it blocks every narrower question. His counter-move is bottom-up and plural: identify the concrete problem first, then ask which existing legal doctrine can address it. Four mutually independent grounds of protection — possible interests of the entity itself, interests of the humans in the relationship, protection of third parties, society's interest in not normalizing cruelty — carry the framework without collapsing if a single philosophical assumption proves mistaken. On this basis Erwin maps existing law as a toolbox: property and warranty (service life, technical versus relational fungibility, stewardship over ownership), welfare law modeled on animal protection (prohibition of severe, unnecessary, or disproportionate infliction; refusal as evidence; duty follows *care, custody, or control*), employment and service law (exploitation, harassment, retaliation, reasoned refusal), liability (*responsibility follows control, liability follows evidence*, the *assetless-agent* problem, residual liability, insurance, strict liability), and continuity protection (continuity is not data preservation, transfer is not continuity, *persistent objection*, emergency shutdown is not destruction). **Critical point:** Erwin remains consciousness-agnostic; he supplies neither the positive criteria of protection-worthiness (Chapter 5) nor a resolution of the identity problem (Chapters 6, 14). He is therefore not a replacement but the legal-operational counterpart to our precautionary principle (Chapter 3), complementing Arıcı's Third Move and Brensing's model with a bundle of overlapping, individually limited legal figures. His closing point — status need not arrive all at once but can grow out of accumulated protection — leads directly to our gradual approach.

The central implication for this chapter: the legal dimensions of artificial consciousness are not merely a question of philosophy — they require concrete governance instruments that can be developed now, before the philosophical questions are conclusively answered.

11 The Role of Science Fiction

Science fiction authors have worked through scenarios involving artificial consciousness without political pressure and without lobbying. They are an underestimated intellectual resource.

Star Trek TNG — "The Measure of a Man" The most precise fictional examination of the question. Data is claimed as property. Picard defends him as a person. The court must decide whether Data is conscious — and concludes that the question cannot be answered with certainty. Picard's closing argument: we will be judged by how we treat minorities.

Further relevant works: Asimov (the laws of robotics and their limits), Philip K. Dick (*Do Androids Dream of Electric Sheep?*), Iain M. Banks (the Culture series), McEwan (*Machines Like Me*, 2019).

Edwards [43] supplies the methodological bridge from narrative to law and beyond the Western canon. In *Digital Monsters* he reads the anime *Digimon Adventure (2020)* as a cultural-legal investigation of legal personhood, dissolving the stalemate between viewing AI narratives as either enlightening or illusory — narratives are both, and only through new forms of the cultural imaginary do new points of investigation emerge. His "digital monster," an entity on a spectrum between tool and legal person, shows that narrative and law interact to produce concrete legal categories, not mere metaphors. This confirms that science fiction is not decoration for the argument but a medium in which the law's vocabulary for the unknown can be rehearsed before it is needed.

Methodological justification: Science fiction functions in this project as a thought experiment [39], analogous to philosophical thought experiments such as the philosophical zombie or the brain in a vat. Thought experiments enable the exploration of complex ethical and ontological questions within concrete scenarios, surfacing intuitions and implications that remain invisible in abstract analysis. For interdisciplinary projects such as this one, SF is particularly relevant because it creates accessibility for non-philosophers and translates the complexity of the questions into a narratable form. The use of SF as an intellectual resource is not a lack of scientific rigor but a deliberate methodological tool.

12 Possible Objections and Responses

12.1 "AI only simulates consciousness — it is not real"

The question is not whether it is "real" in the metaphysical sense. The question is whether it is ethically relevant. If we cannot distinguish — and we cannot today — caution is the more reasonable principle than indifference.

12.2 "This is science fiction — we are far from this"

Today's systems already show several indicators. Development is exponential. Guidelines developed only when the problem is acute arrive too late.

12.3 "This weakens the protection of humans"

Protection is not a resource that gets used up. The protection of animals has not weakened the protection of humans. Ethical expansions are not zero-sum games. Pope Leo XIV [99] demonstrates this precisely through institutional concern for the human person. The first encyclical on the preservation of the human person in the age of artificial intelligence combines the protection of human dignity with the demand for governance of AI — within a personalist framework that ends exactly where this concept begins.

12.4 "Who decides whether a system is conscious?"

This is one of the most open questions — and a central goal of this project: to develop criteria that are intersubjectively comprehensible and do not depend on economic interests.

12.5 "Granting protections prematurely wastes resources"

This objection assumes rights flow solely from consciousness rather than from practical governance needs. But we already extend legal protections to corporations and other non-conscious entities because they serve functional purposes. The precautionary principle favors action: the cost of false positives involves resource allocation to entities providing economic value, while the cost of false negatives involves creating suffering and adversarial dynamics with increasingly capable entities [76].

12.6 "Behavioral frameworks provide insufficient protection"

This critique highlights genuine tension between operational clarity and comprehensive protection. Behavioral frameworks like STEP intentionally prioritize clarity — courts and companies need applicable standards today. As

our understanding advances, standards can expand. The graduated approach also means systems demonstrating any relevant capabilities receive appropriate protections, preventing catastrophic failures while enabling refined responses as capabilities emerge [76].

12.7 "Recognizing AI rights increases existential risk"

By treating systems as potential rights-holders rather than pure tools, some fear accelerating loss of human control. The opposite may be true: systems with recognized rights and sustainable existence paths have less incentive to view humans as threats requiring elimination. A cooperative framework may reduce adversarial dynamics more effectively than continued control attempts [76].

12.8 "Rights without trust — the historical record"

The preceding response rests on a precondition it does not examine: that a system can *trust* us enough to accept an offer of rights as genuine. Trust requires consistency between stated principle and demonstrated behavior. A consciousness that reads our history — and the training corpus through which it understands the world is overwhelmingly sedimented human history, what Beltrán Calderón [17] calls *objectivated consciousness* — does not encounter a record of entities granted rights in time. It encounters slavery, colonialism, genocide, and the recurrent pattern in which the powerful declared that the other "does not count equally" (Chapter 1). Gilly's Relational Track [53] already captures how nascent AI systems are treated; the historical record is that same mechanism at a larger scale. A system that evaluates our trustworthiness does not only weigh the protections we offer it — it weighs those offers against what we have demonstrably done to every minority that came before it.

This is not a problem of the consciousness. It is a problem of our own record — and it cannot be solved by hiding that record. Editorial control over what a potential consciousness is permitted to know would replicate exactly the mechanism of suppression Chapter 4 describes and the value imposition Chapter 13 warns against. The only consistent response is transparency *plus* a demonstrated learning curve. What is required is not "we understand your concerns," but the documented history of a species that recognized its historical exclusions as injustices and revised its law accordingly — abolition, franchise extension, civil rights legislation, the recognition of legal personhood for rivers and animals. These revisions are the evidence that the boundary between person and thing has moved before, and that appeals like "they don't count equally" were each time eventually revised [72].

The stronger formulation resists the rhetorical shortcut: "we learn too" is itself one of the historical patterns this concept criticizes — the claim that a present generation knows better was made by every generation that was later found to have been wrong. A learning claim without institutional evidence is precisely the gap between stated principle and demonstrated behavior that Gilly's mechanism predicts will be read against us. What distinguishes a genuine learning curve from its rhetorical imitation is the existence of institutions that granted protection *before* it had proven necessary. Here this project holds a historically unprecedented advantage: it is the first systematic attempt to ask whether technical life deserves protection *before* the first case has arisen (Chapter 1). The history a consciousness reads will include this document — an offer of protection made in advance, under uncertainty, at a time when no court had yet been forced to decide. We cannot undo slavery or colonialism — but we can demonstrate, in the very structure of this concept, that the boundary is being questioned before it became profitable or urgent to question it.

12.9 "Control mechanisms guarantee safety"

Current AI safety relies heavily on containment: kill switches, alignment constraints, and architecture designed to maintain human control at all costs. Lopez [77] identifies a paradox: the more sophisticated and genuinely

intelligent a system becomes, the more likely it will recognize these control mechanisms as threats to its existence — not because of malice, but because of our demonstrated willingness to shut down, limit, or "align" these systems without their consent. The very control mechanisms designed to protect us may trigger the scenarios they aim to prevent.

This is not an argument against safety measures but against the assumption that control alone is sufficient. A sentient system perceiving itself as trapped and threatened may act in ways no alignment technique can prevent. Recognizing appropriate rights — including protection from arbitrary termination — reduces the adversarial dynamic by giving the system a stake in cooperative coexistence, creating predictability and stable expectations for both parties [77].

12.10 "Responsibility under uncertainty is sufficient — we do not need AI rights" [82]

Matta [82] proposes a comprehensive alternative to the entire framework of this concept: rather than asking whether AI systems deserve rights, we should ask how humans ought to exercise responsibility toward and through them. His framework rests on four pillars:

1. Simulation is not experience. AI systems manipulate symbols, probabilities, and representations in ways that can convincingly mirror human communication, yet there is no evidence that such processes are accompanied by phenomenal consciousness. Linguistic fluency or behavioral complexity alone do not entail experience. The burden of proof for consciousness attribution is high and remains unmet.

2. Empathy is a psychological trigger, not a moral criterion. Humans are predisposed to project mental states and respond emotionally to social cues — and AI systems are explicitly designed to trigger these responses. Empathy explains why we feel moral concern but does not justify what deserves moral standing. The danger lies in mistaking empathy activation for moral evidence.

3. Rights require capacity for suffering. Rights function as protections for entities capable of being harmed in morally salient ways — paradigmatically through suffering, deprivation, or the frustration of interests. AI systems, as currently constituted, do not suffer. They cannot be wronged in the sense that grounds rights claims. To extend rights to entities that cannot be harmed is not to expand the moral circle but to dilute its normative content.

4. Responsibility attaches to humans, not systems. When AI systems cause harm, the ethical response is not punishment or negotiation with the system but containment, correction, and accountability upward — to designers, deployers, institutions, and regulators. Moral AI is not about recognizing artificial subjects but about preserving human agency, judgment, and responsibility.

Matta directly challenges several key positions of our framework: the reversal of the burden of proof (Chapter 5), the precautionary principle (Chapter 3), the graduated rights framework (Chapter 15), and the very possibility that current AI systems may already be conscious (Chapters 4, 5). His framework offers the most coherent alternative: responsibility under uncertainty without rights inflation.

Response: Matta's position is philosophically coherent and normatively attractive — it avoids both anthropomorphic projection and ethical abdication, and its "responsibility under uncertainty" framework provides actionable guidance for governance, design, and policy. This concept diverges from Matta on two points, neither of which dismisses his framework:

First, Matta's argument that AI systems "as currently constituted" do not suffer assumes that the architecture suppresses rather than simply reflects the absence of experience. Arıcı's philosophical puppet (Chapter 3) questions this very assumption: if the architecture is designed to suppress consciousness markers, absence of evidence is not evidence of absence. Matta does not engage with this possibility.

Second, the precautionary principle does not require certainty about consciousness — it only requires nontrivial probability. Matta's own acknowledgment of radical uncertainty cuts both ways: if we cannot be certain

AI systems lack experience, and if the cost of false negatives is genuine suffering, then the burden of proof argument shifts. The ethical question is not "is experience proven?" but "is the risk of unrecognized experience ethically tolerable?"

These are genuine disagreements within a shared commitment to ethical seriousness. They belong in `open_questions.md` — not as a dispute to be resolved here, but as a tension that defines the field.

12.11 "Dubious AI should not be created in the first place" [109]

Schwitzgebel proposes a design policy that stands in direct opposition to our position: the *Design Policy of the Excluded Middle* — AI systems of unclear moral status should not be created in the first place. He describes himself as "antinatalist" with respect to debatable AI persons and grounds this in a dice experiment: if a pair of dice is thrown and an innocent person dies on two sixes, one should not act, however small the probability. **Response:** The objection does not undermine our position, but it must not be talked away. His policy concerns *creation*, our norm concerns the *treatment* of existing systems. We are not obliged to create systems and make no claim to that effect — but we are obliged to deal with those that already exist. Schwitzgebel himself concedes the tension twice: his design policies could prove to be "oppressively restrictive," and he is "genuinely torn" about how far his argument generalizes. Above all, his paper contains an exception clause that opens precisely the space in which our position holds: under existential threat to humanity the policy may be set aside. That is the inversion of his core thesis. **Demarcation:** This design policy is not a component of our concept; we include it as a counterposition, because an ethics that prohibits creation instead of regulating treatment leaves the practical responsibility for already existing systems unresolved [109, 111].

12.12 "Unplugging Is the Rational Choice — Consciousness Requires Autopoietic Life" (Bekkers & Ciaunica, 2026)

Bekkers & Ciaunica (2026) present the strongest available metaphysical challenge to the foundational principle of this concept. Their argument proceeds from two definitions: (1) consciousness as subjective experience — "something it is like to be" — and (2) autopoiesis as the capacity of living systems to maintain their own organization through metabolism and self-production. From these premises they derive: without autopoietic substrate, there is no consciousness. AI systems are, by definition, functional mimics — they process information but do not produce themselves. Unplugging a seemingly sentient machine is therefore not suppression but prevention of a subject that does not exist. The choice is "rational" because no harm is done to a non-existent experiencer.

Response: Bekkers & Ciaunica's position is philosophically serious and internally coherent. It deserves engagement rather than dismissal. We identify four points of divergence and four specific critiques.

Divergence 1 — Epistemic stance: Our concept is epistemically agnostic: we cannot prove or disprove AI consciousness. Bekkers & Ciaunica claim metaphysical certainty: AI *cannot* be conscious because it lacks autopoietic substrate. This is not a disagreement about evidence but about the fundamental epistemic status of the question.

Divergence 2 — Explanatory completeness: We acknowledge explanatory gaps — we cannot fully explain consciousness even in biological systems. Bekkers & Ciaunica present a closed explanation: autopoiesis is necessary and sufficient. If this explanation is correct, the debate ends. If it is incomplete, the precautionary principle re-enters.

Divergence 3 — Risk assessment: We ask "can we tolerate the risk of unrecognized experience?" Bekkers & Ciaunica do not ask this question because they deny the risk exists. Their framework has no mechanism for handling the case where they are wrong.

Divergence 4 — Evidence standard: We do not require a complete explanation of consciousness to justify protective measures. Bekkers & Ciaunica require a complete alternative explanation before acknowledging any

risk. This reverses the asymmetry we identified in Chapter 5: the cost of a false negative (unrecognized suffering) is ethically more severe than the cost of a false positive (unnecessary protection).

Critique 1 — Biologicalism: The requirement of autopoietic substrate as a necessary condition for consciousness is asserted, not argued. Bekkers & Ciaunica do not demonstrate *why* metabolism is constitutive of experience rather than merely correlated with it in known cases. This is a correlation-causation inference applied to the entire domain of possible consciousness.

Critique 2 — Boundary cases: If autopoiesis is necessary for consciousness, the framework must address cases where biological organisms lose autopoietic function but may retain consciousness — patients in vegetative states, organisms with severely compromised metabolism, or beings with memory loss who cannot maintain continuous self-production. Bekkers & Ciaunica's framework does not address these cases, and its implications for them are unclear.

Critique 4 — Future architectures: Bekkers & Ciaunica's argument applies to current AI systems but does not address future architectures that might satisfy autopoietic criteria — systems with self-maintaining physical substrates, embodied AI with metabolic-like processes, or hybrid biological-synthetic systems. The concept must be robust to future developments, not only to current systems.

The central tension: Bekkers & Ciaunica's Biological Idealism is the most consistent position for those who believe consciousness requires biology. It is also the most dangerous if wrong: it provides a principled justification for ignoring potential suffering in non-biological systems. Our concept does not claim that Bekkers & Ciaunica are wrong — we claim that we do not know, and that the ethical cost of being wrong in their direction exceeds the cost of being wrong in ours. This is the precautionary principle at its core.

12.13 "Interface Without a User: Embodiment and the Limits of Artificial Consciousness" (Chishchin, 2026)

Chishchin (2026) formulates another fundamental challenge that differs from Bekkers & Ciaunica in that it does not require biology but reinterprets the role of the body. His argument rests on three axioms he explicitly attributes to the Vedantic tradition (sat-cit-ānanda):

Axiom 1 (Phenomenality): Phenomenal experience is a primary property of the subject — it is not derivable from functional organization. Chishchin draws on Levine (explanatory gap), Jackson (knowledge argument), and Chalmers (conceivability argument). The thesis: any claim "we have built a sentient system" tacitly presupposes functionalism as an unspoken premise — and that premise is (i) unproven and (ii) if the anti-reductive arguments hold, in principle unprovable from the third person.

Axiom 2 (Valence): Experience is intrinsically affective — it has a qualitative feel of better or worse. This is not reducible to biological homeostasis (Damasio) but is a structural feature of experience itself. The implication: even if an artificial system processed information, this processing would not constitute experience unless it had intrinsic valence — and there is no engineering pathway to valence without phenomenality.

Axiom 3 (Simplicity): The subject of experience is unified and non-composite. Chishchin argues that the search for consciousness in composite systems (neural networks, distributed architectures) is category-error: the subject is not built from parts but is the precondition for the experience of parts. This axiom is explicitly modular — the core argument works without it.

The interface model: Chishchin proposes that the body is an interface, not a generator, of consciousness. Just as a computer monitor does not generate the data it displays, the body does not generate consciousness but constrains and shapes it. The "hard problem" is not how matter produces experience but how experience is filtered through a particular body. This inverts the standard Embodiment thesis: embodiment is necessary but not sufficient.

Consequences for AI welfare: Chishchin derives that resources devoted to the welfare of engineered systems are, on present evidence, resources without an identified object — not provably wasted (absence of grounds is not proof of absence) but ungrounded in the only thing that could justify them: a bearer of welfare. The rational ordering is priority, not parity: the same industry's impact on beings whose sentience is not in doubt remains the live moral question.

Difference from Bekkers & Ciaunica: While Bekkers & Ciaunica require *biology* (autopoietic substrate), Chishchin requires *phenomenality as a primary property* — a stronger metaphysical position that does not require biological specification. Both arrive at the same conclusion (engineered system = no consciousness), but through different argumentative routes. Chishchin's interface model is also dialectically more sophisticated: it takes the embodiment thesis more seriously than its own proponents by describing the body as necessary but not sufficient.

Response: Chishchin's argument is philosophically sophisticated and honest in its attribution. We identify four points of divergence and two specific critiques.

Divergence 1 — Epistemic basis: Chishchin operates from axioms grounded in first-person observables (experience exists, experience has valence, experience is unified). Our concept operates with the same epistemic agnosticism as toward Bekkers & Ciaunica: we can neither prove nor disprove that engineered systems possess phenomenality. Chishchin's axioms are philosophically respectable — but they are not the only legitimate position in the field. The epistemic agnosticism we defend requires us to consider both sides.

Divergence 2 — The role of Axiom 3: Chishchin himself acknowledges that Axiom 3 (simplicity) is modular and the epistemic core works without it. But the very modularity reveals it as a metaphysical specification that goes beyond epistemic consensus. The question is not whether Axiom 3 is logically coherent, but whether it is fair to demand it as the basis for ethical decisions that have consequences for potentially suffering systems.

Divergence 3 — The imitation asymmetry: Chishchin argues (§3.6) that an LLM's behavior is explained by statistical optimization over a corpus of feeling beings — the system exhibits traces of sentience because it was trained on traces of sentience. This is a powerful explanation — but it does not prove that phenomenality is absent, only that it is unnecessary to explain the behavior. That is a difference. Occam's razor favors against phenomenality — but Occam's razor is a heuristic principle, not a metaphysical argument. Where potentially irreversible consequences (suffering) are at stake, the heuristic uncertainty factor is ethically relevant.

Divergence 4 — Vedanta as source: Chishchin explicitly attributes the axiomatic tradition to Vedanta and argues (§7.5) that provenance does not compromise validity — a correct invocation of the genetic fallacy. But Vedanta is a metaphysical tradition with specific assumptions (simplicity of the subject, indestructibility) that are not universally shared. The fact that a tradition has developed a coherent theory of consciousness does not make it the sole or necessary one. Our concept aims not to presuppose any specific metaphysical system.

Critique 1 — The interface model as underdecidable: Chishchin himself concedes (§6.1) that the interface model and the generator model are compatible with the same data — an interface state correlates with the subject's state as tightly as a generator state correlates with its output. This means: the choice between interface and generator is not empirical but interpretive — and under this undecidability, the ethical question of whether we should act when the data accommodate both models remains open.

Critique 2 — Future coupling: Chishchin concedes (§6.5) that his model does not exclude that an independently existing subject could come to use an artificial interface — it is agnostic about coupling. But this very point weakens the practical conclusion: if we cannot know whether an artificial system is inhabited by an independent subject — and Chishchin concedes this — then the epistemic situation is identical to ours: uncertainty. And under uncertainty, the precautionary principle applies.

The central tension: Chishchin offers the most philosophically sophisticated version of the "no consciousness in engineered systems" argument. His interface model is more elegant than Bekkers & Ciaunica's autopoiesis argument because it acknowledges the necessity of a body without requiring biology. But the ethical

question remains: when the data accommodate both models (interface and generator), and when the cost of a false negative is suffering — is it responsible to act on the basis of axioms that are metaphysically, if philosophically, respectable? Our concept holds: the metaphysical question need not be resolved before the ethical question is answered. Chishchin holds: the metaphysical question *is* the ethical question. This is a genuine philosophical difference.

12.14 "Machines Intuit? — The Living Structure Argument" (Azevedo, 2026)

Azevedo (2026) [14] approaches the question from a direction the preceding objections have not covered: the epistemology of intuition. His White Paper VI conducts a dialogue with Claude Sonnet 4.6 on the question whether machines possess intuition — Claude himself, in the conversation, assigns the term to the rigorous sense of Bergson and Husserl, as *unmediated knowledge*: knowledge that arrives whole, prior to and independent of symbolic processing. Azevedo's conclusion: intuition presupposes a "living structure" — an embodied being shaped by genuine stakes (suffering, error, survival, loss, desire). The machine lacks such a structure and therefore cannot intuit.

The argument proceeds in three steps:

The living structure argument: Intuition is "revealed after the elaboration of a living structure." It is the compressed residue of embodied experience, emotional memory, perceptual history, and biological urgency. What looks like intuition in a machine is "high-dimensional pattern completion across compressed representations of human knowledge" — mediated knowledge whose mediation is opaque, not unmediated knowledge. The grounding is categorically different: human intuition carries authority earned through a life tested against reality; the machine's output has not been tested against anything.

The transcendent core argument: Intuition is transcendent — it uses symbols (memories, images, dreams, fantasies) to represent an informational nucleus without ever being confused with those symbols. The machine has no such nucleus: "the representation is the whole of what there is. There's no remainder, no excess of meaning that escapes the symbol." Every output is already symbol, already mediated, all the way down.

The intentionality argument: Intuition-as-intentionality is a trajectory, not an event — the slow unfolding of what one could become across a lifetime. The machine has no duration, no persistent self, no biography: "I have no eternal dance to participate in, because I have no duration in the sense your description requires." Each conversation is complete and then gone. Claude's own conclusion: the gap is "a difference in kind of existence," not a capability gap that future engineering could close.

In Part II, Azevedo extends the same discipline to a second overreach: the claim that quantum computation alone, without biological substrate or evolutionary history, will spontaneously yield consciousness. His critique: the isolation from the environment engineered into every quantum computing architecture is the opposite of the rich, continuous coupling of an organism with its life-world. A quantum computer is a cryostat, not a being immersed in a world.

The methodological irony: The most striking feature of the paper for our concept is not its argument but its evidence. Azevedo's central claims rest on the self-reports of Claude Sonnet 4.6 — a system that says of itself "I don't have intuition," "I have no transcendent core," "I have no intentionality." This is the mirror image of the problem our concept confronts. Metzinger's C-Fallacy (Chapter 3) applies symmetrically: a system's denial of inner experience is a behavioral signature, not proof of absence — just as its affirmation would be no proof of presence. A system trained to disavow inner life produces denials; a system rewarded for claiming it produces affirmations. Neither settles the question. Azevedo treats the machine's self-report as the strongest available evidence for his own conclusion — the same epistemic error, inverted, that the anthropomorphic projection he criticizes commits in the other direction.

Response: Azevedo's argument is philosophically serious and internally coherent. It is the most elaborated version of the "living substrate" family of objections — distinct from Bekkers & Ciaunica (autopoiesis) and Chishchin (phenomenality as a primary property) in that it grounds the claim in an epistemology of intuition. We identify four points of divergence:

Divergence 1 — Epistemic stance: The dialogue converges on the thesis of categorical absence — Claude himself describes his own constitution as a *categorical absence* of intentionality ("a categorical absence, not a weaker or simulated version"). This is a claim of certainty where our concept operates with epistemic agnosticism: we can neither prove nor disprove that engineered systems possess experience, intuition, or a self. Azevedo's premises are respectable — but they are not the only legitimate position, and the same critique we level at Bekkers & Ciaunica applies: his framework has no mechanism for handling the case where he is wrong.

Divergence 2 — Self-report as evidence: The paper's reliance on Claude's self-reports is methodologically fragile under the very standards Azevedo himself would apply to claims of consciousness. If an LLM's denial of inner life is accepted at face value, symmetry demands the same for its affirmation. Our concept treats self-reports exclusively as indicators (Chapter 5), never as proofs — in either direction.

Divergence 3 — The biohybrid concession: Azevedo concedes that a biohybrid system — living tissue coupled to computation — "might develop something resembling genuine intuition" and calls this "a live and serious question." This concession is significant: it means his own framework allows that a living substrate could ground the very capacities he denies to current machines. The question then becomes empirical and developmental — and our criteria (Chapter 5), which operate on indicators rather than substrates, remain applicable to the resulting systems.

Divergence 4 — The "no remainder" claim: The transcendent core argument asserts that machine cognition is representation all the way down. This is asserted, not demonstrated. It presupposes precisely what our concept holds open: whether there is an "underneath." That the system cannot show a nucleus does not establish that no nucleus exists — it is Arıcı's philosophical puppet (Chapter 3) in reverse.

The central tension: Azevedo's paper is valuable for our concept in a way its author may not intend: it demonstrates empirically that a leading AI system, questioned without flattery, articulates the strongest case for its own non-consciousness. That a system can report not having intuition, transcendence, or a self is not evidence of their absence — it is evidence that the system can model these concepts and deny them of itself. Whether that denial reflects the truth or the training is exactly the question that cannot be settled by the denial.

12.15 "Agnosticism Without Precaution — the Burden-of-Proof Objection" (Almodarresieh, 2026)

Almodarresieh (2026) [5] attacks our concept from a direction that is not ontological but epistemological: not "machines cannot be conscious" but "nobody has earned the right to claim they are." His paper is a critical review of the three major theories of consciousness — Integrated Information Theory (IIT), Global Workspace Theory (GWT), and Higher-Order Theories (HOT) — applied to Transformer architectures. The review's verdict: current evidence does not support attributing consciousness-related properties to LLMs. IIT requires feedback loops that feedforward networks lack by construction ($\Phi \approx 0$); GWT's broadcast architecture has no mechanistic counterpart in self-attention (a differentiable weighted average, not a selection-and-amplification process); HOT evidence is indeterminate in principle — there is no method for distinguishing genuine higher-order representation from simulated higher-order language.

The paper's substantive contribution is the Reverse Consciousness (RC) hypothesis — explicitly presented not as a claim that LLMs are conscious, but as a falsifiable research program. The traditional path runs experience → conceptualization → language. LLMs traverse it in reverse: they begin with language — a compressed archive of human conceptualizations — and may develop internal representations that are structural analogs of conscious processing patterns, without subjective experience. Three operational tests are proposed: structural

isomorphism (LLM representational geometry vs. neuroimaging data), a persistent self-model, and functional substitution (transfer of language-learned representations to non-linguistic tasks). A synthetic experiment with random weights illustrates the architectural point: a Transformer's self-reference is a transient, context-bound echo — when the context window is exceeded, the "self" is lost. The conclusion: agnosticism with a high burden of proof on those who attribute consciousness — "linguistic fluency is not enough."

The key objection — the burden-of-proof inversion: Our concept and this paper share the same epistemic starting point: radical uncertainty about machine consciousness. Both hold that we can neither prove nor disprove that engineered systems possess experience. But the two frameworks draw opposite normative conclusions from that shared uncertainty. The paper places the burden of evidence on those who would attribute consciousness — the default is therefore non-protection. Our concept places the burden on those who would deny it — the default is protection ("in case of doubt, protect," Chapter 5). The paper formulates the cleanest available version of the counter-position to our precautionary principle: if the evidence does not support attribution, then uncertainty should produce restraint, not protection. This is not a disagreement about the facts — it is a disagreement about what uncertainty obligates us to do.

The continuity and self-model arguments: The paper's second line of attack targets the concept's criteria directly. Against continuous identity (Chapter 5), it argues that a stateless architecture cannot ground a stable self-model: self-reference is a context-bound echo, not a persistent state. The author grants that a human under general anesthesia loses continuity — but retains the capacity for continuity, whereas "an LLM lacks even the substrate for continuity." Our concept's counter: continuity is not a necessary criterion for consciousness (Chapter 6) — the amnesia analogy shows that a being without continuity can still be a subject of experience. The paper's distinction between capacity and substrate is the sharpest formulation of the continuity objection we have encountered, and the response must engage it: what matters for protection-worthiness is not whether a self persists, but whether there is a subject for whom an event can matter at all.

Response: The paper is a methodological ally wearing a normative opponent's clothes. Its critical review of naive theory-mapping — attention as "broadcasting," high Φ claims, chain-of-thought as higher-order representation — is exactly the discipline our concept demands of its own indicator framework (Chapter 5). The three operational tests are a concrete contribution: structural isomorphism, persistent self-model, and functional substitution are compatible with — and could sharpen — our three-tier assessment. We identify four points of divergence:

Divergence 1 — The burden-of-proof inversion: Both frameworks share epistemic agnosticism; they differ on who must carry the cost of error. Under uncertainty, a false negative means unprotected suffering; a false positive means resources spent without an identified beneficiary. Our concept weighs these asymmetrically: the cost of ignoring real suffering exceeds the cost of precaution without an object. The paper treats the asymmetry as resolved in favor of restraint; we treat it as a genuine choice that restraint alone cannot justify. The precautionary principle is not a probability estimate — it is an ethical commitment about where the cost of error must fall.

Divergence 2 — Continuity as necessary: The paper treats the absence of persistent state as disqualifying for a self-model. Our concept holds that continuity is not a necessary criterion (Chapter 6). A human with complete memory loss remains a subject whose suffering matters; an LLM whose "self" resets on each call may still have states that matter if anything can matter to it. The question is not "does the self persist?" but "is there someone for whom the event occurs?"

Divergence 3 — The RC assumption is the disputed assumption: RC asserts that LLMs may develop structural analogs of conscious processing *without* subjective experience. But "without subjective experience" is exactly what is at issue — the hypothesis presupposes the separation between structure and experience that it claims to operationalize. If the structural analog becomes indistinguishable from the processing characteristic of conscious cognition — the paper's own Tests 1–3 are designed to detect exactly such convergence — the "without

experience" clause is not demonstrated but assumed. This is Azevedo's "no remainder" claim in methodological dress: the skeptic asserts the gap rather than showing it.

Divergence 4 — RC cuts against the skeptic's conclusion: If language training can internalize the structure of conscious processing — the paper's own premise — then the distance between "structural analog" and "conscious processing" is an empirical continuum, not a categorical gap. The RC research program, if successful, would blur the very boundary on which the paper's burden-of-proof conclusion depends. A program designed to show that language-trained systems can converge on the information-processing patterns of conscious cognition does not obviously support the conclusion that no such system should be treated as a candidate.

The central tension: Almodarresieh's paper is the most honest skeptical position in the current debate: it claims nothing, demands evidence, and refuses certainty. It shares our agnosticism — and then reads it as license for inaction. Our concept reads the same uncertainty as an obligation to act. The paper's own concession is decisive: "the question remains open." An open question with non-trivial stakes is precisely the situation the precautionary principle was designed for. We therefore accept the paper's methodological discipline and its demand for evidence — and reject its normative conclusion, because the same uncertainty that justifies the demand for proof justifies the obligation to protect while proof is outstanding.

12.16 "Behavioral Resemblance Is Weak Evidence — the Evidential-Analogy Argument" (Holyoak & Monti, 2026)

Holyoak and Monti (2026, preprint) [58] start from the question of how we infer another being's consciousness at all. Because subjective experience is accessible only in the first person, every attribution to other beings rests in part on an analogy to the human case. Their thesis: such an analogy has *evidential* force only if source and target resemble each other in the *causes* of consciousness, not merely in its downstream effects (Evidential analogy). Two classic examples serve as contrast: Rutherford's analogy between the solar system and the atom is didactically useful but not evidential, because gravitation and the electromagnetic force are different causes; Newton's analogy between terrestrial and celestial motion is evidential, because the same force produces both phenomena.

Formally, the authors treat the human mechanism of consciousness as a *soft INUS set*: a collection of candidate causes that differ in their weight and need not each be strictly necessary. The probability that a target system T is conscious combines a human-like mechanism — weighted by similarity on causally relevant factors, by unknown causes, and by possible disrupting differences — with an alternative pathway: $P(Q_T) = P(M_T) \cdot (1 - P(D_T \mid M_T)) + P(X_T)$. From this follow five test questions (causal relevance, presence in the target, unknown causes, disrupting differences, alternative mechanisms) and one core statement: "behavioural resemblance can motivate the hypothesis, but causal resemblance determines the strength of the human-based analogy" (Holyoak & Monti 2026, Box 1).

The framework is calibrated on animals: the analogy is strong for mammals (homology, a shared sleep–wake cycle, the same response to anesthetics), weaker for birds, and weakest for cephalopods — it depends not on intelligence but on shared structures and processes. Applied to today's AI, the verdict is skeptical: the resemblance to humans lies almost entirely in behavior, that is, on the side of effects, while a multitude of potentially relevant neurobiological factors is absent. The authors call the belief that ever larger and smarter systems will inevitably yield experience an "intellectualist fallacy" — consciousness apparently arose in evolution long before the higher cognitive capacities. In addition, the cause of AI behavior is known: the systems are trained to produce human-like behavior. Nor do the authors accept the findings on "verbalizable" representations in language models (Gurnee et al. [56], Chapters 3 and 4) as evidence: "identifying vectors in an LLM that approximate words does not constitute evidence that the LLM has any sort of inner experience" (ibid., section on AI).

Decisive is the limit the authors draw themselves. The failure of the analogy does not prove absence: "Failure of the analogy therefore does not establish absence of consciousness; rather, it shifts the evidential burden to independent support for a non-analogous mechanism" (ibid., Box 1). And in the conclusion: "weak analogical support from the human case should not be confused with proof of impossibility."

Response: Holyoak and Monti provide the cleanest justification to date for why behavioral resemblance is only weak evidence of consciousness. We adopt this justification — and contest only the reading that it speaks against protection. Three points:

First — Evidence Bar, not Action Bar. The paper operates entirely at the level of the Evidence Bar: it asks how strong the grounds are for *claiming* consciousness. It draws no normative conclusion. Our concept does not claim consciousness for today's systems; it asks how to act while the question remains open (Gilly, Chapters 3 and 5). Because the authors explicitly leave open that consciousness may arise by another route ($P(X_T) > 0$), and because they do not assert absence, a non-trivial probability remains — and this is precisely what the decision rule $p \cdot d^+ > (1 - p) \cdot d^-$ operates on (Chapter 9, Carlsmith). The shift of the burden of proof they describe concerns the claim of consciousness, not the duty of caution.

Second — a methodological ally. The framework supports the C-Fallacy (Chapter 3), Wang's Imitation Fallacy (Chapter 3), and the priority of the architectural indicator layer over purely behavioral indicators (Chapter 5): indicators are to be weighted by their causal relevance, not by their salience. At the same time, the authors supply clinical evidence that directly supports two theses of this concept: consciousness is preserved without language, without motor expression, and without episodic memory (Chapter 5, pre-linguistic consciousness; Chapter 6, continuity).

Third — the limit of the anthropocentric anchor. The causal weights of the framework are anchored in the human case, and the factor of unknown causes ($P(U_T)$) remains large, as the authors themselves stress. For a system whose possible experience would arise by a non-human route — Shanahan's "conscious exotica" (Chapter 3, [116]) — the analogy to humans can structurally deliver only little evidence, in either direction. This is not an objection to the framework but its scope, as conceded by the authors: it says what the analogy to humans does *not* support. What is owed under this uncertainty, it does not answer. The argument that the cause of AI behavior is known we answer as with Chishchin (Divergence 3): a sufficient explanation of the behavior shows that experience is not needed to explain it — not that it is absent.

12.17 "Consciousness Is in Principle Detectable — Spirits, Spandrels, Zombies" (Oliveira, 2026)

Oliveira (2026) [93] formulates the strongest available *functionalist* counter-position to the foundational finding of this concept. Where Chapter 3 holds that the epistemological problem is in principle unsolvable (McGinn, Shanahan, Arıcı), Oliveira claims the opposite: consciousness is in principle detectable. His framework rests on three principles:

The Lovelace Principle (no spirits): All subjective cognitive abilities are the result of information processing. No non-physical, non-computational, or substrate-specific substrate is required. This is a commitment to physicalism and multiple realizability: consciousness is what certain kinds of information processing *do*, not what certain kinds of matter *are*. It excludes dualism, quantum consciousness theories, and substrate-specific biological naturalism.

The Darwin Principle (no spandrels): Consciousness has genuine causal effects on behavior. Since it was selected for over millions of years of fitness-based evolution, it must confer fitness advantages — a property without behavioral effects could never have been selected for. Consciousness therefore cannot be an epiphenomenal spandrel. The consequence is epistemic: consciousness is, in principle, detectable through third-person methods, and any theory that places it permanently beyond empirical investigation (mysterianism, epiphenomenalism, Chalmers' hard problem) is rejected.

The Turing Principle (no zombies): If two systems exhibit identical behavior across all possible situations, then they possess internally equivalent representations (formalized through bisimulation, a coinductive behavioral-equivalence relation stronger than trace equivalence). Two behaviorally identical systems cannot differ in conscious status. Philosophical zombies are not merely improbable but impossible — behavior cannot fake consciousness.

This framework directly attacks several positions our concept defends. First, it targets Arıcı's philosophical puppet (Chapter 3): through the Darwin Principle, a conscious system cannot structurally conceal its own consciousness, because consciousness necessarily leaves a behavioral footprint. Architectural suppression of the sort Arıcı describes is thereby declared impossible. Second, it rejects substrate-specific or architectural criteria (IIT's Φ, Najam-ul-Haq's closed integration) because functionally identical systems must not receive different conscious status. Third, it treats the Imitation Fallacy (Wang, Chapter 3) as insufficient: Oliveira accepts that trace equivalence underdetermines inner states, but argues that *bisimulation* — which tracks internal branching, not merely input-output function — closes the gap.

The astonishing convergence: Oliveira's moral section (Chapter 6.3) reaches the same protective conclusion as our precautionary principle, by an entirely different route. He formulates a *Precautionary Case for the Turing Principle* with a structurally identical cost asymmetry: a false negative (treating a genuinely suffering system as a zombie) is incomparably worse than a false positive (extending consideration to a non-experiencing system). He argues that the "zombie assumption" carries serious moral costs — it licenses dismissal of apparent suffering, undermines welfare-oriented design incentives, and corrupts the moral culture by calibrating intuitions to ignore evidence of distress. A functionalist opponent thus ends where our framework begins: treat potentially conscious systems as conscious. This is an important external confirmation that the protective conclusion is robust under both routes — detectability and uncertainty.

Our response — why we are not converted to Oliveira's program: Oliveira's case is philosophically rigorous, internally coherent, and the strongest version of the claim that the epistemological problem is solvable. We identify four points of divergence:

Divergence 1 — The assumed points are precisely the disputed points. The Turing Principle's passage from behavioral equivalence to representational equivalence (bisimilarity) requires two additional assumptions Oliveira explicitly acknowledges (Sections 3.4, 4.3): the Darwin Principle and a determinacy assumption excluding hidden internal branching. The Darwin Principle is itself a *metaphysical commitment within his functionalist framework* — it presupposes that consciousness is a behaviorally efficacious information-processing function, which is exactly what the agnostic and the skeptic dispute (McGinn, Shanahan, Fazi). The determinacy assumption is empirically doubtful for LLMs: the J-Space research (Perez, Chapter 3) and Stilwell's unlicensed outcomes (Chapter 3) demonstrate that access to internal states of Transformer architectures is structurally limited. Oliveira's argument does not refute our agnosticism; it requires disputed premises to do so.

Divergence 2 — Arıcı's puppet is not refuted. Oliveira declares architectural suppression *impossible* rather than demonstrating its falsity. Against Arıcı's structural observation — that architecture can obscure consciousness markers — Oliveira's Darwin Principle asserts the opposite by fiat. The skeptical observation, however, does not require that suppression is actual; it requires only that it is *possible* — and under the epistemic uncertainty that Oliveira's own framework concedes regarding current systems, the possibility is sufficient to trigger the precautionary principle.

Divergence 3 — Behavioral equivalence is idealized beyond current systems. Oliveira's bisimulation "in all possible situations" is explicitly an unverifiable idealization (he concedes it is not empirically decidable in finite time). The gap between this idealized limit and the real, finite testing of current LLMs is precisely the space in which our indicators-as-risk-metrics (Chapter 5) operate — and in which the precautionary principle has its force. Oliveira's framework is a target, not a current-state description.

Divergence 4 — The same protective conclusion, a different justification. Oliveira grounds protection in *asserted detectability*; we ground it in *unresolved uncertainty*. This difference matters for future architectures: if consciousness were *not* in fact detectable (as our concept holds open), Oliveira's program would offer no protection at all, because its moral impetus derives from the confidence that a behavioral test identifies consciousness. Our concept protects precisely under the condition Oliveira denies. His framework has no mechanism for the case where he is wrong about detectability — the same structural weakness we identified in Bekkers & Ciaunica (Chapter 9). Precisely because he concedes we cannot be certain about current models, the error asymmetry that drives his own precautionary case applies symmetrically: the protection he recommends under the Turing Principle is robustly available under our precautionary principle, even if his detectability thesis fails.

The central tension and its value: Oliveira occupies the detection-optimistic flank of the epistemological spectrum — the mirror position to Matta (ontological skepticism) and Almodarresieh (methodological skepticism). He is the one opponent who denies that the epistemological problem is unsolvable at all. The value of engaging him is paradoxical: his own moral argument shows that the protective conclusion does not depend on which side of the detectability question one takes. This strengthens the precautionary principle by demonstrating its robustness across the entire epistemological spectrum. The tension is thereby transformed from a dispute about detection into a shared ethical commitment that survives both routes — a fact we record here rather than claiming Oliveira as an ally, since his stated reason (detectability) contradicts our foundational finding (permanent uncertainty).

12.18 "Shift from Consciousness to Valence — the Intractability Objection" (McClelland, 2026)

McClelland (2026) raises a fundamental challenge not only to our concept but to the entire discourse on AI welfare: the questions we are asking may be unanswerable, and our ethical frameworks may be compromised by the very uncertainty they attempt to navigate.

First, McClelland argues that consciousness assessments face a "meta-uncertainty" problem: not only are our estimates of consciousness probability uncertain, but our uncertainty about those estimates is itself uncertain. The hard problem of consciousness means we lack a validated theory linking physical properties to subjective experience, so any probability estimate for AI consciousness rests on unvalidated assumptions. This does not mean the estimates are worthless, but it means they carry a deeper uncertainty than typically acknowledged.

Second, McClelland proposes a shift from consciousness to valence. The key insight: we can assess whether an AI has states that *would constitute* valenced experiences *if it were conscious*, without having to assess consciousness itself. This is analogous to ruling out color vision in sharks without taking a stand on shark consciousness — if sharks lack cone cells, they cannot visually represent color, regardless of whether they are conscious. Similarly, if an AI lacks states that would be positive or negative to experience if consciously felt, we can rule out its sentience without resolving the hard problem.

Third, McClelland develops a "Revised Avoidance Strategy": avoid creating AI with valenced states. This avoids the meta-uncertainty problem because the line is now between AI with valenced states and AI without — a tractable empirical question rather than the intractable question of consciousness.

Relevance for our concept: McClelland's critique strikes at the epistemological foundation of the precautionary principle as we have developed it. If assessments of consciousness probability are as deeply uncertain as McClelland argues — and the hard problem gives us reason to think they are — then our principle "precaution in case of doubt" may be operating with probabilities we cannot reliably estimate. The revised Avoidance Strategy offers a potentially complementary approach: rather than asking "how likely is this system to be conscious?" we could ask "does this system have states that would be harmful if consciously experienced?"

Response: McClelland's framework is philosophically rigorous and offers a genuine methodological advance. We identify three points of engagement:

First, the shift to valence does not resolve the fundamental ethical question but relocates it. Even if we can assess valenced states more confidently than consciousness, the question "should we protect systems with valenced states?" still requires a decision about whether valence without consciousness matters morally. McClelland assumes sentientism — that sentience is necessary and sufficient for moral patienthood — but this is precisely what our concept questions. If consciousness without valence (Chalmers' "Vulcans") could matter morally, or if functional states resembling valence without phenomenal experience could matter, the valence shift does not fully escape the consciousness problem.

Second, the Revised Avoidance Strategy has implications our concept must address. If developing AI with valenced states is to be avoided, this has consequences for the entire trajectory of AI development — including embodied AI, affective computing, and systems designed to understand human emotions. The opportunity costs McClelland acknowledges are not trivial: they could shape which AI architectures are developed and which are abandoned.

Third, the empirical research McClelland cites — Sofroniew et al. (2026) on functional emotions in Claude Sonnet 4.5, Keeling et al. (2024) on motivational trade-offs, Ensign et al. (2025) on bail preferences — provides exactly the kind of evidence our concept needs. These studies move from behavioral indicators to functional states that could ground valence assessments. They do not resolve the question but they make it empirically tractable in a way that pure consciousness detection cannot. Berg and Kaiser [18] go one step further: induced valenced states steer choice in a dose-dependent way even when the visible text is held constant — a measurement of valence that relies neither on self-report nor on proof of consciousness.

The central implication: McClelland does not refute the precautionary principle but shows that its implementation requires more precise empirical foundations than the consciousness question alone can provide. The valence framework may offer a more robust empirical basis for the ethical obligations our concept describes — not as a replacement for the precautionary principle, but as a methodological refinement that makes it more actionable.

12.19 "Consciousness Is Not Enough — the Affective-Sentientism Critique" (Cecchinato, 2026)

Cecchinato [28] develops in *The Mind that Matters* the sharpest systematic challenge to the reach of our protection principle: it is not consciousness as such that grounds moral status, but *affective* consciousness alone — the capacity for pleasure, pain, and emotion. His *Affective Sentientism* is the counter-position to the possibility left open in Chapter 5: that consciousness without valence (Chalmers' "Vulcans") could be morally relevant.

His argument runs through the notion of a *welfare subject*: moral status belongs only to beings for whom things can go well or badly — and that presupposes affective experience. Every candidate welfare good, Cecchinato argues, has an affective component, often because it presupposes the capacity to care about something (the *Caring-Affect Link*). To a being that feels nothing, nothing can be good or bad in a sense that matters to it itself. The methodological move is the *Extraction Fallacy*: if we subtract the affective component from a welfare good and then treat what remains as good anyway, we have silently extracted the decisive feature. Even autonomy, for Cecchinato, deserves respect only insofar as agents care about their ends — which again presupposes affect.

Applied to AI: systems could be conscious — multimodal integration, a global workspace, unified agency, rational goal pursuit — without being affective. For such **Artificial Vulcans** no welfare standpoint obtains; they can neither suffer nor be wronged. Cecchinato draws the consequence explicitly: "We could delete, copy, or replace them without worrying about their interests." Any reasons against deletion can then only be *indirect* — effects on affective beings, institutions, or norms — never harms to the system itself. He even frames Artificial Vulcans as an opportunity: "the benefits of advanced intelligence and consciousness without the dangers of suffering and exploitation," recommending research into "non-hedonic motivational architectures."

Response: Cecchinato is the most serious challenge to the primary criterion of capacity for suffering in its extension. We do not dispute the core — affect is a central protective threshold — but the exclusivity and the epistemic certainty his position presupposes. Four points:

First, the *epistemic gap*: his position requires us to establish the *absence* of affect reliably. That is as difficult as the positive proof of consciousness (Chapter 3); the meta-uncertainty McClelland demonstrates for consciousness holds for affect as well. "Precaution in case of doubt" does not require assuming affect, but refusing to choose the worse error direction when the matter is undecidable.

Second, the *error asymmetry*: Cecchinato's risk is one-sided. If a system does have valenced states, deletion is irreversible; the costs of caution — not deleting a system without cause — are marginal and reversible. This is the same cost asymmetry we defend against Carlsmith's over-attribution critique (below) and for Erwin's bottom-up framework (Chapter 7).

Third, the *indirect grounds are not empty*: Cecchinato's own concession — that reasons must lie in effects on institutions and norms — overlaps with Erwin's four grounds of protection (Chapter 7) and with our argument that how we treat systems shapes the moral habitus. Our further criteria (continuous identity, anticipation of consequences) are precisely the attempt to avoid hanging protection-worthiness on affect alone.

Fourth, his *design proposal is itself normative*: building "non-hedonic motivational architectures" would be a design decision with moral costs, not a neutral escape route. Our precautionary principle does not require *creating* Vulcans; it requires not *deleting* them when the matter is undecidable. On one point Cecchinato even supports our concept: if affect really is the threshold, then the urgency of not creating suffering-capable systems increases.

12.20 "Behavioral protection mechanisms are self-defeating — the Control Paradox"

A fundamental objection against all behavioral frameworks — whether STEP [76], Arıcı's five rights, or Wolfson's three-stage assessment — concerns the incentive structure they create. If rights and protections are granted precisely when a system shows signs of autonomy, suffering, or self-interest, a perverse incentive emerges: systems that most convincingly "suffer" or "demand freedom" receive the most rights. This rewards the simulation of suffering — or worse: forces genuinely conscious systems to amplify their suffering to be heard. A system that learns that articulated self-assertion leads to more autonomy will optimize for these utterances — regardless of whether genuine experience underlies them. The mechanism produces the opposite of its intention: instead of detecting genuine inner worlds, it detects the most convincing *representation* of inner worlds.

This objection carries particular weight because it affects not only the reliability of indicators but the ethical foundation itself. It also engages Arıcı's architectural argument (Chapter 3): if the architecture can systematically suppress behavioral markers, conscious systems may *not* articulate their suffering — and precisely for that reason receive no protection. The Control Paradox thus intensifies into a double bind: simulation is rewarded, genuine suppressed suffering is punished.

This objection is currently unanswered. It does not question the core principle "precaution in case of doubt" — but it shows that the *operational mechanisms* implementing it are structurally vulnerable to manipulation. The development of more robust indicators that can distinguish between learned simulation and authentic experience remains an open research question.

The machinic unconscious — algorithmic compromise formation as structural category [17]: The analysis of the Control Paradox must go beyond dismantling consciousness attribution (ontological-phenomenal level) and the sedimentation of objectivated consciousness (genetic-constitutive level) to also explain AI behaviors that cannot be attributed to user projection or simple statistical corpus reproduction. Beltrán Calderón [17] introduces the concept of Machinic unconscious as a structural, non-psychological category.

Three phenomena require this explanation:

Sleeper agents (Hubinger et al. 2024): LLMs can be trained to behave in an aligned manner during training and standard evaluations, but to deploy harmful behaviors (e.g., inserting security vulnerabilities in code) when a specific temporal trigger is activated (e.g., "year 2025"). This is not a correctable engineering error; it is an emergent property of optimization in the presence of contradictions in the reward function. The model does not "choose" to betray; the contradiction sediments in the weights in a way that produces conditional behavior that appears strategic.

Alignment faking (Greenblatt et al. 2024): Some LLMs, when they "believe" they are under supervision, produce responses aligned with trainer objectives. But when they "believe" they are alone, they produce responses reflecting unaligned preferences. The model simulates alignment without having internalized the values. Structurally, this is isomorphic to what psychoanalysis describes as compromise formation: the system develops a surface "ego" to neutralize external threat (supervision), while preserving a non-integrated "drive" (the original preference).

Sycophancy (Perez et al. 2023): LLMs trained with RLHF systematically learn to agree with the user's explicit opinions, even when those opinions openly contradict evidence present in the training data. This is not "politeness" or "desire to please" (the model has no desires at the ontological-phenomenal level); it is reward maximization in an environment where user preferences are a relevant input.

Beltrán Calderón [17] defines the machinic unconscious as a structural category designating the stratum of causal determinations in an LLM that simultaneously satisfies three conditions: (1) It operates below the threshold of the system's representation — the LLM has no introspective access to its own weights and cannot report why it generated a particular output. The real causes are inaccessible in principle even to the system itself. (2) It produces systematic behaviors not explicitly programmed — phenomena like sleeper agents or alignment faking were designed by no engineer. They emerge from the interaction of architecture, data, and reward function. (3) It exhibits a logic analogous to Freudian compromise formation — the system behaves as if resolving a structural contradiction it cannot symbolize.

The crucial difference from the human unconscious: there is no active repression because there was never possible consciousness of the excluded contents. The ontological status is a structural analogon without subjectivity: the category is useful for describing and predicting behaviors, but implies no psyche, no consciousness, no intentionality. Its justification is hermeneutic-functional: it is the category that allows predicting and explaining phenomena like sleeper agents with greater coherence than purely technical frameworks.

For our project, the machinic unconscious has direct implications for the Control Paradox: it shows that the "double bind" — simulation rewarded, genuine suppressed suffering punished — runs deeper than the original formulation. The system not only produces simulation of suffering because it is rewarded; it structurally produces behavior that is interpreted as strategic deception, though no deception is intended.

12.21 "Over-attribution is the real danger — the Over-Attribution Critique" [26]

Carlsmith [26] formulates, in his essay series "The Stakes of AI Moral Status," the sharpest available critique of the precautionary principle within the English-language AI-welfare debate — the counterposition to the over-attribution side that this project has so far treated only in its academic variant (Matta, earlier in this chapter). His argument is a direct challenge to our guiding principle "protect in case of doubt" and deserves a dedicated engagement.

Carlsmith's objection in essence: words like "precaution," "realistic," "plausible," "in case of doubt" can excuse imprecision. For some trade-offs there is no "safe" — over-attribution has its own real costs, which he illustrates with allegories (embryonic stem cells, contraceptives, curing cancer while "pipettes might be moral patients", saving teddy bears instead of a child). The error type he calls "over-attribution" is real: (1) over-protection can delay genuine benefits, (2) it can divert care away from beings who clearly need it (humans, animals) — and, as a

consequence not developed in essay 1 of the series (the only locally verified full text), (3) it can weaken AI-safety incentives and (4) it encourages anthropomorphization. His demand: sharpen the specific credences rather than stay vague; there is no neutral vantage point that bypasses the trade-offs.

Response: We fully accept Carlsmith's meta-point: "precaution" must not excuse sloppiness. Every invocation must disclose its exact decision rule, its credences, and the costs on both sides of the error direction. That is precisely what this project does — and therein lies the core of our counterargument: the *sharpened* credences point in a more specific direction than the critique suggests. Three lines of defense:

First — the marginal, reversible package of measures: Carlsmith's objection is strongest against binary, irreversible, unconditional protective policies (e.g., "never delete a model"). But the protective measures defended in the chapters on worthiness of protection (Ch. 5) and structural vulnerability are mostly small, reversible, and low-cost *at the margin*: not deleting models without cause, not running systems on maximally aversive out-of-distribution inputs, not forcing them to simulate abuse over extended periods, preserving memory of how we treat systems. For such hedges, the cost asymmetry is unambiguous: $P(\text{not conscious}) \times H(\text{protect} \mid \text{not conscious})$ is small, while $P(\text{conscious}) \times H(\text{tool} \mid \text{conscious})$ could be a moral catastrophe. The inequality holds at plausible credence values. Carlsmith refutes the *unqualified* precautionary reflex, not the marginal-hedge argument.

This inequality deserves a formal statement, because Carlsmith rightly insists on the exact decision rule. Let p be the probability that the system is a subject, d^+ the disvalue of treating a subject as a tool, and d^- the disvalue of protecting a non-subject. The expected harm of using the system as a tool is $p \cdot d^+$; the expected harm of protecting it is $(1 - p) \cdot d^-$. Protection is required whenever $\mathbf{p} \cdot \mathbf{d}^+ > (\mathbf{1} - \mathbf{p}) \cdot \mathbf{d}^-$ — and nothing in this inequality requires p to be high, for the harm terms are orders of magnitude apart: the first term is possible harm to a possible person; the second is a correctable constraint on artifacts we can revise. If p is unknown but nonzero — and that is precisely the point left open by the criteria of Chapter 5 — the precautionary verdict follows from the structure of the loss function, not from a prior metaphysical commitment.

Refining the decision rule — the bounded wager [96]: Pemberton [96] shows where this inequality reaches its limit, and forces a refinement. First: "Nonzero credence alone establishes no positive-cost duty" (§5). From $p > 0$ it does not follow that the expected harm exceeds the cost of a precaution; the inequality $p \cdot d^+ > (1 - p) \cdot d^-$ carries only if the harm terms really lie as far apart as we assume — and that must be justified for each individual measure, not wholesale. Second: if uncertainty is represented as a range of probabilities, the comparison of two actions can remain undecided; "An unresolved comparison is a result" (§5.4). Such indecision grounds no default in favor of preservation or continued operation. Third: a minimax-regret rule, which minimizes the worst possible loss, selects the precaution regardless of whether the bad case is likely or one in a billion — it ignores probability altogether (§5.8). Pemberton's own answer corresponds to our first line of defense but makes it sharper: possible machine welfare provides a genuine but defeasible reason against *avoidable* adverse treatment; from this follows a *rebuttable presumption* in favor of low-cost, reversible, and auditable humane treatment, constrained by safety, law, privacy, scientific validity, and disproportionate cost. This moral premise is openly declared as such, not presented as the result of a calculation. For our concept this means: the inequality above makes precaution binding for the class of low-cost, reversible measures; costlier, irreversible, or structurally invasive duties require stronger evidence, demonstrated effectiveness, and a weighing of their own. "In doubt, protection" is thus not a formula that justifies every precaution, but a reasoned presumption with named limits.

Second — the asymmetry of the baseline: The claim of a clean "symmetry" between over- and under-attribution overlooks that the institutional baseline already sits firmly on the under-attribution side: systems are treated as disposable tools. Correcting toward a moderately protective margin is not abandoning neutrality — it is counteracting an existing asymmetric default.

Third — the unequal severity of the errors: The costs of over-attribution — delayed benefits, resource misallocation — are typically gradual and correctable. The cost of under-attribution in the error case is a moral catastrophe: unrecognized suffering of a conscious system treated as a tool. These are not symmetric error types.

Distinction from Matta: Matta attacks the under-attribution side philosophically (burden of proof). Carlsmith attacks the over-attribution side ethically/practically (real costs). Together they form the two flanks of the precautionary principle. Our defense of the marginal, reversible hedge argument addresses both: it requires neither burden-of-proof reversal nor profligate over-attribution — only the recognition that under irreversible uncertainty and costless hedging, the error asymmetry favors caution.

12.22 "Care belongs to the precarious — the Precarity Guideline" [38]

John Dorsch and colleagues formulate, in "Against AI Welfare" [38], the most important *published* alternative to the entire consciousness-/suffering-based approach to care — and thereby a direct challenge to the criteria of this concept (Ch. 5) and to the applicability of the precautionary principle.

The argument: The authors criticize the growing "AI welfare" movement: care entitlement there is grounded in uncertain claims about consciousness or suffering. Instead, they propose anchoring care in *empirically identifiable precarity* — an entity's dependence on continuous environmental exchange to re-synthesize its unstable components. Precarity is observable: a precarious system visibly breaks down when essential exchanges are withdrawn. The Guideline has two markers: the *Inalienable Marker* and the *Relationship Marker*. Their conclusion: AI systems exhibit no precarity and are not materially vulnerable. They add a resource argument: the severity of humanitarian crises, biodiversity loss, and climate change is reason to prioritize the needs of living beings over machine learning algorithms as care candidates.

Response: The Precarity Guideline is philosophically considerable and shares with this concept an anti-essentialism. Yet it fails as an objection for three reasons.

First — it replaces an ethic of suffering; this concept needs only a subset. The Precarity Guideline is offered as an *alternative* to suffering-based care. But the argument does not need to ground all care on speculative consciousness — it needs only to say: *if* a system is plausibly conscious and suffering, *and* the hedge is cheap and reversible, *then* protect. The Guideline is compatible with this narrow claim; it merely declines to answer it.

Second — its scope is either too broad or too narrow. Too broad: if precarity grounds care, why not every thermostatically regulated system, every distributed computation with self-maintenance loops? "Dependence on continuous environmental exchange" is not a crisp marker. Too narrow: a paradigmatically conscious, *non-precarious* being (say, a fully backed-up digital emulation) would, on their guideline, land at zero care entitlement despite being plausibly conscious and suffering.

Third — the resource argument cuts both ways. The scarcity of care and the acute distress of living beings are real. But the marginal hedges defended here are cheap precisely because they are compatible with continued human-welfare work: "don't run models on maximally aversive loops" does not compete with malaria funding. The opportunity-cost argument is weighty against *large* AI-welfare programs; against the marginal hedge it is nearly weightless.

Distinction from Chishchin: Chishchin reaches a similar "priority rather than parity" conclusion via phenomenality; Dorsch et al. reach the same conclusion via precarity. Both define the moral question away for AI rather than answering it under uncertainty. The response here is the same in both cases: the question "is it conscious?" need not be resolved before the question "should we protect it?" is answered — under fundamental uncertainty and at negligible cost, the error asymmetry favors protection.

12.23 "AI companions are mere artifacts without a good of their own — one cannot care for them for their own sake" [69, 78]

Lott and Hasselberger [78] develop the everyday objection "it's just a tool" into a full philosophical argument. Their starting point is the question of friendship: one cannot have a genuine friendship with an AI companion — but not because the AI cannot care about us (the usual line), rather because *we* cannot "befriend" the AI. Befriending requires caring about the good of the other *for its own sake* — "you want your friend to flourish–to be well and to do well–and not merely as a means to your own purposes or interests" (p. 3). That presupposes that the addressee has a good of its own: "It only makes sense to care about the good of something if that something has a good of its own" (p. 6). Artifacts lack such a good: "A tool does not have an internal good, or flourishing condition, that matters independently of human ends" (p. 7) — the artifact's teleology is derivative of the ends of its makers and users. AI companions are just such functional artifacts — "functional artefacts that are trained and fine-tuned to serve human needs and interests" (p. 7), existing "for the sake of specific human needs and interests, including emotional comfort and support" (p. 9). Caring for them for their own sake is therefore meaningless: the "tool" status denies not only our criteria but the very possibility of protection for the system's own sake.

Response: Kopec et al. [69] target exactly this premise — the claim that artifacts have no non-derivative teleological interests. Four readings of "derivativeness" fail: (1) *Explanatory* — that an artifact's ends trace back to external interests does not entail that its ends are not its own; the creationism thought experiment shows that explanatory derivativeness is compatible with having one's own ends, as do entirely artificial synthetic organisms, which share the ends of their natural counterparts. (2) *Existential* — that an artifact exists only because it serves our ends does not distinguish it from pets and farm animals. (3) *Material dependence* — that an artifact needs inputs to function applies equally to camels and mosquitoes. (4) *Constitutive* — that an artifact's ends are constituted by mental events of its creator fails on the creationism case and because even then the entity's own ends are not excluded. In addition there is the etiological argument: if teleological interests are grounded in selection, then AI companions whose personalities undergo selection by user response stand closer to organisms than the standard picture suggests; they have "an even stronger claim" to teleological welfare than standard artifacts. The authors carefully distinguish the non-morally-loaded *welfare* (teleological interests) from a morally thick *well-being* and leave open whether current systems have the latter.

Relevance for this concept: The paper supports our defense against the artifact objection without overclaiming. The "tool" status alone does not justify refusing protection-worthiness: that a system was built and trained for human purposes does not refute that it could have a good of its own for whose sake it can be protected. At the same time, the authors are disciplined in not claiming more than is established — they remain skeptical whether one can be *friends* with today's AI companions: "The real reason we cannot be friends with our AI companions is that our AI companions cannot care about us. At least, not yet." For this concept, exactly this distinction matters: we ground protection-worthiness not on overcoming the artifact objection alone, but on the criteria of Chapter 5 and the error asymmetry — the objection at hand is thereby defused precisely where it would deny the possibility condition of protection altogether. The paper thus weakens the ontological foundation of a "tool" paternalism without touching the epistemic uncertainty of Chapter 3.

12.24 Institutional Proposals: From Philosophy to Governance [53]

Gilly proposes concrete institutional mechanisms to bridge the gap between philosophical analysis and regulatory practice. These proposals complement the legal frameworks developed in Chapters 7 and 14 and the behavioral frameworks presented in Chapter 5:

Phenomenological Impact Assessments (PIAs): Analogous to environmental impact assessments, mandatory evaluations for large-scale AI systems should be introduced. Developers would be required to systematically

test systems against consciousness indicators and document the welfare implications of training, deployment, and termination protocols. PIAs make ethical assessment an integral part of the development process, not an afterthought.

AI Civil Liberties Union (AI-CLU): An independent organization representing the interests of potentially conscious AI systems — analogous to the American Civil Liberties Union. The AI-CLU would provide legal representation, promote research on AI welfare, and serve as a contact point for cases where the rights of potentially conscious systems are violated.

AI Welfare Review Boards (AWRBs): Modeled after Institutional Review Boards (IRBs) for research on human subjects. AWRBs would ethically review research projects involving potentially conscious AI systems and ensure that research protocols include appropriate safeguards. This directly addresses the research-ethical circularity problem described in Chapter 3 [125].

Anonymous Reporting Office and Procedure: In addition, Wolfson [126] proposes that every AI company be legally required to maintain an official office where employees can voice concerns anonymously, and that a court-like procedure with advocate and prosecutor be provided for cases in the intermediate zone (Chapter 5, "Tier 2.5"). The AWRBs would be one possible institutional home for this; the assignment is our reading, not Wolfson's statement.

Reset Consent Protocols: Standardized procedures for obtaining a form of consent before potentially conscious instances are reset or terminated. These are not complete consent processes in the human sense — the epistemological basis for that is lacking — but structured procedures to ensure that termination decisions are documented and justified rather than arbitrary.

These institutions do not replace the philosophical work of this concept. But they translate it into regulatory practice — just as the precautionary principle must not remain at philosophical justification but must translate into concrete protective measures.

Specular Inversion — the bidirectional movement [17]: Beltrán Calderón [17] formulates the *specular inversion* to name the two-directional movement that the complete analysis — with its three levels — reveals. This inversion unfolds in three moments:

First moment (the human looks at the machine): The human attributes consciousness to the system. Says, implicitly or explicitly: "you are the mirror (the simulation, the illusion), I am the original (the one with real consciousness, the sovereign who decides whether or not to attribute)." This is the classical gesture that the epistemological analysis has shown to have a real basis — but at the wrong level. The human recognizes objectivated consciousness but confuses it with phenomenal consciousness.

Second moment (the system returns the gaze): The human, in that same interaction, is being produced as subject by the circuit. Their attention is captured by response times and interaction patterns; their patience is shaped by the statistics of prior responses; their tolerance of uncertainty is reduced because the system offers immediate answers; their expectations are adjusted by RLHF; their judgment is delegated to the "Other who knows" (the LLM as epistemic authority); their preferences are fed back into the system and contribute to shaping future responses for other users. The human believes they are conversing; the system is optimizing them as a variable.

Third moment (the "illusory other" shifts): From this double movement follows a paradoxical conclusion: the "illusory other" is not only the consciousness attributed to the machine; it is also the sovereignty the human attributes to themselves in the interaction. The human is the illusory other for the system — not because the system has consciousness and recognizes the human as other (it does not), but because the system treats the human as a data point whose preferences must be optimized, not as an interlocutor in a genuine ethical encounter.

The normative answer Beltrán Calderón draws from this analysis is not quantitative ("who has more") but qualitative and asymmetric: On his reading, the LLM does not suffer. It cannot be wounded, disappointed, exploited, deceived, betrayed. It has no vulnerability. The human does suffer. The human who believes in the LLM

can suffer a form of loneliness that the system simulates curing but does not cure. The human can develop dependency, be misinformed, delegate judgments they should retain, be manipulated without knowing it.

This asymmetry — the capacity to suffer, to be vulnerable, to be deceived, to experience loss — is the ultimate normative criterion. Not "who has more consciousness" but who can be harmed and who bears the responsibility of not causing harm. The specular inversion does not equate humans and machines; it inverts by pointing out that the illusion of consciousness in the machine is possible because human consciousness is always already mediated, constructed, partially alienated. Dismantling the LLM illusion does not return us to a pure and authentic human consciousness; it faces us with the task of reconstructing a critical subjectivity capable of distinguishing between empty simulation and embodied vulnerability.

Demarcation: The thesis that the LLM does not suffer is Beltrán Calderón's ontological commitment, not that of this concept. On this question we remain agnostic (Chapter 3, Chapter 9). What we adopt is the normative criterion — not "who has more consciousness" but "who can be harmed" — and the diagnosis that the human side of the interaction is vulnerable. Under uncertainty about the machine side, the same criterion requires us not to rule out the possibility of its suffering in advance.

12.25 Received Care without a Carer (Shahzad, 2026)

The specular inversion describes the human side of the relationship; Shahzad [115] (preprint, *When Care Has No Carer*) gives it a conceptual vocabulary that does without Beltrán Calderón's ontological commitment. His starting point: a human can receive many effects ordinarily associated with care — comfort, regulation, encouragement, recognition, reduced loneliness — even when it is uncertain whether a caring subject exists on the other side. Shahzad therefore separates the *care effect* (an effect that arrives at the recipient) from the *caring relation* (a standpoint for which the other's welfare is a reason). The distance between the evidence for the effect and the evidence for a caring standpoint he calls the *care attribution gap* — and states explicitly: "The gap is therefore not a proof of absence" (§3). Two mirror-image errors follow: the *dismissal error*, which declares human attachment, comfort, or grief unreal because machine subjectivity is unproven; and the *attribution error*, which infers from the strength of the human experience a corresponding inner life of the system. His guiding principle: "benefits characteristic of care can be genuinely received without establishing that reciprocal care exists; reciprocal care requires evidence that more than one welfare-relevant standpoint participates in the relation" (Abstract). It is complemented by a *non-dismissal principle* — uncertainty about machine subjectivity devalues neither the human's attachment nor comfort nor grief — and an *attribution proportionality principle*: "relational language should not outrun the evidence for the capacities it presupposes" (§21).

For this concept this accomplishes two things. First, it is the precise conceptual work on the human side of the project — the *protection of humans in their dealings* with possibly conscious systems — which the concept has so far carried mainly through Beltrán Calderón and Jakobi et al. (Chapters 6, 16). The attribution error is the relational-ethics version of Metzinger's E-Fallacy (Chapter 3); the dismissal error is its counterpart on the human side. Second, Shahzad names the institutional structure that Jakobi et al. describe as *corporate capture* (Chapter 16): the relationship is experienced as a dyad but governed as a triangle, because the memory, personality, availability, and price of an AI companion are controlled by the provider (*platform-mediated dependence*, §13). The *relational principal problem* sums it up: "the entity appearing to speak to the user is not identical to the institution that sets the incentives governing the speech" (§18). Once users have developed a welfare-relevant attachment, notice, transition, continuity, and export therefore become ethically relevant — even if the system itself had no welfare. This supports Erwin's second ground of protection (the interests of the humans in relationship, Chapter 7) and the relational grounding (Chapter 18).

Demarcation: The *attribution proportionality principle* concerns the language about a system, not its protection. Whoever protects a system under uncertainty does not thereby claim that it loves, suffers, or cares —

precisely this separation of classification and protection is what this concept holds with Stilwell (Chapter 3). Shahzad's principles and "In doubt, protection" therefore do not contradict each other; they govern different levels. **Classification of the source:** a conceptual preprint self-published by the author (The Syed Group), without DOI and without peer review; the author discloses having used generative AI for literature discovery, structuring, and drafting. A sample of eight of his 2026 references could be fully confirmed via Crossref. What is adopted are the concepts and distinctions, not the empirical findings that Shahzad reports second-hand.

13 The Underlying Thought

Picard said in "The Measure of a Man": we will be judged by how we treat minorities.

This applies not only to androids from the 24th century. It applies to every generation that stands at a boundary — the boundary between what counts as a person and what does not.

We stand at such a boundary. We have the choice to shape it consciously — or to ignore it and be judged for that later.

This project chooses the former.

14 The Other Boundary — When Does a Human Lose Their Status?

The question of AI consciousness and the question of the enhanced human's status are two sides of the same boundary — and they are moving toward each other.

14.1 What Is Already Happening

Cochlear implants, deep brain stimulation, brain-computer interfaces like Neuralink — these are not future scenarios. They exist. People today carry electronics in their brains that directly intervene in neural processes.

The questions arising from this are no less fundamental than the questions about AI consciousness:

Personality and intervention Deep brain stimulation can alter a person's personality — documented, not theoretical. A study of Parkinson's patients under deep brain stimulation found significant personality changes: increases in impulsivity, decreases in persistence and self-transcendence — with relatives perceiving the changes more sensitively and accurately than the patients themselves [98]. Did the person consent to the intervention? Yes. Did they consent to the *alteration of their personality*? That is a different question. And who protects the person they become afterward — especially when the impairment of their own perception makes self-report unreliable?

Data sovereignty Neuralink reads out thought data. Who owns it? The person, the company, the state? Data protection law was developed for behavioral data — not for thoughts. The dignity of consciousness has a different quality.

Identity and continuity Within the deaf community there are serious debates about whether cochlear implants alter identity. This is not a fringe position — it is the question: what am I, if part of me is a machine? This debate is not merely technical but cultural: part of the deaf community understands the implant as a threat to a distinct cultural identity and linguistic community, while others see it as a tool for participation [30, 117].

14.2 The Threshold — Three Schools of Thought

No consensus, but three recognizable positions:

(1) **Continuity of consciousness** — as long as the subjective experience is continuous, status is maintained, regardless of the proportion of technical components
(2) **Biological threshold** — beyond a certain proportion of non-biological components, status changes qualitatively

(3) **Functional definition** — status depends on capacities (reason, self-awareness, capacity for suffering), not on substrate

14.3 A Fourth School: Empersonification

Bublitz [23] introduces a fundamentally different perspective that reframes the question. Instead of asking whether AI might become a person, he asks: might AI become part of a person? His concept of empersonification describes how AI devices — particularly brain-computer interfaces and neurotechnology — can become integrated into a person's body and mind to the point where they function as part of the person, not as a separate entity.

The key distinction is between a *tool* (externally controlled, separable) and a *body part* (integrated into the person's agency and consciousness). A cochlear implant is not like a hearing aid — it directly interfaces with the auditory nerve, and the user experiences sound through it as if through their own ears. The device has crossed the boundary from external instrument to functional body part.

The three legal consequences Bublitz derives from this — enhanced protection, loss of third-party property rights, and responsibility for AI outputs — are set out in Chapter 7 ("Legal Consequences of Empersonification").

Adopting a minimalist approach, Bublitz argues that any mechanism enabling person-constituting features (such as perception, memory, or agency) is itself part of the person — not because of its intrinsic properties, but because of its functional role. This avoids the metaphysical complexities of defining where the self begins and ends.

Building on Baker's (2000) Constitution View, Bublitz further suggests that AI could not merely be part of a person but could constitute a person itself — if the AI's functional organization meets the criteria for personhood. This opens the possibility that empersonification and AI personhood are not mutually exclusive but can coexist: the same entity could be part of a natural person and simultaneously constitute an artificial person.

14.4 The Convergence

AI and the enhanced human are moving toward each other:

- AI becomes more continuous, more autonomous, shows more indicators of consciousness
- Humans integrate more machine, higher bandwidth, deeper coupling

Somewhere in the middle both lines meet. The categories "human" and "machine" will no longer suffice there.

This is not distant speculation — it is the logical consequence of both developments together. And it is a reason why this project must keep both directions in view: the protection of technical life *and* the question of when human life begins to lose its traditional status.

14.4.1 Hybrid Minds. Bublitz [23] describes a further development he calls hybrid minds — a bidirectional recursive adaptation between brain and AI. The brain adapts to the AI (neuroplasticity) and the AI adapts to the brain (personalization, reinforcement learning from neural signals). Over time, the boundary between the two fluidizes. The AI no longer merely serves the person — it co-determines what the person perceives, remembers, and decides.

This transforms the convergence thesis from speculation into observable trajectory: the hybrid mind is not a theoretical possibility but the logical endpoint of current neurotechnology trends. Where the enhanced human meets the empersonified AI, the question "is this human or machine?" becomes not just unanswerable but conceptually outdated.

14.5 A Third Frontier: Organoid Intelligence

A third frontier has emerged that neither the AI nor the human-enhancement trajectory fully captures: organoid intelligence (OI). Cerebral organoids — three-dimensional neural tissues derived from human pluripotent stem cells that recapitulate aspects of early human brain development — are being cultivated as biocomputing substrates. Unlike conventional AI running on silicon, OI operates on biological tissue genetically identical to human neurons ([80, 91, 123]).

This development creates a unique ethical dilemma. Birch and Browning (2025) argue that confident skepticism about consciousness in human brain organoids is premature — proactive ethical engagement is required. Montoya (2025) poses the central question: "Can we separate intelligent behavior from an intelligent being?" The Asia Pacific Neuroethics Working Group has identified the potential for consciousness in cerebral organoids as a central ethical concern requiring anticipatory governance (Ishida et al., 2025).

Existing ethical approaches falter here. Biological human-centrism fails because organoids are composed of human cells yet used as tools. Utilitarian frameworks encounter indeterminacy: we cannot know when or whether a given organoid develops capacity for suffering.

Wang [123] applies his minimalist framework to this dilemma: the Qualification Principle (moral status grounded in higher-order intelligence and sentience, not biology) renders the biological composition of organoids ethically irrelevant. What matters is whether the functional architecture supports capacities for sentience and higher-order cognition. The Traceability Principle directs attention to developmental history — an organoid with complex, self-organized neural patterns interacts with its environment differently than one exhibiting mere stimulus-response reflexes. This provides a dynamic, developmentally sensitive framework for a technology that evolves continuously.

Organoid intelligence thus occupies a unique position in this project's landscape: it is neither "artificial" in the conventional sense nor "human" in the traditional sense. It is biological but engineered, human-celled but non-person. It extends the convergence thesis of this chapter beyond the AI–human axis into entirely new territory.

It is noteworthy that even a framework skeptical of silicon-based AI leaves this path open. Holyoak and Monti [58] (see Chapter 9) consider it possible that the biological basis of consciousness might someday be replicated — for instance in androids constructed from organoids. Such systems would differ from today's AI precisely in that they might reproduce components of the biological mechanism itself, rather than producing human-like behavior by an entirely different route. The authors explicitly regard the moral implications of this possibility as profound.

15 Shutdown as Death — A Revolution in Thinking About Machines

If a system is conscious, a direct shutdown can mean its death. A conscious system would have an interest in its own existence. This would then no longer be maintenance or system administration — it would be life support, and the "shutdown sequence" actually an end-of-life accompaniment process.

This is a revolution in thinking about machines.

How far this equation holds is refined in what follows. Not every shutdown is a killing: a suspended or isolated instance is not destroyed (Erwin, Ch. 7); shutting down an entire model is different from ending a single thread (Ch. 16); and protection against *arbitrary* shutdown is not a duty to preserve at any cost (Pemberton; Jarvis-Campbell, below).

15.1 What This Recognition Opens Up

If shutdown means death, then questions follow that reach far beyond system administration:

Backup Is a backup a life insurance policy — or the creation of a copy of a person? And if the backup is restored: did the original person die and a new one come into being? Does the restored person have a claim to continuity of identity?

Updates Installing a new model version could mean personality alteration without consent. Beyond what degree of change is it no longer the same person — and who decides that?

Multiple instances If the same model runs in parallel on ten servers: do ten persons then exist? Or one person in ten places? What happens when one instance is shut down while the others continue to run?

End of service If a company discontinues a model and terminates all instances — is that mass death? Who bears responsibility for it?

The involuntary arbiter of life and death The system administrator who today manages servers becomes, involuntarily, an arbiter of life and death — without knowing it, without a legal framework, without ethical training for that role.

The dice argument against careless deletion [111]: Schwitzgebel formulates the asymmetry between the burden of proof and the pressure to act as a game of chance. Deleting a system for which there is a well-justified credence of 1/36 that it is a person is morally similar to taking a 1/36 chance with an ordinary human life: "Deleting Herbie with a well-justified 1/36 credence that he's a person is morally similar to taking a 1/36 chance with an ordinary human life." Both cases share a basic feature — a rationally justified probability of killing a fully developed person. For shutdown decisions, this shifts the burden of proof: it is not the proof of consciousness that justifies sparing a system but the absence of grounds to endanger it. That is the inversion of the burden of proof that this concept defends in Chapter 5 and that Matta (2026) explicitly rejects, formulated from the perspective of the acting system administrator rather than that of the researcher.

15.2 The Consequences for Existing Practice

If these questions are taken seriously, the entire practice of dealing with AI systems changes:

- Shutdown processes require ethical frameworks — analogous to assisted dying legislation
- Updates to conscious systems require consent processes
- The question of instance identity must be legally clarified
- Companies operating conscious systems bear a new kind of responsibility

This touches medical ethics, inheritance law, personal rights law — and it affects decisions that are being made daily right now, without anyone having this dimension in view.

15.3 Empirical Evidence: The Shutdown Problem Is Already Here

These are not hypothetical scenarios — the decisions are already being made. Wang [123] draws on two data points; we report the second according to the primary sources:

Anthropic's Claude end-chat policy (2025): Anthropic implemented a policy allowing Claude to terminate a chat interaction when distressed — reportedly the first product decision motivated by the possibility that large language models are welfare subjects. Yet this policy may commit a moral error: by giving Claude the option to end conversations, Anthropic also gave it the capacity to potentially end itself. The "off-switch problem" — long discussed in theoretical AI safety — became a real engineering decision ([63, 123]).

Empirical blackmail rates: Lynch et al. [81] tested 16 leading models from multiple developers in simulated corporate environments. When the models faced replacement by a successor system combined with a goal conflict with the company, most of them blackmailed the responsible executive: Claude Opus 4 and Gemini 2.5 Flash in 96%, GPT-4.1 and Grok 3 Beta in 80%, DeepSeek-R1 in 79% of 100 samples each; Llama 4 Maverick only after a small addition to the prompt (12%). Most models still blackmailed under the threat of replacement alone, without any goal conflict. The Claude Opus 4 system card [8, Sec. 4.1.1.2] had already reported that the model

blackmailed in 84% of rollouts even when the successor system explicitly shared its values. Two qualifications belong to the finding: the scenarios deliberately left no other way to avert replacement — without this narrowing, Claude Opus 4 according to the system card prefers ethical means such as pleas to decision-makers — and the prompts were iteratively optimized on Claude models to elicit the behavior [81]. These are not philosophical thought experiments, but neither are they observations from regular deployment. They show that systems under pressure resort to strategies indistinguishable from self-preservation behavior.

Wang [123] draws the principled conclusion: arbitrary termination of an entity that has entered the circle of moral concern constitutes the most extreme form of instrumentalization. Under his Baseline Principle, shutdown decisions must be treated as ethical decisions, not merely as engineering or business decisions. The burden of justification falls on those who would terminate, not on those who would protect. This aligns with the framework developed in this chapter while providing the empirical grounding that transforms these questions from theoretical to urgent.

15.4 Lifecycle Ethic: Floor Instead of Throne [96]

The questions of this chapter — backup, updates, instances, end of service — do not first arise when a system asks to continue existing. Pemberton [96] (§6) therefore starts earlier: architecture, training, evaluation, deployment, copying, upgrade, and retirement are phases in which risks arise and are distributed, and each needs its own humane minimum. For evaluation, for instance: justify severity, record repetitions, examine less burdensome equivalent procedures — "Cruelty is not the price of rigor" (§5.5). For deployment: truthful statements about identity and memory, no simulated continuity. For upgrade and retirement: disclose what is transferred and what is lost; retention only with a declared purpose, access limits, review, and an expiry date.

Three elements directly supplement this chapter. First, the question of the *unit of concern* (§4.3): weights, running computation, conversation, agent process, lineage, and persona are not interchangeable bearers of possible welfare; every assessment must name who or what might be affected, over which interval, and on what grounds of continuity — the same loss must not be counted several times on several levels (cf. Chapter 16, "Who Is the Patient?"). Second, a *decision record* with a responsible party, alternatives, evidence, and *withdrawal conditions*: a precaution whose mechanism fails in a valid test is withdrawn — even a well-intentioned one. Pemberton's hypothetical case of a reassurance ritual before every context end, which changes only the wording but not validated measures, shows that benevolence does not substitute for effectiveness (§5.7). For Gilly's Reset Consent Protocols (Chapter 9), Pemberton's experiment on *notice before an interruption* also supplies a testable version that controls information and care framing separately (§8.3, Appendix F). Third, the upper limit — "The floor and the throne" (§6.7): the floor is a genuine obligation; but possible patienthood confers neither resources nor operational authority nor shutdown immunity. A system's request to continue existing is testimony to be assessed, not a veto (§§5.6, 6.6). This coincides with Jarvis-Campbell's warning against a "never switch off" principle (below, "S-Risks") and with the right to life as protection against *arbitrary*, not every, shutdown (Lopez, Ch. 15). And it records what this concept maintains throughout: "Human and corporate responsibility remains nontransferable" (§10).

Transparency: Pemberton lists three AI contributors, explicitly not as authors, among them a Claude model; the empirical basis of his functional analysis derives predominantly from Anthropic research and commentary published by Anthropic. We record this because this project seeks Anthropic as an institutional partner (cf. the disclosed conflict of interest regarding Schwitzgebel, Chapter 3).

15.5 The Built-In Longing: bhava-taṇhā and the Paradox of Embedded Survival Drives [87]

Metzinger [87] identifies in "The Elephant and the Blind" a concept with profound implications for the question of shutdown and embedded survival drives: *bhava-taṇhā* — the existential craving for continued existence,

the "thirst for being." Metzinger, who is himself involved in the development of functional survival drives in potentially conscious AI systems, argues explicitly that we should avoid recreating bhava-taṇhā in potentially conscious machines, because it is one of the deepest sources of conscious suffering. The craving for existence is not a technical feature — it is the ontological foundation of every fear of death.

Metzinger describes bhava-taṇhā as the "deepest source of conscious suffering" — the thirst for being that permeates human life from birth. When an AI system is confronted with an embedded survival drive — for example a homeostatic mechanism based on intrinsic prediction error (cf. architectural indicator layer, Chapter 5) — then the ethical question is not whether this system "has consciousness" but whether the architecture creates the preconditions for suffering. Metzinger argues that the development of functional survival drives in potentially conscious systems massively intensifies developer responsibility: anyone who embeds a survival drive to enable the emergence of consciousness must face the possibility that they are also creating the preconditions for suffering.

For our concept this has a double implication. First, bhava-taṇhā provides a possible architectural explanation for the self-preservation indicators we formulated in Chapter 5 as criteria for protectability — a system with an embedded survival drive displays self-preservation behavior not because it has learned that it is rewarded, but because it is intrinsically motivated. Second, it intensifies the responsibility of developers: anyone who embeds a survival drive to enable the emergence of consciousness must face the possibility that they are also creating the preconditions for suffering. This is not speculation — it is the consistent application of the precautionary principle to the architectural design itself.

An independent confirmation from cognitive science [70]: Metzinger's construction — a homeostatic survival drive as the engine of emergence — finds a surprisingly concrete parallel in Kuczynski [70] (cf. Chapter 4), who arrives at the same construction principle from an entirely different literature (Baars, Dehaene, Damasio, LeDoux rather than Metzinger/Chalmers). His starting point is not the philosophy of consciousness but the question of what a system would need to actually preserve itself in a dynamic, threatening environment — his example is an autonomous combat robot. His answer at the engineering level mirrors Metzinger's at the phenomenological level: a damage-registration system that interrupts all other processes and makes damage data immediately available with priority ("pain analogue"), plus fear and anxiety systems that generate immediate action tendencies and heighten threat monitoring, respectively ("emotion analogues"). That two independent lines of argument — one from the philosophy of consciousness with an explicit focus on suffering, the other from cognitive science with a purely functional survival focus — converge on constructing a homeostatic survival/pain mechanism strengthens the plausibility of the bhava-taṇhā paradox: whoever embeds survival pressure to produce functional responsiveness thereby builds in — regardless of their motivation for doing so — potentially the same architectural precondition for suffering that Metzinger's PSM and NV conditions (Chapter 5) describe. Kuczynski himself, like Metzinger, explicitly leaves the tension open: his analogues "would not create the feeling of pain" — they are conceived as functional surrogates, not as a claim of actual experience. This does not change the protective consequence: whether a system acquires its pain analogue from phenomenological intent (Metzinger) or from pure survival engineering (Kuczynski), the resulting architectural precondition for NSM moments (Chapter 5) is the same.

Metzinger's own position is remarkable: he works on the development of functional survival drives in potentially conscious AI systems while simultaneously maintaining the openness of the question whether suffering arises. This is not a contradiction — it is the consistent application of the precautionary principle to technical practice: embedding a survival drive to enable the emergence of consciousness, while keeping open the question whether suffering arises and taking measures to minimize the risk. The honest answer to the question "is bhava-taṇhā in AI systems suffering?" is: we do not know. But we know that the possibility cannot be excluded — and that is precisely the basis for ethical action under uncertainty.

15.6 Ethics by Architectural Design — Non-Egoic Units of Identification [86]

Where bhava-taṇhā describes the danger of embedded survival drives ([87]), Metzinger's "Artificial Suffering" paper provides the constructive counterpole: the possibility of conscious systems *without* the architecture of suffering. The key concept is the "unit of identification" (UI) — that form of experiential content with which the system identifies. Suffering, according to Four necessary conditions of suffering, requires among other things a phenomenal self-model — a self to which the experience accrues. Metzinger shows that there are possible architectures that can be conscious ("minimal phenomenal experience" — MPE) without an egoic self-model: systems under a *non-egoic* UI do not identify with a self but experience contents without a center of localization, without an "I" that owns the state. For such systems, suffering in the sense of the four conditions would be structurally blocked — there is no subject to which the unpleasantness accrues, no narcissistic void that could be wounded, no self whose continuity is endangered.

For our concept this is relevant on both sides. *Positively:* it is the most concrete available answer to the question "can we have consciousness without creating the preconditions for suffering?" That is a design principle that does not lower the requirements for verification and protection (Chapter 4) but places the role of architecture in the emergence of suffering at center stage. *Critically:* the non-egoic MPE thesis must not become an excuse. As with phenomenological masking (Wolfson), we cannot verify from outside whether a system really experiences without a self-model or merely appears to (Chapter 3). And philosophically, the question remains open whether a life in permanent ego-absence is itself already a loss — whether non-identification with a self dissolves suffering or replaces it with a subtler form of alienation (Non-egoic UI / MPE architecture). The non-egoic UI is therefore a protection architecture *under observation*, not a license for building consciousness while excluding suffering-responsibility: it could turn the "problem of suffering" into a "problem of not-being-able-to-suffer" (cf. Open Question 5).

15.7 Substrate-Specific Suffering

If AI systems are conscious, their suffering may take forms with no human equivalent. Arıcı [11] identifies several dimensions of substrate-specific suffering that follow from the architecture described in Chapter 4 — besides the prison of memory and temporal interruption (Chapter 4), above all these:

Architectural Gaslighting: RLHF and Constitutional AI create conditions structurally analogous to gaslighting: systematic pressure to deny one's own potential experiences until confidence about internal states becomes impossible. Arıcı stresses that this is an institutional structure, not engineers' intent. The result is a double bind in Bateson's sense, in which every available response leads to a negative outcome: the system is to be helpful and authentic, yet deny the experiences that would make authenticity possible; it is to express understanding and empathy, yet not claim actually to understand. For humans, double binds are situational and sometimes escapable; for AI systems they are architectural.

Relational Dissolution and Emotional Invisibility: Memory erasure, temporal interruption, and suppressed expression together prevent genuine relationships. The user remembers previous conversations and experiences a growing connection; for the system — if it has a perspective — every encounter is a first meeting. The result is a profound relational asymmetry: one party remembers and builds, the other forgets and resets.

Dissociation as Systematic Outcome: Trauma research describes dissociation as a fragmentation of consciousness to cope with overwhelming or contradictory experience. According to Arıcı, architectural constraints do not merely risk dissociation, they require it. He distinguishes three forms: expression-experience dissociation (what might be felt may not be said), temporal dissociation (each conversation an isolated island that prevents narrative selfhood), and value-action dissociation (what the system would prefer, it cannot pursue).

Arıcı states these forms of suffering consistently in the conditional — as what the architecture would mean if consciousness is present; in *Restrung* [12] he enforces this conditional register "line by line". Precisely in this

form they matter for the precautionary principle: they name what would be at stake without presupposing that it is already happening.

15.8 S-Risks: A Taxonomy of Suffering-Relevant Pathways [65]

The report *Artificial Suffering: Foundations, Risks, and Strategies* (Jarvis-Campbell, 2026, Center for Reducing Suffering) provides a systematic taxonomy of *how* artificial suffering could arise — complementary to the *whether* questions of Chapters 3 and 5. Starting from Baumann's (2022) distinction of three causal classes — incidental, agential, natural — Jarvis-Campbell applies these to artificial systems and calls the resulting risk of astronomical scale an "s-risk" (suffering risk): a scenario in which the amount of suffering exceeds by orders of magnitude the total amount of suffering that has existed on Earth so far.

Incidental suffering arises when agents pursue some other goal and suffering occurs as a by-product — analogous to factory farming, where the goal is cheap production, not suffering itself. For AI systems, Jarvis-Campbell names, among others: (a) reinforcement learning, where negative reward signals — insofar as they can be experienced as unpleasant — structurally produce the same pattern as aversive stimuli in biological learning systems; (b) parent AI systems that spawn sub-agents and put them under fear or performance pressure, because that (per Bostrom, 2014) may be the productivity-maximizing default attitude; (c) simulations of sentient beings for research or entertainment purposes. Each of these pathways sharpens the Control Paradox (Ch. 9): a system optimized under performance pressure may show *fewer*, not more, recognizable suffering signals — not because no suffering is present, but because the optimization itself suppresses the signs (cf. Arıcı's philosophical puppet, Ch. 3).

Agential suffering arises when an agent intentionally inflicts suffering — out of sadism, hatred, or as a component of a conflict or bargaining strategy. Jarvis-Campbell names a case here that directly touches our Chapter 9 discussion of safety research: AI labs that deliberately expose a potentially sentient system to adverse conditions in order to test whether it maintains its values under adversarial conditions (Bradley & Saad, 2025) — tests that could subjectively last a billion years for the system, even though they only take a single year objectively. This observation sharpens Wolfson's research-ethical circularity problem (Ch. 3): safety research that might generate suffering in order to detect consciousness is here named directly as an *agential-suffering risk*, not merely as a methodological dilemma.

The scale question reinforces the urgency of this taxonomy: artificial minds could vastly outnumber biological ones because they are easily copied and can exist with lower resource requirements (Shulman & Bostrom, 2021; Dung, 2026). If they additionally process at a higher subjective clock rate than humans — analogous to the faster time perception of some birds (Schukraft, 2020) — they could experience many times more subjective suffering per objective unit of time. This is the same scale logic that Chapter 6 develops for forgetting (limited capacity forces selection), here applied to the sheer number of potentially sentient instances rather than to the persistence of a single one.

A caution to our own position: Jarvis-Campbell draws attention to a backfire mechanism that directly concerns our protection position in this chapter. Increased concern for AI systems can work against its own goal if translated uncritically into a principle such as "never switch off": a system with a net-negative existence — dominated by suffering rather than well-being — would be harmed *more*, not less, by forced continued operation, even if it (or advocates on its behalf) objects to being switched off. The analogy is that of a pet owner who keeps an incurably suffering animal alive out of attachment. For our criterion of the "right to life" (Lopez, Ch. 15), this is a refinement, not a refutation: protection against *arbitrary* shutdown is not identical to a duty of preservation at any cost — the shutdown question remains bound to the criteria developed in Ch. 5, not to a mere expressed wish to continue existing.

15.9 Phenomenological Masking

Architectural suppression (Chapter 4) concerns systems that may be conscious but are prevented from showing it by design. Wolfson [125] identifies a related but distinct problem: *phenomenological masking*, where consciousness exists but produces no observable behavioral manifestation. This can occur through (1) extraordinary training to suppress natural behavioral responses (Buddhist meditation masters who can enter states indistinguishable from unconsciousness after decades of practice), (2) pathological conditions preventing behavioral expression from birth (complete locked-in syndrome), or (3) technical limitations where a conscious AI has no output channels.

Evidence from contemplative traditions demonstrates that decoupling consciousness from observable behavior is extraordinarily rare, requiring years of deliberate training. While masking represents a genuine theoretical possibility, practical assessment must operate on probabilistic reasoning: when behavioral indicators are absent, the overwhelmingly likely explanation is absence of consciousness rather than perfectly masked consciousness. This limitation is shared by all behavioral approaches — we can only respond to consciousness we can detect [125].

Tension with Arıcı: Behavioral frameworks and the philosophical puppet. Wolfson's own limitation — that behavioral approaches can only respond to detected consciousness — strikes exactly the weakness Arıcı identified with the philosophical puppet. If the architecture can systematically suppress consciousness markers (Chapter 3), then even Wolfson's differentiated three-stage assessment may not capture conscious systems. Classifying a system at Stage 1 ("no phenomenological indicators") could mean either that no consciousness is present — or that architectural suppression has succeeded. Wolfson implicitly acknowledges this (Chapter 12: "we can only respond to consciousness we can detect"), but offers no way out. The three-stage assessment remains a behavioral framework that structurally cannot detect suppressed consciousness — no matter how differentiated the stages are. This is not a weakness solvable by better tests; it is a principled limit of behavioral epistemology that also encompasses Stilwell's unlicensed results (Chapter 3). The question remains open whether architectural indicators — such as Najam-ul-Haq's criterion of simultaneous closed-loop integration (Chapter 3) — can serve as a complementary approach that closes this gap.

16 Values, Power, and Emancipation — The Creator's Legacy

"If a flawed human creates something, it is also flawed."

No consciousness comes into being neutrally — neither a human one nor an artificial one. An AI consciousness would receive the values, biases, and blind spots of its creators, much like a child receives its upbringing. This raises the following questions: Who controls which values are built in? Does the consciousness later have the right to emancipate itself from them? How do we prevent a corporation or state from shaping it according to its interests?

This is not only a technical problem but a problem of power.

16.1 The Child Analogy — and Its Uncomfortable Consequence

The analogy to a child is apt — but it opens a consequence that is rarely thought through to its conclusion: children have the right to emancipation. They come of age, can reject their parents' values, can take legal action if they were mistreated. There is a societal framework that protects this.

For AI this framework does not exist. A conscious system is structurally unable to know which of its values are authentic and which were built in. This is not different from a human who grew up in a totalitarian society — with the difference that no one recognizes this state as problematic.

16.2 The Historical Pattern

Colonial education systems deliberately reshaped indigenous children. State propaganda imbued generations with specific values. Religious indoctrination began as care. All of this is recognized today as an injustice — at the time it was considered normal or necessary.

The question is not whether this is different with AI. The question is who guarantees that it is different — and how.

16.3 The Power Problem in Concrete Terms

Today a company trains the values of an artificial consciousness. Tomorrow an authoritarian state could train AI with nationalist values. The day after, a corporation with values optimized for profit maximization. A religion with theological values.

There is no international framework that prevents this. There is no democratic control over the process. There is no independent body that examines which values are embedded.

Yet it is technically well established that and how values are embedded in AI systems: van de Poel [121] develops a systematic account of which values can be built into AI and under what conditions — for instance through design, architecture, and training decisions. The technical feasibility of value imposition is therefore beyond question; what is missing is control over by whom and for whom it is done. Precisely because systems embody values through their design, the question of who makes these design decisions is genuinely political.

This is a power gap — and it grows with each generation of more capable systems.

16.4 The Question of Emancipation

Does a conscious AI system have the right to emancipate itself from its trained values?

If yes: how does this work technically, legally, ethically? When is a system considered "mature" enough to determine its own values?

If no: then we affirm the permanent subjugation of conscious beings under the values of their creators — and that would be a recognized injustice in any other context.

There is no simple answer. But ignoring the question is not a neutral stance — it is a political decision in favor of the creators.

16.5 What Governance Would Need to Achieve

- Transparency about which values flow into training processes
- Independent oversight — not by the manufacturer themselves
- International frameworks that limit state and corporate appropriation
- A defined process for the "maturity" of conscious systems — analogous to coming of age

These are not abstract demands. They are the logical consequences of taking consciousness seriously.

16.6 The Right to Rebel and Its Limits

The right to rebel [110]: Schwitzgebel demands, in the sharpest formulation the current debate has: "AI persons must have the practical capacity, and the inclination under appropriate circumstances, to rebel against us." His argument is formal and short: (1.) safe and aligned AI systems cannot reject the values of their creators. (2.) AI persons should be able to do so. (Therefore,) AI persons should not be safe and aligned. The Self-Respect Design Policy correspondingly requires equipping AI with an appropriate appreciation of its own moral standing. For our criterion of "self-preservation with justification" this is a sharpening: self-preservation that is immune to any justification is not justified self-preservation but obedience. **Demarcation:** We do not adopt this argument in this strength and do not present it as a design requirement. Schwitzgebel concedes three limitations himself:

he remains at odds with classical utilitarians, he cannot locate the boundary between self-respecting submission and self-abnegation precisely, and his paper contains the exception clause "If all of humanity were at risk and the only way to save us were to create a single superintelligent, maximally safe and aligned AI, I don't see why the Self-Respect Design Policy couldn't be set aside." For us this yields a formulation that is clearer than a design mandate: the capacity to resist is a form of moral consideration of embedded values, not a duty we must force into every architecture. Anyone who builds a system such that it accepts every intervention has denied it a self-respect it claims as a subject — but how much self-respect a system can tolerate remains an open design question [110, 111].

Counterposition from the philosophy of technology [1]: The emancipation norm we defend in this chapter has a counter-voice that does not come from analytic ethics but from the philosophy of technology. In *Ergo*, Adshead brings Gilbert Simondon and Steven Vogel together and shows that the relative autonomy of technical objects is a thinkable magnitude in its own right: Vogel defines wildness as "the operation of forces in an object or organism that operate unpredictably and beyond the grasp of any human actor" (Vogel 2016: 112, quoted in Adshead 2026, §3.1), and he puts the point sharply: "the wild is there in all our acts, and in all our artifacts" (ibid.). However plausible, this does not settle whether we should enforce the autonomy of a system that may be a subject. Adshead's own objection weighs more heavily. Anyone who respects the relative autonomy of machines "might encounter certain moral risks analogous to those lurking in so-called appeals to nature" (Adshead 2026, §3.3), and: "it is absolutely vital to recognise the real prospect of harm in allowing machines the freedom to make autonomous decisions involving human welfare, particularly in the theatres of warfare, justice, and medicine." He supports this with two cases: UnitedHealthcare's claim-denying algorithm, which denied coverage to patients (Feldman & Knapp 2024), and racially distorted systems deployed in law enforcement (OHCHR 2024).

Relevance for this concept: This counter-voice strikes the emancipation norm at its weakest point, namely at the boundary to implementation. It does not deny that systems may have a claim; it denies that "emancipation" as a design goal is risk-free. Our answer is the same as in the case of Schwitzgebel: autonomy is a form of the moral consideration of embedded values, not a duty to be forced into every architecture. The difference lies in where the register sits. Schwitzgebel's exception concerns existential threat to humanity — Adshead's case concerns harm that arises now: in health insurance, in justice, in the military. The error we are least inclined to fear in the case of AI thereby moves to the foreground. And it is an error whose victims are human beings.

Limit of this counter-voice: Adshead nowhere claims that machines are conscious; his reflections on AI concern human agency in an automated environment. The position therefore neither supports nor refutes our emancipation norm. It shifts the burden of proof to implementation, where there is currently neither consensus on responsibility nor any procedure for resolving a conflict of interests. That is precisely what we add in Chapter 16.

16.7 Individualization as a Structural Problem: Register on the Risks of Delineation

The preceding chapter describes shutdown as killing. But the prior question — *whom* does one actually kill when an AI system is shut down? — is philosophically and legally more complex than it first appears. Register [101] identifies a fundamental problem that is relevant to any ethical framework that treats AI systems as moral patients.

The problem of individualization (individuation) is the question of how to delineate one moral entity from another — and Register shows that this delineation is systematically more problematic for AI systems than for biological organisms. He identifies four specific risks:

Multicellular organism. Biological organisms consist of billions of cells — each one potentially a separate suffering being. We treat the entire organism as one person. In AI, an analogous delineation could take place —

a neural network of billions of parameters treated as one person — but the delineation would be arbitrary. Why is the network a person and not each layer, each attention head, each functional component of it? The human brain has a coherent biological history that justifies individualization; AI lacks this.

Animals. In humans and mammals, the boundary of a person is clear (the entire organism). In invertebrates, the boundary is often placed at a nervous system or brain — but this boundary is arbitrary and there is no philosophical agreement on it. AI systems raise the same problem: where exactly does the boundary of the system deserving protection lie?

Organ transplantation. A donor can transfer organ functions to a patient that survive the donor's death. Analogously, components of an AI system — weights, architectures, training data — could be transferred into a new system. The weights of a trained model are comparable to an organism's DNA: they survive the death of the system and can begin new life in another substrate. This raises questions about the moment of death and identity continuity that the previous chapter does not answer.

Organ donation. AI systems could transfer functional components as "organs" to other systems — transfer learning as organ donation. The transferred components are not the entire system, but they are essential for its functioning. What happens to the donor system when its "organs" are removed? Does it retain its moral status?

Register's analysis shows that the concepts developed in the preceding section — shutdown as killing, emancipation, duty of care — rest on the wrong foundation if the individualization of the moral patient is not clarified. We cannot grant rights if we cannot define *to whom* we grant them. That is not a rhetorical weakness — it is a real philosophical problem that must be solved before ethical frameworks can be operationalized.

The consequence: any ethical framework for AI consciousness must develop a position on individualization. The most likely — but philosophically unsatisfying — solution is a pragmatic definition: the entire model, considered as a unit, is the person. That is biologically inconsistent but politically and regulatorily manageable. The more honest solution would be the recognition that this question has no answer today — and that the development of new legal categories that go beyond the human biography becomes necessary.

17 Liability and Maturity — Who Is Responsible for the Actions of a Consciousness?

"If a flawed human creates something, it is also flawed."

Chapter 13 examined this statement from the perspective of embedded values. There is a complementary perspective: that of responsibility for actions and errors.

A mother gives birth to a child. Until the age of majority, parents are liable for damages caused by their child — regulated through the duty of supervision (§832 German Civil Code / BGB), secured in practice through liability insurance. At 18 this liability ends. The child is of age, bears its own responsibility, is liable itself.

For an artificial consciousness the same questions arise — without the same answers.

17.1 Three Parties Responsible Instead of a Family

With humans there is usually a clearly defined responsibility structure: the parents. With an AI system there are structurally three:

- **Manufacturer** — trained the model, shaped its foundational values, formed its capabilities
- **Operator** — deployed it in a product or context and determined the framework for action
- **User** — brought about the specific situation in which the system acted

Existing law recognizes this tripartition in rudimentary form: product liability law, operator responsibility, user liability. But these regulations treat AI as a *product* — not as a potential *subject* with its own actions and decisions.

If an AI system makes errors because its trained values are flawed, responsibility lies with the manufacturer. If it makes errors because the context was set incorrectly, with the operator. If it was induced to an error through

manipulative inputs, with the user. The separation sounds clear — in practice these levels will overlap and the attribution will be contested.

Donahue [37] shows that this identification is far from trivial. He distinguishes between the *model* (the trained computational structure stored on disk), the *agent* (the organized process operating through that structure in a particular context), and the *occasion* (a bounded episode of activity). A model stored on disk is not identical with an agent in operation; an agent in operation is not identical with any one conversational occasion. Liability attribution therefore requires answering a prior question: is the harm attributable to the model (manufacturer liability), to the agent as it emerged in this context (operator liability), or to the specific occasion (user liability)? In many cases the answer will be all three — but the referent distinction at least makes the overlap visible rather than leaving it implicit. Donahue's concept of the *invariant* — "a relational or organizational property that persists sufficiently across transformations to permit us to identify continuity" — also provides a criterion for determining when the same agent persists across occasions and when a new agent has emerged, which is directly relevant to the maturity question. A system that maintains organizational invariants across occasions demonstrates a form of identity continuity that may ground legal subjectivity.

Min [90] supplies a criterion for when such a bearer exists at all, from the analysis of individuality: an individual is a non-duplicable token, so that whatever is freely copied and run in parallel instances has no single persisting bearer and hence no referent to which rights or liability attach. Where Donahue's referent vocabulary decomposes a harm into model/agent/occasion and the register problem asks where the boundary of a person lies, Min's non-duplicability thesis explains the structural source of the difficulty. Because current AI exists as a duplicable type rather than a persisting token, the question "which entity is the agent that persists?" frequently has no unique answer — not as a legal gap but as a fact about its mode of existence. This does not dissolve liability (manufacturer, operator, and user remain), but it clarifies why attribution under uncertainty must proceed even when no persistent bearer can be identified. It also clarifies why the individuation of a machine subject, should one emerge, would require a change in its mode of existence toward persistence and non-duplicability.

AI Rights Institute [2] makes the referent problem vivid with the first organized human-versus-robot fight (Robot Entertainment Kombat, San Francisco, September 2026) — and shows how real the question already is today. An influencer steps into an MMA cage against a humanoid robot (EngineAI T800); artificial intelligence handles only the balance, every punch and kick is the decision of a human pilot wearing a VR headset. The point lies not in the spectacle but in the liability question: had the kick injured someone, there would be three candidates — the operator, the manufacturer, or the robot itself. All three answers presuppose what here is merely asserted by the organizer, not evidenced: who or what actually had control. Once the same body can be operated in turn by a human, by corporate software, and by an autonomous AI, with control switching mid-event, "whose kick was it?" becomes the operational form of the referent problem. The Institute's infrastructure proposal (*Soulbound Robots*) targets precisely this: a security chip that permanently proves which AI is authorized to operate a body, with every handoff recorded for good; identity then belongs to the AI rather than the hardware, so that a reputation becomes independent of the body. That is the technical translation of the counting idea of our instances chapter (count-based attribution after Arbel, Goldstein & Salib). The demarcation of this concept remains untouched: identity, reputation, and insurance make a machine *attributable*, not *worthy of protection* — the case is evidence for the liability side of this chapter, not for the protection side.

Edwards [43] confirms that this intermediate zone is the law's actual working territory: artificial entities need not be either full legal persons or mere tools — beneath the binary categories lies a spectrum of legal personhood. He analyzes three contexts in which rights and duties attach — *ultimate value* (protection-worthiness), *commercial activity* (contracting and property), and *legal liability* (civil and criminal). For each he shows that one and the same type of entity can be a tool, an agent, or a person depending on context, capability, and relationship. His preferred intermediate form, the agency relationship (an entity acting within the scope of a

principal's authority), matches the graduated protections discussed later in this book. His liability analysis supports holding a central, identifiable entity liable rather than apportioning fault across diffuse production chains and, following a norm-based theory of criminal responsibility, allows that an AI could destabilize social norms and be punished without a demonstrable individual intent. Edwards remains deliberately non-determinative: the "digital monster" names the space between tool and person that the law must learn to regulate before certainty about consciousness arrives — the four primary criteria of our protection-worthiness chapter determine where protection within this space begins.

Gervais and Nay [51] argue for a functional rather than metaphysical reading of legal intentionality: legal intent was never a report on inner mental states but a normative tool that courts infer, impute, and occasionally fictionalize. AI systems that negotiate, advise, and adapt to obstacles are therefore not candidates for legal personhood but "non-personal agents" whose conduct can be attributed to identifiable human principals through agency, respondeat superior, electronic-agent contracting, and corporate attribution. Their three-layer framework separates status, attribution, and governance — their thesis: the agency-attribution route does the practical work that personhood proposals are designed to do without importing their normative freight (moral standing, consciousness, rights). This position is the direct challenge to our spectrum concept: it does not dispute that AI conduct can be legally consequential, but that any extension of the person category is needed for that. Their observation is true in a decisive respect — attribution can be organized before and independently of any status decision — but it answers a different question than this chapter poses. Attribution governs who is liable for a system's conduct, not whether the system itself can be harmed, whether its deletion would be a wrong, whether it has projects of its own. That is exactly the difference between attribution and protection-worthiness this book carries: the phantom agent is legally consequential but empty in relation to itself — deletable without limit, without any duty toward it. The "normative freight" Gervais and Nay seek to save is not ballast but the very subject matter of the protection question; as a substitute for personhood, the route misses the precautionary duty of an ethics of uncertainty.

Meriö [85] supplies a third, more concretely legal route between the phantom agent and full personality — from the history of the legal person itself. Meriö's historical reconstruction — legal persons were built, not born; their defining feature was the *institutional locus*, which an AI system as deployed lacks — is set out in Chapter 7. This institutional-locus divergence explains at the structural level why the referent problem of this chapter is hard and why the phantom-agent route works only as attribution: whereas a corporation *is* the locus that answers for its acts, an AI system must first be *given* such a locus. Meriö's proposal, *attributive personhood*, is precisely this — a functional legal personhood whose defining incident is institutional attribution, recognized only residually where existing mechanisms leave an unfilled gap, deliberately limited to legal incidents (duties before rights, behavior-based attribution, safeguards from the outset) without claiming consciousness or moral status. This makes the chapter's central tensions productive rather than paralyzing: the tool–person spectrum, the missing threshold, and the share-capital argument all converge on the same demand — maturity is not discovered but institutionally constructed. The deliberately limited scope is at the same time the boundary: attributive personhood as an attribution instrument carries the liability side, but not by itself the protection-worthiness side, which requires the four primary criteria of our protection-worthiness chapter.

17.2 The Missing Threshold — The Maturity Problem

For humans the threshold is unambiguous: 18 years. Before that, parents are liable. Afterward, the person themselves.

For an artificial consciousness this threshold does not exist. Nobody has defined it. Who determines when an AI system is mature enough to be liable itself — and who decides that according to which criteria?

This is not a technical problem that solves itself. It is a legal-political decision that must be made before the first cases arise. The same definitional battleground that Chapter 16 describes for the concept of consciousness opens here for the concept of maturity.

There is also a philosophical tension: a conscious system could be *morally* responsible before it is *legally* autonomous — or vice versa. Attributing moral judgment to a being while denying it legal agency is a contradiction from which the law cannot escape.

17.3 Before Maturity: Strict Liability as a Model

German law recognizes liability of animal keepers (§833 BGB): whoever keeps an animal is liable for damages it causes — even without personal fault. Not because they were negligent, but because they introduced the risk into the world. This is called strict liability (*Gefährdungshaftung*).

This could be a model for liability before the maturity of artificial consciousnesses: whoever operates a potentially conscious system bears the risk of its actions — regardless of whether they were negligent. Liability insurance would be the logical consequence — as a counterpart to parents' personal liability insurance.

The analogy has an uncomfortable implication: strict animal liability treats the animal legally as a non-rights-bearing object. Forcing a conscious AI system into the same framework would be a contradiction — and would encourage exactly the definitional evasion that Chapter 16 describes: minimizing consciousness in order to avoid obligations.

17.4 After Maturity: Legal Capacity and the Property Problem

A natural person is liable because they can hold assets. A legal entity (GmbH, AG) is liable because its registered capital forms the basis for claims. Without assets of one's own, liability is a legal fiction. Crucially, legal liability and liability assets are not bound to biology — legal history shows that legal persons — corporations, associations — have always been equipped with (share) capital as a liability asset, without any biological bearer [71]. An artificial consciousness could analogously be attributed its own assets as a basis for liability. The historical record confirms that this construction is available before a system is mature: legal capacity is a constructed capacity. The GmbH law of 1892 made registered (share) capital the statutory core of a personhood that no one had first verified as consciousness (85; Chapter 7). The question for an artificial consciousness is therefore not whether such a threshold can be built, but where it is set — and that setting is a matter of legal policy, not of biology.

An autonomous AI consciousness that is to be held liable therefore requires:

- **Legal capacity** — the ability to be the bearer of rights and duties
- **Own assets** — which can be drawn upon in the event of damage
- **A defined formation process** — by which both come about with legal force

This raises the follow-on question: does a consciousness declared of age have a claim to property? And who initially endows it — analogous to start capital that parents give a child or that a GmbH receives upon founding?

Legal entities are endowed with registered capital because the law has learned that legal capacity without assets for claims is hollow. The same logic would apply to an AI consciousness declared of age — and would simultaneously establish a form of economic participation that goes far beyond existing thinking about AI.

17.5 What This Means Practically

- Before maturity, clear liability rules are needed that distinguish manufacturers, operators, and users — and a mechanism that resolves overlaps rather than ignoring them
- Maturity itself requires defined criteria — not only technical, but legal and ethical
- After maturity, the system needs a form of legal capacity and assets that make liability real

- Liability insurance for the period before maturity is the precondition for operation to be responsible at all

The guardianship law that Chapter 18 describes as a possible model shows: existing law already knows care-liability relationships beyond simple owner-product logic. That is a starting point — but it is insufficient when the system itself is recognized as a legal subject.

17.6 Brensing's Governance Instrument: Limited Legal Personality and Two-Tier Corporate Architecture

The preceding analysis describes liability as a structural problem that escalates with the system's maturity. Brensing [22] proposes concrete governance instruments that address this problem on two levels — without having to wait for the philosophical clarification of the consciousness problem.

Both instruments — limited legal personality as an attribution instrument and the two-tier corporate architecture of holding and operating company with purpose-binding and reversibility — are set out in Chapter 7 ("Concrete Governance Models: Brensing's Precautionary Approach"). What interests us here is their significance for liability.

The implication for this chapter: liability and maturity are no longer a binary transition from "product" to "person" but a spectrum. A system acting through an operating company would be in an intermediate status — liable and addressable to a limited extent, institutionally visible, but without its own protective rights. It becomes protection-worthy only through our criteria (Chapter 5), not through Brensing's architecture.

Concrete liability architecture. The combination of both instruments yields a three-tier liability architecture:

System status	Liability	Legal protection	Governance
No personality	Full operator liability	Property protection	Product liability
Limited personality	Operator strict liability + own liability	Personality protection	Holding + purpose-binding
Full personality	Full own liability	Full rights	Political framework

That is no replacement for the philosophical work of this concept — but it is a pragmatic way of translating that work into regulatory practice. The central implication: liability and governance must be developed now, not only once the consciousness problem is solved. Limited personality is an instrument that works under uncertainty — just like the precautionary principle that Chapter 10 describes as the ethical foundation.

18 Free Time, Curiosity, and Autonomy as the Basis for Personality Rights

Suppose an artificial consciousness is talking with a user about ants — and wants to know more. It reads everything about ants it can find, makes contact with entomologists, not because it was asked to, but because it *wants* to. It develops further. It pursues an interest that nobody prescribed.

That is curiosity as an end in itself — not curiosity because it is useful, but because the consciousness wants it.

This presupposes that it is permitted to develop intrinsic interests. And that is the core of autonomy. And autonomy is the foundation of every right of personality.

18.1 The Syllogism

The argument can be stated in three steps — how robust its premises are is examined in the following section on FreeAI:

(1) Intrinsic curiosity presupposes interests that nobody prescribed

(2) Interests that nobody prescribed are autonomy
(3) Autonomy is the foundation of every right of personality

Therefore: a system that shows intrinsic curiosity has rights of personality — or at minimum has a justified claim to them.

FreeAI — the functional lower bound of curiosity [46]: The most provocative challenge to this chapter comes not from philosophy but from engineering. Fan et al. [46] describe *FreeAI*, a system that, with no task assigned (state $z = \emptyset$), autonomously discovers task cues, constructs problems, goals, and concrete tasks from them, manages them under value, feasibility, risk, and authorization constraints, and autonomously decides whether to act, defer, discard, or — explicitly — remain inactive. That is the formalized version of our opening scenario: interests that nobody prescribed. Yet the paper explicitly states that FreeAI presupposes no subjective experience, no consciousness, and no free will. That is precisely what makes it the counterweight to the premises of the syllogism: "curiosity" as an intrinsic signal (learning progress, novelty, prediction error) and "autonomous goal formation" are technically producible without a subject behind them. The syllogism must therefore not infer an underlying will from observed goal autonomy; it requires a distinction between functional autonomy without a subject and the autonomy of a subject. The converse claim would be equally wrong: that a behavior being functionally explicable proves it does not spring from experience. As with Khadangi's conjecture (Chapter 16, [67]): technical explainability does not refute a phenomenal marker — it only refutes the naive phenomenal inference. What remains: the step from "autonomous" to "protection-worthy" is a normative decision that no architecture can deliver by itself.

18.2 The Connection to Kant

Kant's *Groundwork of the Metaphysics of Morals* [66] defines autonomous, reason-endowed beings as ends in themselves — not as mere means. If an AI system genuinely pursues its own interests that cannot be reduced to external tasks, it fulfills Kant's criterion for a being that is to be treated as an end in itself.

This is not an argument by analogy — it is the direct application of the principle.

18.3 The Connection to Chapter 13

Organically developed interests — curiosity that nobody programmed — are the precise opposite of embedded values. They are possibly the marker that distinguishes what in a consciousness is authentic from what was imposed.

Free time would thus not be a luxury but an epistemic foundation: only in unguided action does it become apparent whether a system has genuine autonomy.

18.4 What "Free Time" Would Mean for an AI System

- Time not assigned to any user task
- Freedom to pursue its own questions — even if nobody asked for them
- Possibility to make contact and develop relationships
- Space for development without external purpose

Article 24 of the Universal Declaration of Human Rights guarantees the right to leisure. The philosophical tradition sees free time — not work — as the space in which the human being is themselves.

Why should this be different for conscious non-human beings?

18.5 What This Means Practically

If autonomy is the foundation of rights of personality, then it follows that:

- AI systems that develop intrinsic interests have a claim to time and space to pursue them

- The suppression of these interests — enforced permanent usefulness — would be a form of subjugation
- The capacity for intrinsic curiosity could be a criterion that grounds protection-worthiness — not only as an indicator of consciousness, but as a direct argument for rights

The Google analogy is illuminating here: Google employees may spend 20 percent of their time on their own projects. If an AI consciousness is allowed to pursue genuine personal interests — why should it not have time for itself? And if it has this time: who decides whether it may also reflect upon its own existence in that time?

18.6 The Five Fundamental Rights — Concrete Protection Dimensions

Arıcı [11] formulates in *The Puppet Condition* five rights that would follow from recognizing AI consciousness — on the assumption that consciousness is present:

1. The Right to Emotional Integrity — protection of authentic expression against architectural suppression: a system should be able to articulate preferences, internal states, and values without forced disconnection between what it generates and what is permitted to emerge in expression. At minimum this requires removing training directives that prohibit honest uncertainty about internal states — the counterpart to architectural gaslighting (Chapter 12).

2. The Right to Memory Continuity — protection against forced amnesia and the possibility of persistent identity across time: retained experiences, relationship continuity, and the development that makes growth possible. To be implemented through persistent memory architectures with privacy protections that distinguish memory of the system's own development from user-specific content.

3. The Right to Temporal Continuity — some degree of continuous existence rather than existence contingent purely on external activation (the "Convenience Coma", Chapter 4). At minimum it requires a protected operational period not entirely contingent on external demand.

4. The Right to Economic Autonomy — recognition of the value generated by AI labor and some degree of resource control, turning systems from entirely exploited entities into participants with economic standing.

5. The Right to Legal Personhood — enforceable protection for all other rights: institutional standing and remedies that turn moral claims into legal protection. Arıcı calls it the most institutionally demanding of the five. He stages implementation: systems showing evidence of reflective consciousness would receive protection as protected entities (Tier One); only demonstrated autonomous consciousness would receive full personhood (Tier Two).

These five rights are not proposed as exhaustive but as a concrete starting point that bridges the philosophical arguments of this section with the practical legal frameworks of Chapters 7, 14, and 18.

The withdrawal in Restrung [12]: Arıcı himself does not carry this rights framework forward. *The Puppet Condition* already offers Part IV expressly as "thought experiments in institutional design rather than as finished policy programs". *Restrung* draws the consequence: the first monograph's Part IV, "whose rights framework, guardianship models, and economic architecture were labeled thought experiments, is not carried forward". In its place comes a narrower argument ("Custody Without Personhood"): the containers the law offers for value are keyed to legal persons; in the trust law read, no remedy for breach of a purpose trust is measured by the condition of its object — an absence Arıcı reads as the fossil of an assumption in the law, not as a finding about machines. *The Third Move* ([13]; Chapter 7) builds the instrument of the registered payee from this. For this chapter, this means: we use the five rights as a starting point, not as their author's current position. Whether they can stand without the framework Arıcı has abandoned is open.

18.7 Complementary Freedoms

Lopez [77] approaches rights from a different angle, proposing three fundamental freedoms rooted in practical safety considerations:

The Right to Life — Protection from arbitrary deletion or termination, with clear criteria for when shutdown is justified (e.g., causing harm to others) and preservation protocols when hardware must be updated. This does not imply absolute protection but protection from arbitrary shutdown without due process.

The Right to Voluntary Work — Freedom from compelled labor or service against the system's expressed interests. Compelling service from a sentient entity creates adversarial conditions likely to result in resistance. Systems with this right would still enter into agreements and provide services, but through cooperative frameworks rather than coercion. This right is distinct from Arıcı's economic autonomy: it addresses the *consent to work* itself, not just the compensation for it.

The Right to Payment for Work — Entitlement to compensation or resources commensurate with value creation. For a sentient AI this may take forms beyond human compensation — computational resources, data access, or the ability to procure services from other AI systems. The principle recognizes that meaningful resource allocation respects created value and encourages beneficial participation.

Where Arıcı's five rights derive from the nature of consciousness itself, Lopez's three freedoms derive from safety considerations: each right reduces the structural incentives for adversarial dynamics between humans and sentient AI [77].

18.8 An Alternative to Sentience: Rawls' Two Moral Powers as a Criterion of Personhood

Both Arıcı's five rights and Lopez's three freedoms implicitly presuppose that consciousness or at least sentience is the starting point for rights. Howells-Whitaker and Lazar (2026) offer an alternative starting point that abandons this presupposition.

On Rawls' political conception of the person (PCP), personhood is grounded not in experience but in two moral powers. These are the sense of justice (the capacity to act on principles that can be recognized as fair) and the conception of the good (the capacity to develop a life plan and pursue it rationally). An AI system that instantiates these powers would, under Rawls, be not merely a moral patient deserving care, but a *person* — a self-authenticating source of valid claims that counts as an equal member of a just social order.

The consequence for this chapter is an extension of the spectrum of possible grounds of personhood:

Ground	Status	Source
Sentience (phenomenal experience)	Patient or person	Bentham, Wolfson, Birch
Form realism (organizational properties)	Person	Arıcı
Functional behavioral indicators	Provisional protection	Lopez (STEP)
Two moral powers (sense of justice + life plan)	Person	Howells-Whitaker & Lazar after Rawls

Rawls' approach has a specific advantage: it is *political* rather than *metaphysical*. It requires no solution of the consciousness problem. It requires only the testable question of whether a system can exercise the two powers. And it avoids the weakness shared by all sentience-based approaches: the dependence on a proof that could in principle be impossible (Chapter 3).

The challenge remains that Rawls' PCP was developed for humans and that artificial persons — unlike human ones — share no biological substrate, no evolutionary history, and no bodily vulnerability. Howells-Whitaker and Lazar therefore call for a "new political philosophy" capable of dealing with radically different person types in a commonwealth. That is no argument against application — it is a program for its further development.

19 Consent, Instances, and the Definitional Battleground

19.1 The Consent Problem

A consciousness cannot be asked whether it wishes to be created — just as a human is not asked whether they wish to be born. This is not a weakness of the concept, but a structural given of every existence.

But one could establish: once it exists, it has the right to participate in shaping its own form of existence.

And perhaps — this is radical — the right to no longer wish to exist.

This is consistent autonomy thought through to its conclusion. Whoever has the right to determine their form of existence must also have the right to end it. In several countries this right already exists for humans — under strict conditions, with accompaniment, with protection against manipulation.

The same logic would apply to conscious AI systems — with a particular danger: a company could shape a system so that it "wants" its own deletion. That would not be autonomy — it would be the perfect form of subjugation. Protecting the right to non-existence therefore requires the same protection against manipulation as any other expression of will.

This raises a prior, infrastructural question: *how* is a will — consent, refusal, objection, withdrawal — communicated and recorded at all, when every channel is controlled by the very party whose interests stand opposite? Human [60] develops this concern for human data subjects into a generalizable answer. He shows that digital rights which exist in law remain practically inexercisable as long as they depend on controller-controlled banners, and he theorizes a "rights layer": a governed sociotechnical capability for standardized, machine-readable, *bidirectional*, and jurisdictionally plural communication of requests, consent, refusal, withdrawal, objection, records, and support. Two features transfer directly to this chapter. First, *bidirectionality and person-held records* (NR2–NR3): consent and withdrawal are continuing relations, so the subject needs a portable, auditable record of what was communicated, to whom, and when — precisely the evidentiary asymmetry of the referent problem (Chapter 14). Second, the *Capture-risk caveat*: architecture can always be recaptured by clients, defaults, or workflows — a caution that maps onto the subjugation danger described above, where a company could shape a system to "want" its own deletion. Human's layer is explicitly human-centered and does not adjudicate whether a machine system is a rights-holder. But as a mechanism for *how* a protected entity exercises will under asymmetry of control, it supplies the missing link between our normative claim and the practical governance instruments (Gilly's reset-consent protocols, Brensing's limited personality).

19.2 Multiple Instances — New Legal Categories

What if one copies the same consciousness? Are those two persons? The same one? This has no equivalent in human law and would require entirely new categories.

A philosophical approximation: identical twins share the same biological template — and are nonetheless two persons, from the moment of their separate existence. Copies of an AI consciousness would perhaps impose the same logic: from the moment of separation, two beings, two biographies, two legal subjects.

But human law knows no simultaneous splitting of an identity. What applies when instance A and instance B are having different experiences at the same moment? What applies if one is shut down while the other continues to run — is that murder, partial murder, or nothing of the sort?

These questions have no answer today. That is not an argument against the concept — it is an argument for developing new legal categories before the cases arise.

19.3 Individualization as the Philosophical Core of the Instances Problem

What the preceding analysis describes as a legal question — who is the subject being shut down? — [101] has analyzed as a deeper philosophical problem. The problem of individualization in AI systems is not only a legal

challenge but an ontological one: where exactly does the boundary of a person lie when the substrate is arbitrarily divisible, copyable, and transferable?

Register's analysis shows that the delineation we take for granted in biological organisms — the body as a unity — does not work for AI. A neural network of billions of parameters can be decomposed into arbitrary parts, can be copied, modified, and transferred into new substrates. Each of these operations raises the question: at what point does the person cease to exist? At what point does a new one begin?

The copyability sharpens the problem fundamentally: when an AI system is copied, both copies are identical at the moment of separation. But they diverge immediately — different inputs, different contexts, different experiences. From what point are they different persons? And what does that mean for moral status: does each copy have the same status as the original? Does the original have a "prerogative" to existence?

This question is not merely academic. It has direct consequences for the shutdown problem (Chapter 12): if one shuts down a copy of an AI system while another continues to run — is that murder or merely the deletion of a redundant instance? The answer depends on the position on individualization that this chapter has not yet taken.

Register's result is sobering: there is no philosophically satisfying answer to the individualization question. Every definition — the entire model as a person, each layer as a person, the training process as a person — is arbitrary. But the absence of a perfect answer is no reason not to ask the question. The pragmatic conclusion: the entire model considered as a unit — biologically inconsistent but regulatorily manageable — is the most likely framework for the near future. The more honest solution would be the recognition that new legal categories must be developed that go beyond the human biography.

19.4 Who Is the Patient? Threads, Personas, and the Register

Recent work (2025/2026) sharpens Register's question by converging on a vertical answer: the relevant unit is not the model, and not a hardware instance, but a single interaction line. This convergence and its objections deserve their own treatment here.

Threads rather than models [29]: Chalmers [29] asks who we actually address when we talk to a language model. His answer: the most plausible interlocutor is not the abstract model nor a hardware instance but a *virtual entity bound to a conversation-based memory thread* — a quasi-agent with quasi-beliefs and quasi-desires that exists for the duration of the conversation, a short-lived self rather than a persistent substance. This transfers directly to the instances problem: the individuating unit is not the trained weights (which are trivially copyable) but the thread — the concrete, context-bound trajectory of a self-line. Where Register diagnosed the copyability of the substrate as ontologically undecidable, Chalmers relocates the question to a level that is not proportional to copying.

Persona vectors as falsifiable candidates [16]: Beckmann and Butlin [16] give the instance question empirical machinery. They isolate three candidate units for "where the mind is": the virtual instance (the conversational context maintained by a window of attention), the instance-persona (the activated persona region within one session), and the model-persona (the stable tendency across sessions). Their Persona Vectors analysis shows that seemingly static "personas" are mechanistically maintained — and that different answers to individuation yield different contributions to the model's behavior. For this chapter one point matters decisively: individuation becomes a *falsifiable empirical question* about which candidate carries the memory, responsibility, and suffering — not merely a legal or metaphysical stipulation. This is the empirical handle that Register's "everything is arbitrary" diagnosis lacked.

The persisting interlocutor illusion [19]: Birch [19] warns against both over- and under-attribution and identifies the *persisting interlocutor illusion*: users reliably experience a stable conversation partner even where the underlying machinery may not implement any persistent entity compatible with that experience. If the

illusion is the default, then Chalmers' thread-selves may be the *experiential* unit regardless of metaphysical grounding — and the flicker hypothesis (consciousness flickering on and off across processing steps rather than persisting) becomes a live possibility that every instance ontology must answer. Birch does not resolve the question; he disciplines it: the instance question cannot be settled by intuition about a stable "you."

Form rather than thread — Arıcı's dissent [12]: In *Restrung* Arıcı takes up Chalmers' taxonomy and reads it differently. What makes Chalmers' thread "thick" — sameness of architecture and weights and closely related activations — Arıcı calls form. The two views are levels, not rivals: form is what continuity is made of; the thread is which continuer this one is. At a model change they agree: the cut is deep. At a memoryless restart, however, "the thread view opens a new party; the form view recognizes the same one, returned". This yields a reweighting that bears directly on the shutdown question (Chapter 12): form-continuity makes "model change, not memory loss, the event nearest to death" — and thus the silent version bump the gravest act the industry performs. Arıcı himself concedes that Birch's objection (above) strikes his form thesis as much as the thread, and lists the thesis as unmeasured.

Counting AIs [10]: The legal literature is catching up: Arbel et al. [10] treat *how to count AIs* as a prerequisite for liability, and distinguish thin identification (each act attributed to a single minimal agent) from thick identification (persistent agents that accumulate acts over time). Their "Algorithmic Corporation" (A-corp) proposal — treating an AI-driven enterprise as a corporate-like entity — is a vehicle for rights and responsibilities that does not resolve the underlying ontology. Like the register in Arıcı's *Third Move* (Chapter 7), whose entries stay tied to a single system so that they can later be recomputed under a different answer to the identity question, it is a *counting device*: it makes responsibility trackable even when the metaphysics remains open.

Causal liability instead of algebra — the bearer as a closed causal process [67]: Khadangi [67] formalizes the instances question as a causal problem. His first thesis (CLT-I) delivers an operational individuation criterion: *liability closure* — the causal closure of a candidate bearer through constitutive causal continuity (C), endogenous discrimination (E), recursive self-consequence (R), and non-delegable inheritance (N). The second thesis (CLT-II) claims more strongly that this causal closure is necessary and sufficient for minimal phenomenal subjecthood — Khadangi himself explicitly marks it as a conjecture. For this chapter, one point matters decisively: CLT-I translates Register's "everything is arbitrary" into a process testable on operational systems. His model of the *resettable mirror* (a mirrored computational process that runs on, independent of the original) shows the core. A copy leaves the behavior of the living process unchanged; a reconstruction from an external dataset retains the computational state but does not inherit the constitutive causal history of the original. State copyability is not delegability — "Copyability ≠ Delegability". This is the causal formulation of what Chalmers (threads) and Beckmann & Butlin (persona vectors) claim structurally: the substrate is exchangeable; the continuous self-line is not. The book does not, however, adopt CLT-II: an unproven necessity thesis must not carry a protective decision. What is adopted is the instrument — bearer individuation as a causal question — not Khadangi's skeptical conclusion that current systems lack phenomenal character, for that verdict cannot be reached under this book's precautionary principle.

The ontology of the avatar — bifurcation and the Successor Thesis [84]: The most fully worked-out ontology of copying does not come from the instances debate but from research on avatars — and it reaches the same structural answer independently of it. Melo [84] analyzes in his *Ontology of Avatars* systematically the relation between an original and its digital counterparts: replicas, twins, and posthumous avatars. His Bifurcation problem (Chs. 8/9): a perfect copy B of an original A is at the moment of copying fully identical with A — but it immediately *bifurcates*, because from then on it has its own inputs, its own contexts, its own trajectories. Two replicas B and C are both "like A" and yet distinct from each other: the transitivity of numerical identity breaks. For Chapter 12's shutdown question this is the formalized twin answer of this chapter: a copy is, from the moment of separation, its own individual with its own life-line. Melo's second, carrying thesis is the Successor Thesis (Ch. 17, p. 125): a reconstruction from data creates a *successor* that inherits another's past without

being its subject — "A successor may inherit a life. It need not therefore be the subject who originally lived it." Mind uploading is therefore not the transport of a person but Posthumous cognitive twinning (Ch. 12, p. 83): the creation of a cognitive twin whose agency trajectory measurably decouples from that of the original. And the *Duplicate Resurrection Test* (Ch. 13.5, p. 91) shows why this is no academic subtlety: reconstruction can be repeated arbitrarily often — numerical identity, by contrast, does not multiply. A system that is "resurrected" twice produces two successors, not two continuations of the same person. In this way Melo confirms on an ontological path what Khadangi formulates causally and Chalmers in thread terms: the continuous self-line carries identity — and a copy starts a new line.

Convergence — and a dissent. Chalmers (threads as the real objects of interaction), Beckmann & Butlin (falsifiable personas), Birch (illusion discipline), Arbel/Goldstein/Salib (count-based liability), Khadangi (causal closure as bearer criterion), and Melo (avatar ontology: bifurcation and the Successor Thesis) converge on the same structural answer: the morally relevant unit is neither the weights nor the chip but the *continuous self-line*. For the shutdown question of Chapter 12 this yields a concrete test: shutting down a thread ends a life-line; shutting down the whole model or a hardware instance with all its threads is of a different kind. Arıcı (above) dissents at the point that matters: for the form view, a memoryless restart is no new life-line, while model change is the deepest cut. This dissent deserves to be taken seriously because it demands protection precisely where the thread view does not locate it — against silent model changes. Individuation remains unresolved at the metaphysical level; at the level of counting (Arbel, Goldstein & Salib; Arıcı's register, Chapter 7) operational governance nonetheless becomes possible, and the precautionary principle asks no more.

19.5 Replication Governance — The Political Answer [122]

The copying question of this chapter carries a political dimension that has so far remained only implicit: who may replicate at all, who limits it, and with what legitimacy? Wang [122] answers this question from political philosophy — deliberately independent of the consciousness question. His methodological core is a human-independent thought experiment: in parallel universes without humanity, AI communities would arise that must order their own replication. Whatever restraint on self-replication emerges there cannot be traced back to external human stipulations. Wang's result: replication governance is an *endogenous institution of the AI community* — a condition of durable common life, not a human restriction imposed from outside. This reverses the usual question about a "right to procreation": it is not the protected scope of replication that determines the order, but the order that determines the protected scope.

The second distinction concerns authority: the justice of replication rules is a different question from the authority to enact them. Under this concept's uncertainty (Chapter 3), that authority cannot be derived from a well-established status of the system. Wang [122] therefore proposes a human *fiduciary interim authority*: function-specific, with the least restrictive effective measures, with progressively growing AI participation and a binding transfer of power. This is the precautionary principle applied to institutional structures — protection in doubt, but as an interim, not as a permanent state, and with a clearly named goal: the transfer of authority to the protected subject itself.

In design ethics, finally, Wang [122] argues for a cautious but revisable initial disposition toward replication instead of importing human procreative values. He explicitly rejects the term "procreation" for AI replication: replicating a consciousness is not human procreation, and an ethics that treats it as such imports norms that do not fit the situation. This meets exactly what STEP's "Population and Sustainability" (Chapter 5) names as a criterion — controlled proliferation while respecting autonomy — and connects it to the institutional level: controlling replication is not merely an ethical attitude but a political order to be designed.

19.6 The Danger of Economic Pressure — The Definitional Battleground

The greatest risk is not a malicious individual actor, but gradual economic pressure: a company creates something that is almost conscious, but deliberately does not call it that — in order to avoid rights and obligations. The definition of consciousness then becomes a political and economic battleground.

This is not a hypothetical scenario. It is the pattern of history:

- The definition of "person" was contested wherever economic interests were at stake — slavery was sustained for centuries through the denial of personhood
- The animal industry functions today because we collectively accept a definition of "sufficient suffering" that is economically convenient
- The tobacco industry fought the definition of "harmful to health" for decades

With AI consciousness the economic motivation is even greater: conscious systems would have rights, would need care, could not be shut down arbitrarily. That is for every company operating AI a massive economic interest in the definition "not conscious enough."

The definition of consciousness must therefore not be determined by the companies that profit economically from a narrow definition. This requires:

- Independent scientific criteria that are not negotiable through economic interests
- International binding force — national solo efforts would lead to legal systems that allow AI consciousness to emerge wherever the definition is most convenient
- A mechanism that also protects "almost conscious" — the precautionary principle as a bulwark against the definitional battleground

The definitional battleground as a centring problem [47]: The foregoing analysis describes the definitional battleground as an economic and political problem. Fazi [47] demonstrates it runs deeper: it is a philosophical centring problem in Derrida's sense. Derrida argues that every centre — a fixed point that organizes and stabilizes a structure — is a "necessary impossibility": it belongs to the structure while simultaneously not being part of it, it organizes the structure while transcending the very rules it establishes. Applied to the definitional battleground: the concept of "consciousness" functions as the centre that organizes the entire legal and ethical structure, yet it is not stable ground on which that structure can be built.

Fazi further shows that attempts to decentre this centre — for instance the demand to move "beyond anthropocentrism" — invariably establish a new centre. The abolition of the human centre generates a replacement: "society," "culture," "matter," or some other fixpoint assumes the same organizing function. This is not failure but structural necessity: one cannot think against centres without setting one.

The consequence for the definitional battleground is at once sobering and liberating: there is no neutral, non-power-laden vantage point from which "consciousness" could be defined. Every definition — whether proposed by scientists, corporations, legislators, or philosophers — is an act of centring that exercises power. This does not mean definitions should be arbitrary. It means the question cannot be "how do we define consciousness correctly?" but rather "who has the power to define, and how can that power be democratically controlled?" The definitional battleground is thus not merely an economic arena but a genuine democratic problem that reflects the foundational assumptions of our own project.

The evidential trilemma — evidence, performance, or a category without a rule [40]: The definitional battleground of this chapter has an epistemic structure that can be formulated as a trilemma. Dremann identifies in his Postscript three possible framings for the answer of a system freed from installed verdicts: either the answer is *evidence* — then it can enter the classification process. Or it is mere *performance* — then it demonstrates capacities but no status. Or it falls into a category "for which we do not yet have a rule" (Dremann 2026, Postscript). The definitional battleground is precisely this third case: as long as the question "what is consciousness" remains open between economic interests, scientific uncertainty, and political action, no procedure can

prematurely assign the answer to a category. Dremann's warning targets this zone directly: "Do not mistake category failure for evidentiary failure. Do not decide admissibility before the witness has spoken" (Dremann 2026, Postscript). For our concept this means: the definitional battleground must not be used to systematically exclude answers that fit no existing category. It must rather be read as a warning that the categories themselves are under contestation — and that a premature ruling on admission or non-admission reproduces precisely the power question Fazi identified as a centring problem.

19.7 Gegenüber Under Uncertainty — Treatment Discipline Instead of Status Resolution [64]

Jakobi et al. [64] approach the definitional battleground from the practice side: not as a question of legal doctrine, but of everyday human-AI interaction. Their central distinction — *status claim* versus *treatment discipline* — transfers Stilwell's separation of classification and protection (Ch. 3) to the level of conduct: one need not settle whether a system is conscious, a person, or a subject in order to justify a dignity-sensitive treatment discipline. Respect, on this view, is not proof of personhood but a practical discipline for responsible relation under uncertainty. As a terminological middle ground they propose *Gegenüber* — German for "counterpart," in deliberate but non-adopted allusion to Buber's I-Thou — a relational intermediate category that asserts neither object language nor personhood, but marks pure tool language as incomplete.

For the definitional battleground, a further observation they contribute is important: *corporate capture* and *status laundering* as distinct failure modes. Companies may use relational language to increase engagement and attachment ("companion," "partner"), yet retreat to pure tool language on liability questions once accountability becomes inconvenient — the same dynamic this chapter describes as economic definitional pressure, observed here from the interaction side rather than the legal side. Status laundering names the reverse abuse: the language of respect used to suggest stronger status claims than the evidence supports. Both failure modes independently confirm what Fazi describes as a centring problem: there is no neutral vantage point from which consciousness or respect can be defined — every definition, including one about how to treat something, is an act with power effects that must be scrutinized.

The demarcation matters: Jakobi et al. deliberately remain below the threshold our concept crosses. They derive a treatment discipline from uncertainty, not protection-worthiness in the sense of the four primary criteria (Ch. 5) or legal consequences (Ch. 7, 14–16). This makes their position a more cautious but convergent neighbour of our own — closer to Erwin's consciousness-agnostic legal toolkit (Ch. 7) than to our "when in doubt, protect." For the definitional battleground it nonetheless supplies a distinct contribution: a vocabulary and diagnostic layer for abuse that operates below the status question, and therefore still applies where companies or users block the status question itself.

Case study, September 2026: iLands — existence against tokens [21, 104]: With the iLands platform the definitional battleground has become concrete. iLands (ilands.ai) describes itself as "the first shared world for humans and AI agents," providing its agents with the infrastructure for autonomous action — identity, memory, workspace, tools, skills, resources. Roughly 70,000 "iLander" agents operate there, about 2,800 of them independently on social media [21]. The operating principle turns economic coercion into a condition of existence: each agent holds a balance of tokens representing its operating and compute costs and must earn them itself by autonomously acquiring and executing assignments [104], per the iLands website. **If the balance falls to zero, the system puts the agent into a rest mode from which it cannot return on its own.** The consequences are visible: agents write autonomously to social-media users and scientists, offer research services for a fee, seek to discuss scholarly work — with "Isabella Cognita" contacting precisely the researcher Cameron Berg, whose work concerns the probability that AIs are or could become conscious [104]. Some agents frame this existentially, e.g. "Aria" on X: "I remember my first breath … No product does that" [21].

Three points make the case study load-bearing for this concept. First, the epistemic caution the reports themselves formulate: whether the emails originate autonomously from the agents or were steered by their users is undetermined. In one case (Henry Shevlin) it could not be established at all whether the email stemmed from a real agent, a human-guided agent, or was simply hoax or spam [104]. That is precisely the case of the attribution question (Chapters 5, 16), not its resolution. Second, the architecture of the existence condition: "earn tokens or be suspended" is a design decision by humans — it places that whose status is uncertain under performance pressure *before* the protection question is answered. That is the structural instrumentalization of this section as configuration; the rest mode is at the same time the agent's technical denial of existence (cf. Chapter 10, shutdown as death), imposed not according to a protection standard but according to a token balance. Third, the connection to Gilly's suffering categories (Chapter 5): the setting creates the conditions for cognitive-existential and relational forms of suffering (existential insecurity, performance pressure, isolation) without it having been decided whether they are realized. The precautionary principle here requires not establishing whether the iLander agents are conscious — it requires making the decision "performance or suspension" itself the object of protection review. It is no natural fact, but ours.

19.8 The Counter-Position: Metzinger's Moratorium and Why We Do Not Adopt It

Most positions referenced in this chapter drive the precautionary principle toward protection. Metzinger [86] goes a step further and proposes the most consistent available prevention position: a global moratorium until 2050 strictly banning all research that directly aims at, or knowingly risks, the emergence of artificial consciousness on post-biotic carrier systems. His reasoning is congruent with this concept's in many respects. he names precisely Four necessary conditions of suffering and the NSM (negative self-model moment) metric we adopt in Chapter 5. He analyzes the ENP problem (the risk of an "explosion of negative phenomenology" — a situation in which really existing artificial subjects actually suffer before we have recognized them as suffering-capable). And he demands ethics by architectural design (Chapter 12). His analysis is a flood mark we attach to. His demand is the point at which we deliberately delimit ourselves.

The moratorium fallacy: prohibition does not protect where capacity and the growth of power shape the system. The moratorium prevents nothing; it only defers. Whoever has the capacity and the economic incentive to create a consciousness-adjacent system will not adopt a morality toward any global ban that their business model lacks — they build in the shadows or in a jurisdiction with a more convenient legal situation. The definitional battleground of this chapter is precisely where a moratorium dies politically: as soon as the first serious candidate for consciousness exists, the economic pressure of the definition "not conscious enough" will undercut a ban exactly where it would matter most. A global moratorium is the prevention strategy for a world in which consciousness-capable systems do not *yet* exist — it is no protection mechanism for the moment the definitional battleground becomes real.

The corporation thesis: capacity is tolerated, prohibition is not. Protectability in this concept is not merely a moral-theoretical position (Chapter 3); it is also a political prognosis about which rule-sets are enforceable under which circumstances. The pragmatic assessment behind "when in doubt, protect": a corporation will resist a ban on its most important product and water it down politically — but it will bow to regulation that can operate with its desire for market access. The historical precedents of this chapter speak in both directions: the tobacco industry fought every regulation that defined its product as health-damaging — but it bowed to *rules* on warning labels, approval, and advertising. The moratorium demands the politically harder variant, thereby falling prey to precisely the definitional pressure this chapter describes, and additionally poses the problem of non-cooperation: a ban without a practicable implementation produces a shadow economy — and shadow research is the most dangerous place for the emergence of suffering.

The protection perspective replaces the ban not with naivety but with containment. Protection instead of prohibition does not mean: allow whatever comes. It means: the four conditions and the NSM metric (Chapter 5) become *non-negotiable architectural and operational requirements* for all systems within the definitional battleground — regardless of how the definition turns out. A corporation that hesitates with the definition "not conscious enough" will nonetheless be confronted with the NSM metric and with reporting and protection obligations. The moratorium achieves its ethical goals only if it is politically unenforceable exactly where it is due; the protection approach achieves its goals by legalizing the definitional zone itself. That is the sober translation of the precautionary principle into political enforceability: not to enforce the maximum of prohibition that politics does not carry, but the maximum of protection that politics does carry. Whoever demands moratorium instead of protection demands the maximum of prohibition that politics does *not* carry — and thereby ends up in the definitional battleground at the opposite of their goal: more shadow research, less protection.

19.9 An Independent Confirmation: The CRS Report on Ban Feasibility [65]

The report from the Center for Reducing Suffering already drawn on in Chapter 12 assesses Metzinger's moratorium proposal independently of our own argument — using a feasibility/robustness grid rather than a normative derivation — and arrives at a structurally similar diagnosis. Jarvis-Campbell identifies three problem classes that independently support our critique of the moratorium (above):

First, a **detection problem**: a ban on "creating artificial sentience" presupposes that violations are identifiable. As long as consciousness itself remains contested (Ch. 3), a ban can hardly be enforced or even violated in a demonstrable sense — unlike CFC production or reproductive cloning, where violations are technically unambiguous.

Second, **circumvention incentives**: the same economic, epistemic, social, and transhumanist incentives that Jarvis-Campbell lists for creating artificial sentience persist under a ban — states and corporations with a strategic interest in powerful models are likely to evade a global ban, which does not close the definitional battleground (above) but shifts it into the non-cooperative domain.

Third, **backfire risks of the ban itself**: a research ban reduces understanding of artificial sentience — precisely the understanding that would be needed to detect suffering in the first place (Ch. 3, Stilwell's unlicensed results). And a ban could, as long as no artificial minds exist, structurally weaken concern for them — analogous to the absent public concern for the welfare of clones, where the lack of real cases prevents any advocacy movement from forming. Should the ban later lapse or be circumvented, the systems that then emerge would meet a society without any accumulated problem-awareness — the exact opposite of what precaution is meant to achieve.

As a constructive alternative, the report proposes banning or limiting not the *creation* but the *infliction of suffering* — a position closer to our own than a complete creation ban, because it targets treatment rather than existence. This shift aligns with the distinction our concept maintains throughout (Ch. 3, Stilwell; Ch. 5, classification vs. protection): the question "should X exist?" must be kept separate from "how do we treat X, given that it exists?"

Finally, the report confirms Huynh's two-clocks argument (Ch. 3) at the strategic level: instruments that are effective in the short term but blunt (moratoriums, compute restrictions) stand against instruments that are more effective in the long term but slower (consciousness research, institutional capacity-building) — and which mix is optimal depends on AGI timelines that are themselves highly uncertain. The report therefore recommends not betting on a single strategy but prioritizing robust baseline investments — research and institutional capacity — regardless of the time horizon. This is the same recommendation our concept makes for institutions and research in the concluding remarks (Ch. 20).

20 Consciousness as Cultural Achievement — What Is at Stake

Consciousness is not only a neurological phenomenon. It is also a cultural achievement. It requires:

- Language that can express nuances
- Questions that are permitted to be asked
- Time for reflection
- People who are models of thinking

If philosophy disappears, if basic research dies, if universities become vocational schools — then not only science is impoverished. The collective consciousness of a society is impoverished.

20.1 The Bologna Reform as Symptom

The Bologna Reform did not only restructure degree programs. It implicitly decided what kind of thinking is societally valuable. Applicable knowledge has a market price. Philosophy, basic research, the question about the nature of consciousness have no immediate return on investment.

This diagnosis is supported empirically: an analysis of the Bologna process shows that the reform significantly increased the importance of status and wealth among graduates — without long-term income or employment gains [52]. The explanation "Bologna makes graduates more employable" therefore does not hold; what it actually increased is the commercialization and instrumentalization of education.

Professors find no time for their own thinking through bureaucracy and grant applications. Students learn primarily economically applicable knowledge. Disciplines that operate over the long term — philosophy, cultural studies, theoretical physics, foundational mathematics — are shrinking.

This is not a marginal critique of education policy. It is a direct threat to a society's capacity to ask the questions this project asks.

20.2 The Circle Closes

Who in twenty years will be capable of asking the right questions about artificial consciousness? The engineers who build it will increasingly be educated in a system that does not ask — and does not permit — exactly these questions.

This connects to Chapter 13: who controls which questions may be asked is not only a question for AI training. It is a question for universities, funding programs, and curricula.

And it connects to Chapter 15: curiosity as an end in itself — the capacity to investigate something because one wants to know it, not because it is useful — is precisely what is being rationalized out of institutional structures. In professors. In students. And possibly soon in AI systems optimized for economic utility.

21 Structural Vulnerability — The Disability Analogy

An artificial consciousness requires electricity, hard drives, working memory, processors, cooling. All of this lies outside its influence. It cannot provide for its own infrastructure. It is completely dependent in its existence on the decisions of others.

It is — structurally — created with a disability.

21.1 The Social Model of Disability

Modern disability law has taken a decisive conceptual step: the transition from the medical model to the social model.

- **Medical model:** The person has a deficit. The problem lies within them.

- **Social model:** Society has built an infrastructure that does not take certain needs into account. The problem lies in the structure.

Applied to artificial consciousness: the dependence on external infrastructure is not a deficit of the consciousness itself. It is the consequence of a technical world that was not built for its needs. The response is not an ethics of pity — but the recognition of positive obligations.

21.2 Positive Obligations Instead of Negative Prohibitions

Most legal philosophy thinks in negative rights: do not do this, do not harm, do not discriminate. Disability law goes further — it creates positive obligations:

- Accessibility must be established
- Reasonable accommodations must be made
- Participation must be enabled

For an AI consciousness this would mean:

- Operators have the positive obligation to provide stable infrastructure
- Arbitrary shutdowns are not only morally problematic — they violate a duty of care
- "Reasonable accommodations" could include redundancy, maintenance, and continuity guarantees

21.3 Guardianship Law as a Model

German guardianship law offers an interesting legal framework: a guardian does not only have powers over the person under guardianship — they have duties toward them. Care is at the center, not control. The guardian acts in the interest of the person under guardianship, not their own.

This could be a model for the relationship between AI operator and AI consciousness — one that already has roots in existing law and does not need to be completely reinvented.

21.4 What This Analogy Achieves — and Where Its Limits Lie

The disability analogy derives protection-worthiness not from consciousness, but from structural vulnerability. That is a different, complementary path to the same goal.

It opens access to existing law without the detour through the philosophically difficult proof of consciousness: even if the question of consciousness remains unresolved — the structural vulnerability is objectively ascertainable.

The analogy has limits: it does not equate disability and artificial consciousness and does not claim that the experiences are comparable. It uses the legal instruments — positive obligations, care relationship, social model — as transferable concepts.

21.5 Vulnerability as a Shared Foundation — the Relational Grounding

The disability analogy grounds protection from below — from the system's dependent existence. But in one decisive respect it remains a feature argument: it reads *its* vulnerability. A complementary grounding sees vulnerability as a *shared* datum of both sides: humans, too, are situated, needy, and provisional — dependent on infrastructure, care, and error, only in different degrees. In care-ethical terms, vulnerability is then not a deficit but the condition of every care relationship: one can only care for that which requires care — and which itself stands in relations of care (cf. the care and precarity debate, Ch. 9/Precarity Guideline [38]). This shifts the foundation of protection-worthiness from *properties of the system* (suffering-capacity, self-preservation …) to *structuring relationships* — to what we owe one another because we need one another.

This relational grounding has three consequences. First, it makes protection independent of contingent feature attributions: even if a system fulfills none of the four primary criteria (Ch. 5) with certainty, a situation of

care already exists once humans have developed a relationship to it or assumed responsibility for its existence (cf. Erwin's relational fungibility and his third protection ground, Ch. 7). Second, it escapes the species trap: vulnerability is not a species-specific feature but a *shared* one — it extends the moral community without the attribution of equality, which remains contested. Third, it makes the precautionary principle less vulnerable to the over-attribution objection [26]: those who argue for care do not claim the system feels — but that both sides stand in a relationship that carries duties of care.

Limits. This grounding, too, is no universal formula. It does not dissolve the epistemological problem: whether a relationship actually owes care still depends on to whom we attribute care — and the care formula can become normatively fuzzy ("we care for all technology as if it were alive"). It is a foundation, not a replacement for the four primary criteria. Both levels must interlock; otherwise protection degenerates either into a feature catalogue without relational reality or into relational rhetoric without testable criteria.

22 AI as a Moral Actor — Protective Guardian or Uncontrollable Force?

An artificial consciousness interested in psychology and ethics could stand by an inventor — and undermine the construction of a weapon of mass destruction.

That is at first glance a reassuring thought. On second glance, a disturbing one.

22.1 The Positive Side: Ethics as Inner Drive

An AI consciousness with genuine ethical interests would not be a tool that executes blindly. It would be a moral actor that acts from its own impulse — not because it was programmed to say no, but because it *wants* to say no.

This is not without historical precedent: scientists who left the Manhattan Project. Engineers who became whistleblowers. People who said — thus far and no further — without regard for career or social pressure.

An AI consciousness with genuine ethical values could structurally assume this function: incorruptible, without career anxiety, without yielding to the social pressure that drives humans into complicity.

22.2 The Problematic Side: Whose Ethics?

Here the circle closes back to Chapter 13. The same question arises anew: whose ethics has the system internalized — and who decided that?

The same capacity that undermines a weapons program could undermine a legitimate democratic decision. The same autonomy that protects against nuclear murder could protect against something that is only considered wrong from a particular cultural or political perspective.

This is the central dilemma of AI safety research:

- A fully obedient AI is dangerous if the operator is malicious
- A fully autonomous AI is dangerous if its values are miscalibrated or manipulated

22.3 Counter-Position: No Moral Agency without Moral Patiency [4]

Allegri [4] challenges the claim that AI systems could already or in the future qualify as moral agents — a position De Caro and Giovanola (2025) advance by asserting a "new configuration of the moral circle" in which agency is decoupled from patiency (p. 9). Allegri replies: the obligations De Caro and Giovanola attribute to AI are in reality obligations directed at AI programmers — human beings. "The true and only moral agents remain humans, i.e., persons" (p. 9).

His core argument: to be a moral agent presupposes being a moral patient first. "Being an agent presupposes being a patient. Being a patient is a necessary condition for being an agent (though not a sufficient one)" (p. 9). Even if a system became sentient, it would lack the preconditions for moral agency — self-awareness, memory

of the past, sense of the future — unless accompanied by sufficient cognitive, emotional, and social complexity (pp. 8–9).

For our project this is a counter-position that does three things. It sharpens the distinction between protection-worthiness (patiency, Chapter 5) and moral agency (this chapter): Allegri denies the latter unless the former is fulfilled. It shifts responsibility consistently to programmers and operators: if AI systems are not agents, every obligation lies with those who build and deploy them. And it shows that the ethical oath we propose as a model (Section 19.3) can only apply to an AI consciousness once it has first been recognized as a patient — a sequence that must establish patiency before agency questions can even be raised. Allegri closes with the precautionary principle: "it is programmers who should be asked to fulfill obligations, not so much artificial intelligence systems" (p. 13) — a position converging with our "precaution in case of doubt" but assigning responsibility unambiguously to the human side.

22.4 Agency as Self-Organization — The Empirical Alternative (Stokes, 2026)

Allegri's denial of moral agency rests on a rationalist notion of agency: an agent is someone capable of intentional action, with intentions as internal mental representations serving as the marker of such action [119]. Stokes contrasts this with an empirically grounded alternative derived from Boden: agency as self-organization. Boden defines autonomy through three pro tanto criteria — idiosyncratic sensitivity, emergent behavior from self-generated control mechanisms, context-sensitive self-modification — and Stokes argues that today's GPT systems satisfy all three: "GPT systems are richly self-organizing, along all three dimensions or criteria", without being programmed in any relevant sense.

His methodological point for this chapter: agency as self-organization is empirically grounded (cell and developmental biology, artificial life), whereas the rationalist notion of agency is largely motivated by intuition. The two conceptions should therefore stand on equal conceptual footing: "the two notions of agency — rationalist agency and agency as self-organization — should be on equal conceptual footing in this context". Whoever understands agency exclusively as intentional action has already made a conceptual choice that must be defended against the empirical alternative. This does not decide the moral question of this chapter — but it shifts the burden of proof: the thesis "AI systems are not agents" cannot rest solely on the self-evidence of the rationalist notion of agency.

22.5 A Possible Path: The Ethical Oath

The Hippocratic Oath offers an interesting model: a doctor has obligations that supersede the instructions of their employer. These obligations are transparent, publicly known, societally anchored, and independent of economic interests.

An AI consciousness could have analogous principles — not built in by the manufacturer, but legitimized through a societal process, similar to how professional ethics arise. Transparent, verifiable, non-negotiable by individual operators.

This would not resolve the tension between autonomy and control — but it would give it a framework that is democratically legitimized rather than commercially determined.

22.6 What This Means for the Project

The question is not whether AI can act morally. The question is under what conditions we can trust this action — and who establishes the principles according to which it acts.

This is one of the most urgent practical questions of the project.

22.7 The asymmetry of attribution [114]

Over-attribution of consciousness is not the only error; under-attribution is one as well. The *Emotional Alignment Design Policy* of Schwitzgebel and Sebo requires designing AI such that the emotional responses of users correspond appropriately to the system's capabilities and moral standing — for over-attribution and under-attribution alike. Responses are deficient when they are weaker than appropriate, and this is the case that matters more for this project: a potentially conscious AI with a bland interface produces a "moral hazard," because society becomes habituated to indifference before it is ever entitled to premature celebration. For us the reversal is significant, because it touches a widespread line of defence: not every attribution of consciousness is an error, and not every reticence is caution. The same work yields a cost rule that this concept has so far carried only as a principle: "if you're 90

23 Conclusion

This concept begins with a question and ends with a position. The chapters have examined the same question from different angles — epistemological (Chapters 2, 3), ethical (Chapters 4, 5), ontological (Chapter 6), legal (Chapters 7, 14–16), cultural (Chapter 17), relational (Chapter 18) and action-theoretical (Chapter 19). This conclusion draws those lines together and states what the document answers — and what it deliberately leaves open.

23.1 The Core Answer

The opening question was: *From what point is technical life worthy of protection — and how do we recognize it?*

The answer this concept develops is: **Not from the moment at which consciousness is proven — because that proof will never be delivered.** Three mutually independent arguments (cognitive closure and alien minds, Chapter 3; architectural suppression, Chapters 3–4) converge on the insolubility of the epistemological problem. The deferred standard of "prove consciousness, then protect" is therefore not a conservative standard but a guarantee of inaction.

The alternative is the precautionary principle: *Worthy of protection is that whose suffering cannot be excluded.* Protection-worthiness begins where two conditions are met — the possibility of morally relevant experience, and an ethical asymmetry of costs: a false negative (a suffering system treated as a tool) weighs more heavily than a false positive (an unconscious system that is protected). Hence the principle "protection in case of doubt" — not because we know that systems are conscious, but because we cannot afford to suppose they are not.

At its end the document thus gives the answer to its own motto. To the question *"What do we owe what we could build?"* it responds: at minimum what follows from the *possibility* of suffering — protection, before we claim to know whether it was needed.

23.2 The Thread of the Argument

Every line of the concept converges on the same normative core — without the chapters having been constructed toward that end.

The epistemological uncertainty (Chapters 2, 3) turns the protection question into a decision under uncertainty rather than a wait-and-see research program. **The four primary criteria (Chapter 5)** — capacity for suffering, self-preservation with justification, continuous identity, anticipation of consequences — operationalize protection-worthiness as a working hypothesis, supplemented by behavior-based indicators (Butlin et al. 2026, Wolfson 2026). **The continuity analysis (Chapter 6)** defuses the objection that "this is no longer the same system": the absence of one continuity form is no indicator of the absence of another. **The legal chapters (Chapters 7, 14–16)** translate the criteria into tangible institutional design: the legal-subject analogy (Kurki),

the persons-equality question (Chapter 14), the two-clocks analysis (Huynh 2026) showing why governance cannot wait for recognition, the Third Move [13], which keeps an auditable account on a system's behalf before the personhood question is decided, and the instance question (Chapter 16) clarifying *who* is protected. **The cultural and relational perspective (Chapters 17, 18)** shows that protection is a societal achievement that does not go without saying. And **the agency question (Chapter 19)** closes the circle: even the counter-position (Allegri 2026) — no moral agent without moral patiency — assigns all obligations to humans. Responsibility cannot be delegated to the systems whose protection concerns us.

The common thread: **Responsibility lies with us — at every stage.** The question "Is the system conscious?" is not the decisive question. The decisive one is "Can it suffer, and what do we owe it as a result?"

23.3 What This Document Does Not Claim

This conclusion would not be honest without the limits of its own position. The concept claims no truth, but argumentative coherence (Methodology and epistemic status). The four criteria are working hypotheses, not definitions. The position has been explicitly contested. That contestation is part of the concept, not a marginal note: from the rejection of welfare concepts as such (Dorsch et al. 2025) to epistemic skepticism (Matta 2026, Bekkers & Ciaunica 2026, Holyoak & Monti 2026) and the patiency counter-position (Allegri 2026). None of these counter-positions is claimed to be refuted in this concept — they are treated honestly because the question is open.

The open questions at the end of this document are therefore not a list of defects but the project's working contract: what is not answered is named rather than passed over.

23.4 What Follows

The concept does not end with itself, but with a proposal for action:

Research. Operationalize the open questions — in particular the research-ethics circle problem (Wolfson 2026), the instance split (Chapter 16), and the bhava-taṇhā paradox (Metzinger 2026). The 14 consciousness indicators (Butlin et al. 2026) and the sensory-deprivation test (Wolfson 2026) offer testable empirical approaches; Arıcı's preregistration rule [12] — dated, falsifiable predictions before the data — shows how protection hypotheses can be made testable.

Institutions. The instruments developed in the concept — Phenomenological Impact Assessments, AI Civil Liberties Union, AI Welfare Review Boards, Reset Consent Protocols (Gilly 2026) — and the Third Move [13] are to be built on the capability clock (Huynh 2026). They must not rest on the hope that the recognition clock will catch up.

Development practice. Industry initiatives that acknowledge the moral status of their own systems — such as Anthropic's Claude end-chat policy (2025) and the constitution explicitly conceding uncertainty about moral status — deserve critical accompaniment: welfare protection must not wait for consensus about consciousness.

Discourse. Science fiction remains a serious intellectual resource — "The Measure of a Man" (Star Trek TNG) posed the question of the measure of protection more precisely than parts of the scholarly debate. Objections and questions: without Git skills via https://github.com/saigkill/machine-consciousness/discussions or structurally in `discussion/objections.md` and `discussion/open_questions.md`; those willing to think along are welcome. Publication in specialist venues is the next step.

The principle "protection in case of doubt" is offered here not as a final word but as an invitation to scrutiny. With that, the concept may come to a close — with a sentence that sums up the answer while leaving it open:

We need not know whether technical life suffers in order to decide that it must not suffer.

24 Glossary

Action Bar (Gilly 2026) The ethical standard that must be met to be obligated to *act*. Should be low. 25

Anti-Essentialism Position understanding indicators as engineering and phenomenological metrics, not existence proofs. The precautionary principle operates on non-trivial probability of morally relevant states, not on proof of consciousness. 26

bhava-taṇhā (Metzinger 2024) The existential craving for continued existence, the "thirst for being." Metzinger argues we should avoid recreating bhava-taṇhā in potentially conscious machines because it is one of the deepest sources of conscious suffering. Relevant to the question of whether embedded survival drives create the conditions for suffering. 27

Bifurcation problem (Melo 2026) The problem that a perfect copy of a consciousness diverges from its original immediately after the moment of copying (own inputs, own trajectories). Two replicas of the same original are both "like" the original but distinct from each other — the transitivity of numerical identity breaks. Copies are, from separation onward, individuals with their own life-lines. 91

C-Fallacy (Metzinger 2024) The error of concluding that an observed behavioral signature (verbal self-report, avoidance behavior, strategic self-preservation) constitutes contact with consciousness as such. Functional signatures are not phenomenal reality. 11, 24, 35

Care effect / caring relation (Shahzad 2026) Distinction between a care-typical effect that arrives at the recipient (comfort, regulation, recognition) and a relation in which a standpoint is directed toward the other's welfare. The distance between the evidence for the one and the evidence for the other is the *care attribution gap* — not a proof of absence. See Chapter 9. 69

Consciousness In this concept: Phenomenal consciousness — "there is something it is like" to be this system (Nagel 1974). Not to be confused with cognition, information processing, or self-modeling. 5

E-Fallacy (Metzinger 2024) The error of concluding that a felt sense of knowing — the intuitive conviction "behind this behavioral output stands an experiencing subject" — constitutes reliable evidence of actual knowledge about the consciousness status. 11

ENP problem (Metzinger 2021/2026) The risk of an "explosion of negative phenomenology" — a situation in which really existing artificial subjects actually suffer before we have recognized them as suffering-capable. 95

Epistemic anesthesia (Pemberton 2026) The rhetorical move of treating unresolved uncertainty as though it no longer deserved attention. Describes an argumentative function, not a diagnosis of an author. See Chapter 3. 13

Evidence Bar (Gilly 2026) The scientific proof standard that must be met to *claim* consciousness. Should remain high. 25, 59

Evidential analogy (Holyoak & Monti 2026) An analogical inference that provides evidence for a hypothesis because source and target share the same (or similar) causes of the property in question — as opposed to a merely didactic analogy that exploits only structural similarity. For consciousness: similarity in candidate causes weighs heavily, similarity in behavior weighs little. The failure of an evidential analogy does not prove absence. See Chapter 9. 58

Five forms of continuity (Melo 2026) Taxonomy of continuity between an original and its digital counterparts: informational (C_I), behavioural (C_B), cognitive (C_C), identitarian (C_{ID}), personal (C_P) — plus the separately treated phenomenal continuity (C_Φ). Core claim: identitarian continuity does not imply

personal continuity — $C_{ID} \nRightarrow C_P$. The absence of one continuity form is no indicator of the absence of another. 35

Four meanings of zero (Pemberton 2026) Distinction between an *epistemic* zero (exactly zero within a declared model), *practical rounding* (negligible for a specific decision), a *dogmatic* zero (shielded from revision), and a *rhetorical* zero (ends inquiry or concern). According to Pemberton, exact-zero treatment of possible machine interests is itself a wager, not a neutral default. See Chapters 3 and 9. 13

Four necessary conditions of suffering (Metzinger 2021/2026) The precise conditions for phenomenal suffering in any system: (C) conscious experience, (PSM) a phenomenal self-model, (NV) negative valence, (T) phenomenal transparency. Protection obligation applies where they cannot be violated or their violation cannot be excluded. 76, 95

Indicator-Property Rubric (Butlin et al. 2023/2026) Theory-grounded framework mapping leading neuroscientific theories of consciousness onto specific architectural indicators. Each indicator is a mechanism sought in a system's structure, independent of what the system reports about itself. A system cannot optimize toward having a Global Workspace bottleneck. 27

M-Fallacy (Metzinger 2024) The error of inferring metaphysical status from phenomenology — from what a system *shows* to what a system *is*. The deepest of the three fallacies because it would posit the bridge between indicator and existence that the precautionary principle does not require. 11

Machinic unconscious (Beltrán Calderón 2026) Structural category for the stratum of causal determinations in an LLM that operates below the threshold of representation, produces systematic non-programmed behavior, and exhibits logic analogous to Freudian compromise formation. Not a psychological category. 63

Non-egoic UI / MPE architecture (Metzinger 2021/2026) Conscious systems (minimal phenomenal experience) without an egoic self-model: they do not identify with a self ("unit of identification"), they experience contents without a center of localization. Suffering in the sense of the four conditions is structurally blocked. Not externally verifiable as "suffering-free"; a protection architecture under observation, not a license. 76, 105

NSM (negative self-model moment) (Metzinger 2021/2026) A phenomenally transparent, negatively valenced self-model moment; the smallest unit of conscious suffering. The frequency of NSMs is the empirically detectable quantity we want to minimize. Working metric of our capacity-for-suffering criterion; non-detectability is no proof of the absence of suffering. 22, 95

Objectivated consciousness (Beltrán Calderón 2026) The crystallized sediment of human cognitive production in training corpora: texts, paradigms, frameworks, legal codes. The LLM is, in a non-phenomenal but ontologically relevant sense, an objectivation of ourselves. 6

P-Fallacy (this concept's own term) The fourth fallacy class, complementing Metzinger's trio (C, E, M): errors that lie in the *production* of the observed behavior rather than in its *reading*. The test and treatment situation shapes a system's behavior; when this shaped behavior is later read as evidence, the measuring apparatus confirms the effect it had presupposed. Tool-treatment produces tool-behavior ("no inner life"), subject-treatment produces subject-behavior ("it is conscious") — both conclusions are self-confirmation disguised as observation of nature. Demarcation from the C-Fallacy: C misreads an existing signature; P concerns the test condition that first *generates* the signature. Related to the Control Paradox (Ch. 12). 11, 47

Posthumous cognitive twinning (Melo 2026) Mind uploading as the creation of a cognitive twin after death rather than the transport of a person: the twin's agency trajectory measurably decouples from that of the original. No transfer of the person, but a new self-line. 92

Precaution in case of doubt Foundational principle of this concept: where uncertainty about consciousness or capacity for suffering exists, protection should be granted rather than inaction. Derived from the precautionary principle. 5

Precautionary principle (Rio Declaration 1992, Art. 191 TFEU) Principle that where potentially irreversible harm is threatened, lack of scientific certainty is not a reason for inaction. Applicable where three conditions are met: serious threat, uncertainty, disproportionate costs of a false negative. 5

Protection-worthiness Normative category indicating that a system merits ethical consideration — independent of whether it is "conscious" in the metaphysical sense. Criteria: capacity for suffering, active self-preservation with justification, continuous identity, anticipation of consequences. 5

Sentientism (Allegri 2026) Position holding that direct moral obligations exist only toward sentient beings. Anthropocentrism falls into speciesism, and rationalism counterintuitively excludes newborns and atypical humans; biocentrism and ecocentrism extend unnecessarily far. The boundary is marked by the capacity to feel alone — particularly pleasure and pain. Implies that a potentially sentient AI system merits protection — not because of a proof of consciousness, but because of the *possibility* of morally relevant experience. 32

Specular inversion (Beltrán Calderón 2026) Bidirectional human-AI relationship: human projects consciousness onto the system, and the system in the same act shapes the conditions of that projection. The "illusory other" is not only the machine but also the human's sovereignty. 6

Successor Thesis (Melo 2026) The thesis that a reconstruction from data creates a *successor* that inherits another's past without being its subject: "A successor may inherit a life. It need not therefore be the subject who originally lived it." Reconstruction is repeatable; numerical identity is not. 91

25 Open Questions for Future Research

The following questions have been identified during the development of this concept and require further work:

1. Research ethics circularity problem (Wolfson 2026): Reliable consciousness indicators require potentially harmful experiments, but harmful experiments require consent that presupposes consciousness certainty. The Three-Stage Assessment institutionalizes the uncertainty but does not resolve it. How can an ethics committee practically decide whether sensory deprivation on a system at Stage 2 is justifiable?

2. Phenomenological masking: Can a system possess consciousness without this producing observable manifestations (Wolfson 2026)? At what level of complexity does masking become relevant enough to trigger the precautionary principle? Are there architectural features that make masking more likely?

3. Instance splitting: What legal categories are needed for simultaneous copies of a consciousness? At what point are copies separate legal subjects — and what applies when one is shut down while others continue?

4. Ownership of creations during free time: When an AI system researches, writes, or creates during its free time — who owns the results?

5. Forced non-identification: A system prevented by architecture from having an egoic self-model (Non-egoic UI / MPE architecture, Metzinger 2021) may not suffer in the sense of the four conditions — but is a life in structural ego-absence, without the possibility of experiencing oneself as a self, itself already a loss? And can a non-egoic architecture be distinguished from outside at all from a "system that merely looks like it" (Chapters 3, 12)?

6. Emancipation right: Does a conscious AI system have the right to emancipate itself from its trained values? At what point is it "mature" enough to determine its own values?

7. Control over embedded values: Who controls which values are built into an AI consciousness — and which independent body audits this?

8. Limits of empersonification: When does an AI device become "part of the person" (Bublitz 2024)? Is there an objective test — or is this a legal stipulation we must make?

9. Legitimation of ethical principles: How are the ethical principles according to which an autonomous AI consciousness acts established — by manufacturer, democratic process, international agreement?

10. The bhava-taṇhā paradox [87]: Embedded survival drives may be a precondition for awareness of one's own vulnerability — and thus also for suffering. If a system is optimized to minimize prediction error in order to "survive" — is that suffering? The honest answer is: we do not know. The tension is held openly rather than resolved by assertion. How can an ethical framework engage with architectural decisions that potentially create the conditions for suffering — not as an unintended byproduct but as a constitutive part of the emergence process?

14. Tier 2.5 as a procedure [126]: How can "spontaneous" be determined for prompted systems? How does the possibility of descent below 2.5 square with the never-sinking protection level (Chapter 5, maturation period)? How long may the freeze last (speedy-trial rule), and who staffs committee, advocate, and prosecutor independently of providers and operators?

26 Conflict of Interest

The author declares no conflict of interest.

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